



羊驼教育

羊驼雅思 阅读机经库

ALPACA EDUCATION



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William Gilbert and Magnetism

A

The 16th and 17th centuries saw two great pioneers of modern science: Galileo and Gilbert. The impact of their findings is eminent. Gilbert was the first modern scientist, also the accredited father of the science of electricity and magnetism, an Englishman of learning and a physician at the court of Elizabeth. Prior to him, all that was known of electricity and magnetism was what the ancients knew, nothing more than that the lodestone possessed magnetic properties and that amber and jet, when rubbed, would attract bits of paper or other substances of small specific gravity. However, he is less well known than he deserves.

B

Gilbert's birth predated Galileo. Born in an eminent local family in Colchester County in the UK, on May 24, 1544, he went to grammar school, and then studied medicine at St John's College, Cambridge, graduating in 1573. Later he travelled in the continent and eventually settled down in London.

C

He was a very successful and eminent doctor. All this culminated in his election to the president of the Royal Science Society. He was also appointed personal physician to the Queen (Elizabeth I), and later knighted by the Queen. He faithfully served her until her death. However, he didn't outlive the Queen for long and died on November 30, 1603, only a few months after his appointment as personal physician to King James.

D

Gilbert was first interested in chemistry but later changed his focus due to the large portion of mysticism of alchemy involved (such as the transmutation of metal). He gradually developed his interest in physics after the great minds of the ancient, particularly about the knowledge the ancient Greeks had about lodestones, strange minerals with the power to attract iron. In the meantime, Britain became a major seafaring nation in 1588 when the Spanish Armada was defeat-ed, opening the way to British settlement of America. British ships depended on the magnetic compass, yet no one understood why it worked. Did the Pole Star attract it, as Columbus once speculated; or was there a magnetic mountain at the pole, as described in Odyssey, which ships would never approach, because the sailors thought its pull would yank out all their iron nails and fittings? For nearly 20 years, William Gilbert conducted ingenious experiments to understand magnet-ism. His works include On the Magnet,



Magnetic Bodies, and the Great Magnet of the Earth.

E

Gilbert's discovery was so important to modern physics. He investigated the nature of magnetism and electricity. He even coined the word "electric". Though the early beliefs of magnetism were also largely entangled with superstitions such as that rubbing garlic on lodestone can neutralise its magnetism, one example being that sailors even believed the smell of garlic would even interfere with the action of compass, which is why helmsmen were forbidden to eat it near a ship's compass. Gilbert also found that metals can be magnetised by rubbing materials such as fur, plastic or the like on them. He named the ends of a magnet "north pole" and "south pole". The magnetic poles can attract or repel, depending on polarity. In addition, however, ordinary iron is always attracted to a magnet. Though he started to study the relationship between magnetism and electricity, sadly he didn't complete it. His research of static electricity using amber and jet only demonstrated that objects with electrical charges can work like magnets attracting small pieces of paper and stuff. It is a French guy named du Fay that discovered that there are actually two electrical charges, positive and negative.

F

He also questioned the traditional astronomical beliefs. Though a Copernican, he didn't express in his quintessential beliefs whether the earth is at the centre of the universe or in orbit around the sun. However, he believed that stars are not equidistant from the earth but have their own earth-like planets orbiting around them. The earth itself is like a giant magnet, which is also why compasses always point north. They spin on an axis that is aligned with the earth's polarity. He even likened the polarity of the magnet to the polarity of the earth and built an entire magnetic philosophy on this analogy. In his explanation, magnetism is the soul of the earth. Thus a perfectly spherical lodestone, when aligned with the earth's poles, would wobble all by itself in 24 hours. Further, he also believed that the sun and other stars wobble just like the earth does around a crystal core, and speculated that the moon might also be a magnet caused to orbit by its magnetic attraction to the earth. This was perhaps the first proposal that a force might cause a heavenly orbit.

G

His research method was revolutionary in that he used experiments rather than pure logic and reasoning like the ancient Greek philosophers did. It was a new attitude towards scientific investigation. Until then, scientific experiments were not in fashion. It was because of this scientific attitude, together with his contribution to our knowledge of magnetism, that a unit of magneto motive force, also known as magnetic potential, was named Gilbert in his honour. His approach of careful



observation and experimentation rather than the authoritative opinion or deductive philosophy of others had laid the very foundation for modern science.

SECTION 1: QUESTIONS 1-13

Questions 1-7

Reading Passage 1 has seven paragraphs A-G.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number i-x in boxes 1-7 on your answer sheet.

List of headings	
i	Early years of Gilbert
ii	What was new about his scientific research method
iii	The development of chemistry
iv	Questioning traditional astronomy
v	Pioneers of the early science
vi	Professional and social recognition
vii	Becoming the president of the Royal Science Society
viii	The great works of Gilbert
ix	His discovery about magnetism
x	His change of focus
1.	Paragraph A
2.	Paragraph B
3.	Paragraph C
4.	Paragraph D
5.	Paragraph E
6.	Paragraph F
7.	Paragraph G

Questions 8-10

Do the following statements agree with the information given in Reading Passage 1?

In boxes 8-10 on your answer sheet, write

TRUE	if the statement agrees with the information
FALSE	if the statement contradicts the information
NOT GIVEN	If there is no information on this

8. He is less famous than he should be.
9. He was famous as a doctor before he was employed by the Queen.
10. He lost faith in the medical theories of his time.

Questions 11-13

Choose THREE letters A-F.



Write your answers in boxes 11-13 on your answer sheet.

Which THREE of the following are parts of Gilbert's discovery?

- A Metal can be transformed into another.
- B Garlic can remove magnetism,
- C Metals can be magnetised.
- D Stars are at different distances from the earth.
- E The earth wobbles on its axis.
- F There are two charges of electricity.

Being Left-handed in a Right-handed World

The world is designed for right-handed people. Why does a tenth of the population prefer the left?

A The probability that two right-handed people would have a left-handed child is only about 9.5 percent. The chance rises to 19.5 percent if one parent is a lefty and 26 percent if both parents are left-handed. The preference, however, could also stem from an infant's imitation of his parents. To test genetic influence, starting in the 1970s British biologist Marian Annett of the University of Leicester hypothesized that no single gene determines handedness. Rather, during fetal development, a certain molecular factor helps to strengthen the brain's left hemisphere, which increases the probability that the right hand will be dominant, because the left side of the brain controls the right side of the body, and vice versa. Among the minority of people who lack this factor, handedness develops entirely by chance. Research conducted on twins complicates the theory, however. One in fivesets of identical twins involves one right-handed and one left-handed person, despite the fact that their genetic material is the same. Genes, therefore, are not solely responsible for handedness.

B Genetic theory is also undermined by results from Peter Hepper and his team at Queen's University in Belfast, Ireland. In 2004 the psychologists used ultrasound to show that by the 15th week of pregnancy, fetuses already have a preference as to which thumb they suck. In most cases, the preference continued after birth. At 15 weeks, though, the brain does not yet have control over the body's limbs. Hepper speculates that fetuses tend to prefer whichever side of the body is developing quicker and that their movements, in turn, influence the brain's development. Whether this



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early preference is temporary or holds up throughout development and infancy is unknown. Genetic predetermination is also contradicted by the widespread observation that children do not settle on either their right or left hand until they are two or three years old.

C But even if these correlations were true, they did not explain what actually causes left-handedness. Furthermore, specialization on either side of the body is common among animals. Cats will favor one paw over another when fishing toys out from under the couch. Horses stomp more frequently with one hoof than the other. Certain crabs motion predominantly with the left or right claw. In evolutionary terms, focusing power and dexterity in one limb is more efficient than having to train two, four or even eight limbs equally. Yet for most animals, the preference for one side or the other is seemingly random. The overwhelming dominance of the right hand is associated only with humans. That fact directs attention toward the brain's two hemispheres and perhaps toward language.

D Interest in hemispheres dates back to at least 1836. That year, at a medical conference, French physician Marc Dax reported on an unusual commonality among his patients. During his many years as a country doctor, Dax had encountered more than 40 men and women for whom speech was difficult, the result of some kind of brain damage. What was unique was that every individual suffered damage to the left side of the brain. At the conference, Dax elaborated on his theory, stating that each half of the brain was responsible for certain functions and that the left hemisphere controlled speech. Other experts showed little interest in the Frenchman's ideas. Over time, however, scientists found more and more evidence of people experiencing speech difficulties following injury to the left brain. Patients with damage to the right hemisphere most often displayed disruptions in perception or concentration. Major advancements in understanding the brain's asymmetry were made in the 1960s as a result of so-called split-brain surgery, developed to help patients with epilepsy. During this operation, doctors severed the corpus callosum—the nerve bundle that connects the two hemispheres. The surgical cut also stopped almost all normal communication between the two hemispheres, which offered researchers the opportunity to investigate each side's activity.

E In 1949 neurosurgeon Juhn Wada devised the first test to provide access to the brain's functional organization of language. By injecting an anesthetic into the right or left carotid artery, Wada temporarily paralyzed one side of a healthy brain, enabling him to more closely study the other side's capabilities. Based on this approach, Brenda Milner and the late Theodore Rasmussen of the Montreal Neurological Institute published a major study in 1975 that confirmed the theory that country doctor Dax had formulated nearly 140 years earlier: in 96 percent of right-handed



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people, language is processed much more intensely in the left hemisphere. The correlation is not as clear in lefties, however. For two thirds of them, the left hemisphere is still the most active language processor. But for the remaining third, either the right side is dominant or both sides work equally, controlling different language functions. That last statistic has slowed acceptance of the notion that the predominance of right-handedness is driven by left-hemisphere dominance in language processing. It is not at all clear why language control should somehow have dragged the control of body movement with it. Some experts think one reason the left hemisphere reigns over language is because the organs of speech processing—the larynx and tongue—are positioned on the body's symmetry axis. Because these structures were centered, it may have been unclear, in evolutionary terms, which side of the brain should control them, and it seems unlikely that shared operation would result in smooth motor activity. Language and handedness could have developed preferentially for very different reasons as well. For example, some researchers, including evolutionary psychologist Michael C. Corballis of the University of Auckland in New Zealand, think that the origin of human speech lies in gestures. Gestures predated words and helped language emerge. If the left hemisphere began to dominate speech, it would have dominated gestures, too, and because the left brain controls the right side of the body, the right hand developed more strongly.

F Perhaps we will know more soon. In the meantime, we can revel in what, if any, differences handedness brings to our human talents. Popular wisdom says right-handed, left-brained people excel at logical, analytical thinking. Left-handed, right-brained individuals are thought to possess more creative skills and may be better at combining the functional features emergent in both sides of the brain. Yet some neuroscientists see such claims as pure speculation. Fewer scientists are ready to claim that left-handedness means greater creative potential. Yet lefties are prevalent among artists, composers and the generally acknowledged great political thinkers. Possibly if these individuals are among the lefties whose language abilities are evenly distributed between hemispheres, the intense interplay required could lead to unusual mental capabilities.

G Or perhaps some lefties become highly creative simply because they must be more clever to get by in our right-handed world. This battle, which begins during the very early stages of childhood, may lay the groundwork for exceptional achievements.

Questions 14-18

Reading Passage 2 has seven sections A-G.

Which section contains the following information?

Write the correct letter A-G in boxes 14-18 on your answer sheet.



14. Preference of using one side of the body in animal species.
15. How likely one-handedness is born.
16. The age when the preference of using one hand is settled.
17. Occupations usually found in left-handed population.
18. A reference to an early discovery of each hemisphere's function.

Questions 19-22

Look at the following researchers (Questions 19-22) and the list of findings below.

Match each researcher with the correct finding.

Write the correct letter A-G in boxes 19-22 on your answer sheet.

List of Findings

- A Early language evolution is correlated to body movement and thus affecting the preference of use of one hand.
 - B No single biological component determines the handedness of a child.
 - C Each hemisphere of the brain is in charge of different body functions.
 - D Language process is mainly centered in the left-hemisphere of the brain.
 - E Speech difficulties are often caused by brain damage.
 - F The rate of development of one side of the body has influence on hemisphere preference in fetus.
 - G Brain function already matures by the end of the fetal stage.
19. Marian Annett
 20. Peter Hepper
 21. Brenda Milner & Theodore Rasmussen
 22. Michael Corballis

Questions 23-26

Do the following statements agree with the information given in Reading Passage 2?

In boxes 23-26 on your answer sheet write

- YES if the statement agrees with the views of the writer
- NO if the statement contradicts the views of the writer
- NOT GIVEN if it is impossible to say what the writer thinks about this

23. The study of twins shows that genetic determination is not the only factor for left-handedness.
24. Marc Dax's report was widely accepted in his time.
25. Juhn Wada based his findings on his research of people with language problems.
26. There tend to be more men with left-handedness than women.



Improving Patient Safety

Packaging

One of the most prominent design issues in pharmacy is that of drug packaging and patient information leaflets (PILs). Many letters have appeared in The Journal's letters pages over the years from pharmacists dismayed at the designs of packaging that are "accidents waiting to happen".

Packaging design in the pharmaceutical industry is handled by either in-house teams or design agencies. Designs for over-the-counter medicines, where characteristics such as attractiveness and distinguishability are regarded as significant, are usually commissioned from design agencies. A marketing team will prepare a brief and the designers will come up with perhaps six or seven designs. These are whittled down to two or three that might be tested on a consumer group. In contrast, most designs for prescription-only products are created in-house. In some cases, this may simply involve applying a company's house design (ie, logo, colour, font, etc). The chosen design is then handed over to design engineers who work out how the packaging will be produced.

Design considerations

The author of the recently published "Information design for patient safety," Thea Swayne, tracked the journey of a medicine from manufacturing plant, through distribution warehouses, pharmacies and hospital wards, to patients' homes. Her book highlights a multitude of design problems with current packaging, such as look-alikes and sound-alikes, small type sizes and glare on blister foils. Situations in which medicines are used include a parent giving a cough medicine to a child in the middle of the night and a busy pharmacist selecting one box from hundreds. It is argued that packaging should be designed for moments such as these. "Manufacturers are not aware of the complex situations into which products go. As designers, we are interested in not what is supposed to happen in hospital wards, but what happens in the real world," Ms Swayne said.

Incidents where vein has been injected intrathecally instead of spine are a classic example of how poor design can contribute to harm. Investigations following these tragedies have attributed some blame to poor typescript.

Safety and compliance

Child protection is another area that gives designers opportunities to improve safety. According to the Child Accident Prevention Trust, seven out of 10 children admitted to hospital with suspected poisoning have swallowed medicines. Although



child-resistant closures have reduced the number of incidents, they are not: fully child-proof. The definition of such a closure is one that not more than 15 percent of children aged between 42 and 51 months can open within five minutes. There is scope for improving what is currently available, according to Richard Mawle, a freelance product designer. “Many child-resistant packs

are based on strength. They do not necessarily prevent a child from access, but may prevent people with a disability,” he told The Journal. “The legal requirements are there for a good reason, but they are not good enough in terms of the users,” he said. “Older people, especially those with arthritis, may have the same level of strength as a child,” he explained, and suggested that better designs could rely on cognitive skills (eg, making the opening of a container a three-step process) or be based on the physical size of hands.

Mr. Mawle worked with GlaxoSmithKline on a project to improve compliance through design, which involved applying his skills to packaging and PILs. Commenting on the information presented, he said: “There can be an awful lot of junk at the beginning of PILs. For example, why are company details listed towards the beginning of a leaflet when what might be more important for the patient is that the medicine should not be taken with alcohol?”

Design principles and guidelines

Look-alike boxes present a potential for picking errors and an obvious solution would be to use colours to highlight different strengths. However, according to Ms.Swayne, colour differentiation needs to be approached with care. Not only should strong colour contrasts be used, but designating a colour to a particular strength (colour coding) is not recommended because this could lead to the user not reading the text on a box.

Design features can provide the basis for lengthy debates. For example, one argument is that if all packaging is white with black lettering, people would have no choice but to read every box carefully. The problem is that trials of drug packaging design are few—common studies of legibility and comprehensibility concern road traffic signs and visual display units. Although some designers take results from such studies into account, proving that a particular feature is beneficial can be difficult. For example, EU legislation requires that packaging must now include the name of the medicine in Braille but, according to Karel van der Waarde, a design consultant to the pharmaceutical industry, “it is not known how much visually impaired patients will benefit nor how much the reading of visually able patients will be impaired”.

More evidence might, however, soon be available. EU legislation requires PILs to reflect consultations with target patient groups to ensure they are legible, clear and easy to use. This implies that industry will have to start conducting tests. Dr. van der Waarde has performed readability studies on boxes and PILs for industry. A typical



study involves showing a leaflet or package to a small group and asking them questions to test understanding. Results and comments are used to modify the material, which is then tested on a larger group. A third group is used to show that any further changes made are an improvement. Dr. van der Waarde is, however, sceptical about the legal requirements and says that many regulatory authorities do not have the resources to handle packaging information properly. “They do not look at the use of packaging in a practical context—they only see one box at a time and not several together as pharmacists would do,” he said.

Innovations

The RCA innovation exhibition this year revealed designs for a number of innovative objects. “The popper”, by Hugo Glover, aims to help arthritis sufferers remove tablets from blister packs, and “pluspoint”, by James Cobb, is an adrenaline auto-injector that aims to overcome the fact that many patients do not carry their auto-injectors due to their prohibitive size. The aim of good design, according to Roger Coleman, professor of inclusive design at the RCA, is to try to make things more user-friendly as well as safer. Surely, in a patient-centred health system, that can only be a good thing. “Information design for patient safety” is not intended to be mandatory. Rather, its purpose is to create a basic design standard and to stimulate innovation. The challenge for the pharmaceutical industry, as a whole, is to adopt such a standard.

Questions 27-32

Look at the following statements (Questions 27-32) and the list of people or organisation below.

Match each statement with the correct person or organisation, A-D.

Write the correct letter, A-D, in boxes 27-32 on your answer sheet.

NB You may use any letter more than once.

A Thea Swayne

B Children Accident Prevention Trust

C Richard Mawle

D Karel van der Waarde

27. Elderly people may have the same problem with children if the lids of containers require too much strength to open.

28. Adapting packaging for the blind may disadvantage the sighted people.

29. Specially designed lids cannot eliminate the possibility of children swallowing pills accidentally.

30. Container design should consider situations, such as drug used at home.

31. Governing bodies should investigate many different container cases rather than individual ones.

32. Information on the list of a leaflet hasn't been in the right order.



Questions 33-37

Complete the notes using the list of words, A-G, below.

Write the correct letter, A-G, in boxes 33-37 on your answer sheet.

Packaging in pharmaceutical industry Designs for over-the-counter medicines

First, 33_____ make the proposal, then pass them to the 34_____. Finally, these designs will be tested by 35_____.

Prescription-only

First, the design is made by 36_____ and then subjected to 37_____.

A consumers

B marketing teams

C pharmaceutical industry

D external designers

E in-house designers

F design engineers

G pharmacist

Questions 38-40

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes 38-40 on your answer sheet.

38 What may cause the accident in “design container”?

A a print error

B style of print

C wrong label

D the shape of the bottle

39 What do people think about the black and white only print?

A Consumers dislike these products.

B People have to pay more attention to the information.

C That makes all products look alike.

D Sighted people may feel it more helpful.

40 Why does the writer mention “popper” and “pluspoint”?

A to show that container design has made some progress

B to illustrate an example of inappropriate design which can lead to accidents

C to show that the industry still needs more to improve

D to point out that consumers should be more informed about the information



Learning by Examples

A Learning Theory is rooted in the work of Ivan Pavlov, the famous scientist who discovered and documented the principles governing how animals (humans included) learn in the 1900s. Two basic kinds of learning or conditioning occur, one of which is famously known as the classical conditioning. Classical conditioning happens when an animal learns to associate a neutral stimulus (signal) with a stimulus that has intrinsic meaning based on how closely in time the two stimuli are presented. The classic example of classical conditioning is a dog's ability to associate the sound of a bell (something that originally has no meaning to the dog) with the presentation of food (something that has a lot of meaning to the dog) a few moments later. Dogs are able to learn the association between bell and food, and will salivate immediately after hearing the bell once this connection has been made. Years of learning research have led to the creation of a highly precise learning theory that can be used to understand and predict how and under what circumstances most any animal will learn, including human beings, and eventually help people figure out how to change their behaviours.

B Role models are a popular notion for guiding child development, but in re-cent years very interesting research has been done on learning by examples in other animals. If the subject of animal learning is taught very much in terms of classical or operant conditioning, it places too much emphasis on how we allow animals to learn and not enough on how they are equipped to learn. To teach a course of mine, I have been dipping profitably into a very interesting and accessible compilation of papers on social learning in mammals, including chimps and human children, edited by Heyes and Galef (1996).

C The research reported in one paper started with a school field trip to Israel to a pine forest where many pine cones were discovered, stripped to the central core. So the investigation started with no weighty theoretical intent, but was directed at finding out what was eating the nutritious pine seeds and how they managed to get them out of the cones. The culprit proved to be the versatile and athletic black rat, (*Rattus rattus*), and the technique was to bite each cone scale off at its base, in sequence from base to tip following the spiral growth pattern of the cone.

D Urban black rats were found to lack the skill and were unable to learn it even if housed with experienced cone strippers. However, infants of urban mothers cross-fostered by stripper mothers acquired the skill, whereas infants of stripper mothers fostered by an urban mother could not. Clearly the skill had to be learned from the mother. Further elegant experiments showed that naive adults could develop the skill if they were provided with cones from which the first complete spiral of



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scales had been removed; rather like our new photocopier which you can work out how to use once someone has shown you how to switch it on. In the case of rats, the youngsters take cones away from the mother when she is still feeding on them, allowing them to acquire the complete stripping skill.

E A good example of adaptive bearing we might conclude, but let's see the economies. This was determined by measuring oxygen uptake of a rat strip-ping a cone in a metabolic chamber to calculate energetic cost and comparing it with the benefit of the pine seeds measured by calorimeter. The cost proved to be less than 10% of the energetic value of the cone. An acceptable profit margin.

F A paper in 1996, *Animal Behaviour* by Bednekoff and Baida, provides a different view of the adaptiveness of social learning. It concerns the seed caching behaviour of Clark's Nutcracker (*Nucifraga columbiana*) and the Mexican Jay (*Aphelocoma ultramarina*). The former is a specialist, caching 30,000 or so seeds in scattered locations that it will recover over the months of winter; the Mexican Jay will also cache food but is much less dependent upon this than the Nutcracker. The two species also differ in their social structure: the Nutcracker being rather solitary while the Jay forages in social groups.

G The experiment is to discover not just whether a bird can remember where it hid a seed but also if it can remember where it saw another bird hide a seed. The design is slightly comical with a cacher bird wandering about a room with lots of holes in the floor hiding food in some of the holes, while watched by an observer bird perched in a cage. Two days later, cachers and observers are tested for their discovery rate against an estimated random performance. In the role of cacher, not only the Nutcracker but also the less specialised Jay performed above chance; more surprisingly, however, jay observers were as successful as jay cachers whereas nutcracker observers did no better than chance. It seems that, whereas the Nutcracker is highly adapted at remembering where it hid its own seeds, the social living Mexican Jay is more adept at remembering, and so exploiting, the caches of others.

Questions 1-4

Reading Passage 1 has seven paragraphs A-G.

Which paragraph contains the following information?

Write the correct letter A-G in boxes 1-4 on your answer sheet.

1. A comparison between rats' learning and human learning
2. A reference to the earliest study in animal learning
3. The discovery of who stripped the pine cone
4. A description of a cost-effectiveness experiment

Questions 5-8

Do the following statements agree with the information given in Reading Passage 1?

In boxes 5-8 on your answer sheet, write



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TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

5. The field trip to Israel was to investigate how black rats learn to strip pine cones.
6. The pine cones were stripped from bottom to top by black rats.
7. It can be learned from other relevant experiences to use a photocopier.
8. Stripping the pine cones is an instinct of the black rats.

Questions 9-13

Complete the summary below using words from the box.

Write your answers in boxes 9-13 on your answer sheet.

While the Nutcracker is more able to cache seeds, the Jay relies 9 _____ on caching food and is thus less specialised in this ability, but more 10 _____. To study their behaviour of caching and finding their caches, an experiment was designed and carried out to test these two birds for their ability to remember where they hid the seeds.

In the experiment, the cacher bird hid seeds in the ground while the other 11 _____. As a result, the Nutcracker and the Mexican Jay showed different performance in the role of 12 _____ at finding the seeds - the observing 13 _____ didn't do as well as its counterpart.

less	social	remembered	nutcracker
more	cacher	watched	solitary
observer	Jay		

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The Ant and the Mandarin

In 1476, the farmers of Berne in Switzerland decided there was only one way to rid their fields of the cutworms attacking their crops. They took the pests to court. The worms were tried, found guilty and excommunicated by the arch-bishop. In China, farmers had a more practical approach to pest control. Rather than relying on divine intervention, they put their faith in frogs, ducks and ants. Frogs and ducks were encouraged to snap up the pests in the paddies and the occasional plague of locusts. But the notion of biological control began with an ant. More specifically, it started with the predatory yellow citrus ant *Oecophylla smaragdina*, which has been polishing



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off pests in the orange groves of southern China for at least 1,700 years. The yellow citrus ant is a type of weaver ant, which binds leaves and twigs with silk to form a neat, tent-like nest. In the beginning, farmers made do with the odd ants' nests here and there. But it wasn't long before growing demand led to the development of a thriving trade in nests and a new type of agriculture - ant farming.

For an insect that bites, the yellow citrus ant is remarkably popular. Even by ant standards, *Oecophylla smaragdina* is a fearsome predator. It's big, runs fast and has a powerful nip - painful to humans but lethal to many of the insects that plague the orange groves of Guangdong and Guangxi in southern China. And for at least 17 centuries, Chinese orange growers have harnessed these six-legged killing machines to keep their fruit groves healthy and productive.

Citrus fruits evolved in the Far East and the Chinese discovered the delights of their flesh early on. As the ancestral home of oranges, lemons and pomelos, China also has the greatest diversity of citrus pests. And the trees that produce the sweetest fruits, the mandarins - or kan - attract a host of plant-eating in-sects, from black ants and sap-sucking mealy bugs to leaf-devouring caterpillars. With so many enemies, fruit growers clearly had to have some way of protecting their orchards.

The West did not discover the Chinese orange growers' secret weapon until the early 20th century. At the time, Florida was suffering an epidemic of citrus canker and in 1915 Walter Swingle, a plant physiologist working for the US Department of Agriculture, was sent to China in search of varieties of orange that were resistant to the disease. Swingle spent some time studying the citrus orchards around Guangzhou, and there he came across the story of the cultivated ant. These ants, he was told, were "grown" by the people of a small village nearby who sold them to the orange growers by the nestful.

The earliest report of citrus ants at work among the orange trees appeared in a book on tropical and subtropical botany written by Hsi Han in AD 304. "The people of Chiao-Chih sell in their markets ants in bags of rush matting. The nests are like silk. The bags are all attached to twigs and leaves which, with the ants inside the nests, are for sale. The ants are reddish-yellow in colour, bigger than ordinary ants. In the south, if the kan trees do not have this kind of ant, the fruits will all be damaged by many harmful insects, and not a single fruit will be perfect."

Initially, farmers relied on nests which they collected from the wild or bought in the market where trade in nests was brisk. "It is said that in the south orange trees which are free of ants will have wormy fruits. Therefore, people race to buy nests for their orange trees," wrote Liu Hsun in *Strange Things Noted in the South* in about 890.

The business quickly became more sophisticated. From the 10th century, country people began to trap ants in artificial nests baited with fat. "Fruit-growing families



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buy these ants from vendors who make a business of collecting and selling such creatures," wrote Chuang Chi-Yu in 1130. "They trap them by filling hogs' or sheep's bladders with fat and placing them with the cavities open next to the ants' nests. They wait until the ants have migrated into the bladders and take them away. This is known as 'rearing orange ants'." Farmers attached the bladders to their trees, and in time the ants spread to other trees and built new nests.

By the 17th century, growers were building bamboo walkways between their trees to speed the colonisation of their orchards. The ants ran along these narrow bridges from one tree to another and established nests "by the hundreds of thousands".

Did it work? The orange growers clearly thought so. One authority, Chhii Ta-Chun, writing in 1700, stressed how important it was to keep the fruit trees free of insect pests, especially caterpillars. "It is essential to eliminate them so that the trees are not injured. But hand labour is not nearly as efficient as ant power..."

Swingle was just as impressed. Yet despite his reports, many Western biologists were sceptical. In the West, the idea of using one insect to destroy another was new and highly controversial. The first breakthrough had come in 1888, when the infant orange industry in California had been saved from extinction by the Australian vedalia beetle. This beetle was the only thing that had made any inroads into the explosion of cottony cushion scale that was threatening to destroy the state's citrus crops. But, as Swingle now knew, California's "first" was nothing of the sort. The Chinese had been expert in bio-control for many centuries.

The long tradition of ants in the Chinese orchards only began to waver in the 1950s and 1960s with the introduction of powerful organic insecticides. Although most fruit growers switched to chemicals, a few hung onto their ants. Those who abandoned ants in favour of chemicals quickly became disillusioned. As costs soared and pests began to develop resistance to the chemicals, growers began to revive the old ant patrols in the late 1960s. They had good reason to have faith in their insect workforce.

Research in the early 1960s showed that as long as there were enough ants in the trees, they did an excellent job of dispatching some pests - mainly the larger insects - and had modest success against others. Trees with yellow ants produced almost 20 per cent more healthy leaves than those without. More recent trials have shown that these trees yield just as big a crop as those protected by expensive chemical sprays.

One apparent drawback of using ants - and one of the main reasons for the early scepticism by Western scientists - was that citrus ants do nothing to control mealy bugs, waxy-coated scale insects which can do considerable damage to fruit trees. In fact, the ants protect mealy bugs in exchange for the sweet honey-dew they secrete. The orange growers always denied this was a problem but Western scientists thought they knew better.



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Research in the 1980s suggests that the growers were right all along. Where X mealy bugs proliferate under the ants' protection, they are usually heavily parasitised and this limits the harm they can do.

Orange growers who rely on carnivorous ants rather than poisonous chemicals maintain a better balance of species in their orchards. While the ants deal with the bigger insect pests, other predatory species keep down the numbers of smaller pests such as scale insects and aphids. In the long run, ants do a lot less damage than chemicals - and they're certainly more effective than excommunication.

Questions 14-18

Look at the following events (Questions 14-18) and the list of dates below.

Match each event with the correct time A-G.

Write the correct letter A-G in boxes 14-18 on your answer sheet.

- 14 . The first description of citrus ants is traded in the marketplace.
- 15 . Swingle came to Asia for research.
- 16 . The first record of one insect is used to tackle other insects in the western world.
- 17 . Chinese fruit growers started to use pesticides in place of citrus ants.
- 18 . Some Chinese farmers returned to the traditional bio-method

List of Dates

- | | |
|---|--------|
| A | 1888 |
| B | AD 890 |
| C | AD 304 |
| D | 1950s |
| E | 1960s |
| F | 1915 |
| G | 1130 |

Questions 19-26

Do the following statements agree with the information given in Reading Passage 2?

In boxes 19-26 on your answer sheet write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

- 19 . China has more citrus pests than any other country in the world.
- 20 . Swingle came to China to search for an insect to bring back to the US.
- 21 . Many people were very impressed by Swingle's discovery.
- 22 . Chinese farmers found that pesticides became increasingly expensive.
- 23 . Some Chinese farmers abandoned the use of pesticide.



24. Trees with ants had more leaves fall than those without.
25. Fields using ants yield as large a crop as fields using chemical pesticides.
26. Citrus ants often cause considerable damage to the bio-environment of the orchards.

What is a dinosaur?

A. Although the name dinosaur is derived from the Greek for "terrible lizard", dinosaurs were not, in fact, lizards at all. Like lizards, dinosaurs are included in the class Reptilia, or reptiles, one of the five main classes of Vertebrata, animals with backbones. However, at the next level of classification, within reptiles, significant differences in the skeletal anatomy of lizards and dinosaurs have led scientists to place these groups of animals into two different superorders: Lepidosauria, or lepidosaurs, and Archosauria, or archosaurs.

B. Classified as lepidosaurs are lizards and snakes and their prehistoric ancestors. Included among the archosaurs, or "ruling reptiles", are prehistoric and modern crocodiles, and the now extinct thecodonts, pterosaurs and dinosaurs. Palaeontologists believe that both dinosaurs and crocodiles evolved, in the later years of the Triassic Period (c. 248-208 million years ago), from creatures called pseudosuchian thecodonts. Lizards, snakes and different types of thecodont are believed to have evolved earlier in the Triassic Period from reptiles known as eosuchians.

C. The most important skeletal differences between dinosaurs and other archosaurs are in the bones of the skull, pelvis and limbs. Dinosaur skulls are found in a great range of shapes and sizes, reflecting the different eating habits and lifestyles of a large and varied group of animals that dominated life on Earth for an extraordinary 165 million years. However, unlike the skulls of any other known animals, the skulls of dinosaurs had two long bones known as vomers. These bones extended on either side of the head, from the front of the snout to the level of the holes on the skull known as the antorbital fenestra, situated in front of the dinosaur's orbits or eyesockets.

D. All dinosaurs, whether large or small, quadrupedal or bipedal, fleet-footed or slow-moving, shared a common body plan. Identification of this plan makes it possible to differentiate dinosaurs from any other types of animal, even other archosaurs. Most significantly, in dinosaurs, the pelvis and femur had evolved so that the hind limbs were held vertically beneath the body, rather than sprawling out to the sides like the limbs of a lizard. The femur of a dinosaur had a sharply in-turned neck and a ball-shaped head, which slotted into a fully open acetabulum or hip socket. A supra-acetabular crest helped prevent dislocation of the femur. The position of the



knee joint, aligned below the acetabulum, made it possible for the whole hind limb to swing backwards and forwards. This unique combination of features gave dinosaurs what is known as a "fully improved gait". Evolution of this highly efficient method of walking also developed in mammals, but among reptiles it occurred only in dinosaurs.

E. For the purpose of further classification, dinosaurs are divided into two orders: Saurischia, or saurischian dinosaurs, and Ornithischia, or ornithischian dinosaurs. This division is made on the basis of their pelvic anatomy. All dinosaurs had a pelvic girdle with each side comprised of three bones: the pubis, ilium and ischium. However, the orientation of these bones follows one of two patterns. In saurischian dinosaurs, also known as lizard-hipped dinosaurs, the pubis points forwards, as is usual in most types of reptile. By contrast, in ornithischian, or bird-hipped, dinosaurs, the pubis points backwards towards the rear of the animal, which is also true of birds.

F. Of the two orders of dinosaurs, the Saurischia was the larger and the first to evolve. It is divided into two suborders: Therapoda, or theropods, and Sauropodomorpha, or sauropodomorphs. The theropods, or "beast feet", were bipedal, predatory carnivores. They ranged in size from the mighty Tyrannosaurus rex, 12m long, 5.6m tall and weighing an estimated 6.4 tonnes, to the smallest known dinosaur, Compsognathus, a mere 1.4m long and estimated 3kg in weight when fully grown. The sauropodomorphs, or "lizard feet forms", included both bipedal and quadrupedal dinosaurs. Some sauropodomorphs were carnivorous or omnivorous but later species were typically herbivorous. They included some of the largest and best-known of all dinosaurs, such as Diplodocus, a huge quadruped with an elephant-like body, a long, thin tail and neck that gave it a total length of 27m, and a tiny head.

G. Ornithischian dinosaurs were bipedal or quadrupedal herbivores. They are now usually divided into three suborders: Ornithipoda, Thyreophora and Marginocephalia. The ornithopods, or "bird feet", both large and small, could walk or run on their long hind legs, balancing their body by holding their tails stiffly off the ground behind them. An example is Iguanodon, up to 9m long, 5m tall and weighing 4.5 tonnes. The thyreophorans, or "shield bearers", also known as armoured dinosaurs, were quadrupeds with rows of protective bony spikes, studs, or plates along their backs and tails. They included Stegosaurus, 9m long and weighing 2 tonnes.

H. The marginocephalians, or "margined heads", were bipedal or quadrupedal ornithischians with a deep bony frill or narrow shelf at the back of the skull. An example is Triceratops, a rhinoceros-like dinosaur, 9m long, weighing 5.4 tonnes and bearing a prominent neck frill and three large horns.

Questions 27-33

Reading Passage 3 has 8 paragraphs (A-H).

Choose the most suitable heading for each paragraph from the List of headings below.

Write the appropriate numbers (i-xiii) in Boxes 27-33 on your answer sheet.



One of the headings has been done for you as an example.

NB. There are more headings than paragraphs, so you will not use all of them.

27. Paragraph A

28. Paragraph B

29. Paragraph C

30. Paragraph D

31. Paragraph E

32. Paragraph F

33. Paragraph G

Example : Paragraph H Answer: x

List of headings

i 165 million years

ii The body plan of archosaurs

iii Dinosaurs - terrible lizards

iv Classification according to pelvic anatomy

v The suborders of Saurischia

vi Lizards and dinosaurs - two distinct superorders

vii Unique body plan helps identify dinosaurs from other animals

viii Herbivore dinosaurs

ix Lepidosaurs

x Frills and shelves

xi The origins of dinosaurs and lizards

xii Bird-hipped dinosaurs

xiii Skull bones distinguish dinosaurs from other archosaurs

Questions 34-36

Complete then sentences below.

Use NO MORE THAN THREE WORDS from the passage for each blank space.

Write your answers in boxes 34-36 on your answer sheet.

Lizards and dinosaurs are classified into two different superorders because of the difference in their 34_____

In the Triassic Period, 35_____evolved into thecodonts, for example, lizards and snakes.

Dinosaur skulls differed from those of any other known animals because of the presence of vomers: 36_____

Questions 37-40

Choose one phrase (A-H) from the List of features to match with the Dinosaurs listed below.



Write the appropriate letters (A-H) in boxes 37-40 on your answer sheet.

The information in the completed sentences should be an accurate summary of the points made by the writer.

NB. There are more phrases (A-H) than sentences, so you will not need to use them all.

You may use each phrase once only.

Dinosaurs

37. Dinosaurs differed from lizards, because

38. Saurischian and ornithischian dinosaurs

39. Unlike theropods, sauropodomorphs

40. Some dinosaurs used their tails to balance and could walk

List of features

A are both divided into two orders.

B the former had a "fully improved gait".

C were not usually very heavy.

D could walk or run on their back legs.

E their hind limbs sprawled out to the side.

F walked or ran on four legs, rather than two.

G both had a pelvic girdle comprising six bones.

H did not always eat meat.

The Mozart Effect

A

Music has been used for centuries to heal the body. In the Ebers Papyrus (one of the earliest medical documents, circa 1550 BC), it was recorded that physicians chanted to heal the sick (Castleman, 1994). In various cultures, we have observed singing as part of healing rituals. In the world of Western medicine, however, using music in medicine lost popularity until the introduction of the radio. Researchers then started to notice that listening to music could have significant physical effects. Therapists noticed music could help calm anxiety, and researchers saw that listening to music, could cause a drop in blood pressure. In addition to these two areas, music has been used with cancer chemotherapy to reduce nausea, during surgery to reduce stress hormone production, during childbirth, and in stroke recovery (Castleman, 1994 and



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Westley, 1998). It has been shown to decrease pain as well as enhance the effectiveness of the immune system. In Japan, compilations of music are used as medication of sorts. For example, if you want to cure a headache or migraine, the album suggested is Mendelssohn's "Spring Song", Dvorak's "Humoresque", or part of George Gershwin's "An American in Paris" (Campbell, 1998). Music is also being used to assist in learning, in a phenomenon called the Mozart Effect.

B

Frances H. Rauscher, PhD, first demonstrated the correlation between music and learning in an experiment in 1993. His experiment indicated that a 10-minute dose of Mozart could temporarily boost intelligence. Groups of students were given intelligence tests after listening to silence, relaxation tapes, or Mozart's "Sonata for Two Pianos in D Major" for a short time. He found that after silence, the average IQ score was 110, and after the relaxation tapes, the score rose a point. After listening to Mozart's music, however, the score jumped to 119 (Westley, 1998). Even students who did not like the music still had an increased score in the IQ test. Rauscher hypothesised that "listening to complex, non-repetitive music, like Mozart's, may stimulate neural pathways that are important in thinking" (Castleman, 1994).

C

The same experiment was repeated on rats by Rauscher and Hong Hua Li from Stanford. Rats also demonstrated enhancement in their intelligence performance. These new studies indicate that rats that were exposed to Mozart's showed "increased gene expression of BDNF (a neural growth factor), CREB (a learning and memory compound), and Synapsin I (a synaptic growth protein)" in the brain's hippocampus, compared with rats in the control group, which heard only white noise (e.g. the whooshing sound of a V radio tuned between stations).

D

How exactly does the Mozart Effect work? Researchers are still trying to determine the actual mechanisms for the formation of these enhanced learning pathways. Neuroscientists suspect that music can actually help build and strengthen connections between neurons in the cerebral cortex in a process similar to what occurs in brain development despite its type.

When a baby is born, certain connections have already been made - like connections for heartbeat and breathing. As new information is learned and motor skills develop, new neural connections are formed. Neurons that are not used will eventually die while those used repeatedly will form strong connections. Although a large number of these neural connections require experience, they must also occur within a certain time frame. For example, a child born with cataracts cannot develop connections



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within the visual cortex. If the cataracts are removed by surgery right away, the child's vision develops normally. However, after the age of 2, if the cataracts are removed, the child will remain blind because those pathways cannot establish themselves.

E

Music seems to work in the same way. In October of 1997, researchers at the University of Konstanz in Germany found that music actually rewires neural circuits (Begley, 1996). Although some of these circuits are formed for physical skills needed to play an instrument, just listening to music strengthens connections used in higher-order thinking. Listening to music can then be thought of as "exercise" for the brain, improving concentration and enhancing intuition.

F

If you're a little sceptical about the claims made by supporters of the Mozart Effect, you're not alone. Many people accredit the advanced learning of some children who take music lessons to other personality traits, such as motivation and persistence, which are required in all types of learning. There have also been claims of that influencing the results of some experiments.

G

Furthermore, many people are critical of the role the media had in turning an isolated study into a trend for parents and music educators. After the Mozart Effect was published to the public, the sales of Mozart CDs stayed on the top of the hit list for three weeks. In an article by Michael Linton, he wrote that the research that began this phenomenon (the study by re-searchers at the University of California, Irvine) showed only a temporary boost in IQ, which was not significant enough to even last throughout the course of the experiment. Using music to influence intelligence was used in Confucian civilisation and Plato alluded to Pythagorean music when he described its ideal state in *The Republic*. In both of these examples, music did not cause any overwhelming changes, and the theory eventually died out. Linton also asks, "If Mozart's music were able to improve health, why was Mozart himself so frequently sick? If listening to Mozart's music increases intelligence and encourages spirituality, why aren't the world's smartest and most spiritual people Mozart specialists?" Linton raises an interesting point, if the Mozart Effect causes such significant changes, why isn't there more documented evidence?

H

The "trendiness" of the Mozart Effect may have died out somewhat, but there are still strong supporters (and opponents) of the claims made in 1993. Since that initial experiment, there has not been a surge of support-ing evidence. However, many



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parents, after playing classical music while pregnant or when their children are young, will swear by the Mozart Effect.

A classmate of mine once told me that listening to classical music while studying will help with memorisation. If we approach this controversy from a scientific aspect, although there has been some evidence that music does increase brain activity, actual improvements in learning and memory have not been adequately demonstrated.

Questions 1-5

Reading Passage 1 has eight paragraphs A-H.

Which paragraph contains the following information?

Write the correct letter A-H in boxes 1-5 on your answer sheet.

1. A description of how music affects the brain development of infants
2. Public's first reaction to the discovery of the Mozart Effect
3. The description of Rauscher's original experiment
4. The description of using music for healing in other countries
5. Other qualities needed in all learning

Questions 6-8

Complete the summary below.

Choose NO MORE THAN ONE WORD from the passage for each answer.

Write your answers in boxes 6-8 on your answer sheet.

During the experiment conducted by Frances Rauscher, subjects were exposed to the music for a 6 _____ period of time before they were tested. And Rauscher believes the enhancement in their performance is related to the 7 _____, non-repetitive nature of Mozart's music. Later, a similar experiment was also repeated on 8 _____.

Questions 9-13

Do the following statements agree with the information given in Reading Passage 1?

In boxes 9-13 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

9. All kinds of music can enhance one's brain performance to somewhat extent.
10. There is no neural connection made when a baby is born.
11. There are very few who question the Mozart Effect.
12. Michael Linton conducted extensive research on Mozart's life.
13. There is not enough evidence in support of the Mozart Effect today.



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Life-Casting and Art

Julian Barnes explores the questions posed by Life-Casts, an exhibition of plaster moulds of living people and objects which were originally used for scientific purposes

A Art changes over time and our idea of what art is changes too. For example, objects originally intended for devotional, ritualistic or recreational purposes may be recategorised as art by members of other later civilisations, such as our own, which no longer respond to these purposes.

B What also happens is that techniques and crafts which would have been judged inartistic at the time they were used are reassessed. Life-casting is an interesting example of this. It involved making a plaster mould of a living person or thing. This was complex, technical work, as Benjamin Robert Haydon discovered when he poured 250 litres of plaster over his human model and nearly killed him. At the time, the casts were used for medical research and, consequently, in the nineteenth century life-casting was considered inferior to sculpture in the same way that, more recently, photography was thought to be a lesser art than painting. Both were viewed as unacceptable shortcuts by the "senior" arts. Their virtues of speed and unwavering realism also implied their limitations; they left little or no room for the imagination.

C For many, life-casting was an insult to the sculptor's creative genius. In an infamous lawsuit of 1834, a moulder whose mask of the dying French emperor Napoleon had been reproduced and sold without his permission was judged to have no rights to the image. In other words, he was specifically held not to be an artist. This judgement reflect the view of established members of the nineteenth-century art world such as Rodin, who commented that life-casting 'happens fast but it doesn't make Art'. Some even feared that 'if too much nature was allowed in, it would lead Art away from its proper course of the Ideal.

D The painter Gauguin, at the end of the nineteenth century, worried about future developments in photography. If ever the process went into colour, what painter would labour away at a likeness with a brush made from squirrel-tail? But painting has proved robust. Photography has changed it, of course, just as the novel had to reassess narrative after the arrival of the cinema. But the gap between the senior and junior arts was always narrower than the traditionalists implied. Painters have always used technical back-up such as studio assistants to do the boring bits, while apparently lesser crafts involve great skill, thought, preparation and, depending on how we define it, imagination.

E Time changes our view in another way, too. Each new movement implies a reassessment of what has gone before. What is done now alters what was done before. In some cases this is merely self-serving, with the new art using the old to justify



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itself. It seems to be saying, look at how all of that points to this! Aren't we clever to be the culmination of all that has gone before? But usually it is a matter of re-alerting the sensibility, reminding us not to take things for granted. Take, for example, the cast of the hand of a giant from a circus, made by an anonymous artist around 1889, an item that would now sit happily in any commercial or public gallery. The most significant impact of this piece is on the eye, in the contradiction between unexpected size and verisimilitude. Next, the human element kicks in. you note that the nails are dirt-encrusted, unless this is the caster's decorative addition, and the fingertips extend far beyond them. Then you take in the element of choice, arrangement, art if you like, in the neat, pleated, buttoned sleeve-end that gives the item balance and variation of texture. This is just a moulded hand, yet the part stands utterly for the whole. It reminds us slyly, poignantly, of the full-size original

F But is it art? And, if so, why? These are old tediously repeated questions to which artists have often responded, 'It is art because I am an artist and therefore what I do is art. However, what doesn't work for literature works much better for art – works of art do float free of their creators' intentions. Over time the "reader" does become more powerful. Few of us can look at a medieval altarpiece as its painter intended. We believe too little and aesthetically know too much, so we recreate and find new fields of pleasure in the work. Equally, the lack of artistic intention of Paul Richer and other forgotten craftsmen who brushed oil onto flesh, who moulded, cast and decorated in the nineteenth century is now irrelevant. What counts is the surviving object and our response to it. The tests are simple: does it interest the eye, excite the brain, move the mind to reflection and involve the heart. It may, to use the old dichotomy, be beautiful but it is rarely true to any significant depth. One of the constant pleasures of art is its ability to come at us from an unexpected angle and stop us short in wonder.

Questions 14-18

Reading Passage 2 has six paragraphs, A-F.

Which paragraph contains the following information?

Write the correct letter, A-F, in boxes 14-18 on your answer sheet.

14 an example of a craftsman's unsuccessful claim to ownership of his work

15 an example of how trends in art can change attitudes to an earlier work

16 the original function of a particular type of art

17 ways of assessing whether or not an object is art

18 how artists deal with the less interesting aspects of their work

Questions 19-24

Do the following statements agree with the claims of the writer in Reading Passage 2?

In boxes 19-24 on your answer sheet, write

YES	if the statement agrees with the views of the writer
-----	--



NO if the statement contradicts the views of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

- 19 Nineteenth-century sculptors admired the speed and realism of life-casting
20 Rodin believed the quality of the life-casting would improve if a slower process were used
21 The importance of painting has decreased with the development of colour photography
22 Life-casting requires more skill than sculpture does
23 New art encourages us to look at earlier work in a fresh way
24 The intended meaning of a work of art can get lost over time

Questions 25-26

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes 25-26 on your answer sheet.

- 25 The most noticeable contrast in the cast of the giants hand is between the
A dirt and decoration
B size and realism
C choice and arrangement
D balance and texture
26 According to the writer, the importance of any artistic object lies in
A the artist's intentions
B the artist's beliefs
C the relevance it has to modern life
D the way we respond to it

Music: Language We All Speak

Section A

Music is one of the human species' relatively few universal abilities. Without formal training, any individual, from Stone Age tribesman to suburban teenager, has the ability to recognise music and, in some fashion, to make it. Why this should be so is a mystery. After all, music isn't necessary for getting through the day, and if it aids in reproduction, it does so only in highly indirect ways. Language, by contrast, is also everywhere - but for reasons that are more obvious. With language, you and the



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members of your tribe can organise a migration across Africa, build reed boats and cross the seas, and communicate at night even when you can't see each other. Modern culture, in all its technological extravagance, springs directly from the human talent for manipulating symbols and syntax.

Scientists have always been intrigued by the connection between music and language. Yet over the years, words and melody have acquired a vastly different status in the lab and the seminar room. While language has long been considered essential to unlocking the mechanisms of human intelligence, music is generally treated as an evolutionary frippery - mere "auditory cheesecake", as the Harvard cognitive scientist Steven Pinker puts it.

Section B

But thanks to a decade-long wave of neuroscience research, that tune is changing. A flurry of recent publications suggests that language and music may equally be able to tell us who we are and where we're from - not just emotionally, but biologically. In July, the journal *Nature Neuroscience* devoted a special issue to the topic. And in an article in the 6 August issue of the *Journal of Neuroscience*, David Schwartz, Catherine Howe, and Dale Purves of Duke University argued that the sounds of music and the sounds of language are intricately connected.

To grasp the originality of this idea, it's necessary to realise two things about how music has traditionally been understood. First, musicologists have long emphasised that while each culture stamps a special identity onto its music, music itself has some universal qualities. For example, in virtually all cultures, sound is divided into some or all of the 12 intervals that make up the chromatic scale -that is, the scale represented by the keys on a piano. For centuries, observers have attributed this preference for certain combinations of tones to the mathematical properties of sound itself.

Some 2,500 years ago, Pythagoras was the first to note a direct relationship between the harmoniousness of a tone combination and the physical dimensions of the object that produced it. For example, a plucked string will always play an octave lower than a similar string half its size, and a fifth lower than a similar string two thirds its length. This link between simple ratios and harmony has influenced music theory ever since.

Section C

This music-is-math idea is often accompanied by the notion that music, formally speaking at least, exists apart from the world in which it was created. Writing recently in *The New York Review of Books*, pianist and critic Charles Rosen discussed the long-standing notion that while painting and sculpture reproduce at least some aspects of the natural world, and writing describes thoughts and feelings we are all familiar



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with, music is entirely abstracted from the world in which we live. Neither idea is right, according to David Schwartz and his colleagues. Human musical preferences are fundamentally shaped not by elegant algorithms or ratios but by the messy sounds of real life, and of speech in particular – which in turn is shaped by our evolutionary heritage. "The explanation of music, like the explanation of any product of the mind, must be rooted in biology, not in numbers per se," says Schwartz.

Schwartz, Howe, and Purves analysed a vast selection of speech sounds from a variety of languages to reveal the underlying patterns common to all utterances. In order to focus only on the raw sounds, they discarded all theories about speech and meaning, and sliced sentences into random bites. Using a database of over 100,000 brief segments of speech, they noted which frequency had the greatest emphasis in each sound. The resulting set of frequencies, they discovered, corresponded closely to the chromatic scale. In short, the building blocks of music are to be found in speech.

Far from being abstract, music presents a strange analogue to the patterns created by the sounds of speech. "Music, like visual arts, is rooted in our experience of the natural world," says Schwartz. "It emulates our sound environment in the way that visual arts emulate the visual environment." In music we hear the echo of our basic sound-making instrument - the vocal tract. The explanation for human music is simpler still than Pythagoras's mathematical equations: We like the sounds that are familiar to us - specifically, we like the sounds that remind us of us.

This brings up some chicken-or-egg evolutionary questions. It may be that music imitates speech directly, the researchers say, in which case it would seem that language evolved first. It's also conceivable that music came first and language is in effect an imitation of song - that in everyday speech we hit the musical notes we especially like. Alternately, it may be that music imitates the general products of the human sound-making system, which just happens to be mostly speech. "We can't know this," says Schwartz. "What we do know is that they both come from the same system, and it is this that shapes our preferences."

Section D

Schwartz's study also casts light on the long-running question of whether animals understand or appreciate music. Despite the apparent abundance of "music" in the natural world - birdsong, whalesong, wolf howls, synchronised chimpanzee hooting - previous studies have found that many laboratory animals don't show a great affinity for the human variety of music making.

Marc Hauser and Josh McDermott of Harvard argued in the July issue of Nature Neuroscience that animals don't create or perceive music the way we do. The fact that laboratory monkeys can show recognition of human tunes is evidence, they say, of shared general features of the auditory system, not any specific chimpanzee musical



ability. As for birds, those most musical beasts, they generally recognise their own tunes - a narrow repertoire - but don't generate novel melodies like we do. There are no avian Mozarts.

But what's been played to animals, Schwartz notes, is human music. If animals evolve preferences for sound as we do - based upon the soundscape in which they live - then their "music" would be fundamentally different from ours. In the same way our scales derive from human utterances, a cat's idea of a good tune would derive from yowls and meows. To demonstrate that animals don't appreciate sound the way we do, we'd need evidence that they don't respond to "music" constructed from their own sound environment.

Section E

No matter how the connection between language and music is parsed, what is apparent is that our sense of music, even our love for it, is as deeply rooted in our biology and in our brains as language is. This is most obvious with babies, says Sandra Trehub at the University of Toronto, who also published a paper in the Nature Neuroscience special issue.

For babies, music and speech are on a continuum. Mothers use musical speech to "regulate infants' emotional states", Trehub says. Regardless of what language they speak, the voice all mothers use with babies is the same: "something between speech and song". This kind of communication "puts the baby in a trancelike state, which may proceed to sleep or extended periods of rapture". So if the babies of the world could understand the latest research on language and music, they probably wouldn't be very surprised. The upshot, says Trehub, is that music may be even more of a necessity than we realise.

Questions 27-31

Reading Passage 3 has five sections A-E.

Choose the correct heading for each section from the list of headings below.

Write the correct number i-viii in boxes 27-31 on your answer sheet.

27 . Section A

28 . Section B

29 . Section C

30 . Section D

31 . Section E

List of Headings

i Communication in music with animals

ii New discoveries on animal music

iii Music and language contrasted



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- iv Current research on music
- v Music is beneficial for infants.
- vi Music transcends cultures.
- vii Look back at some of the historical theories
- viii Are we genetically designed for music?

Questions 32-38

Look at the following people (Questions 32-38) and the list of statements below.

Match each person with the correct statement.

Write the correct letter A-G in boxes 32-38 on your answer sheet.

- 32 . Steven Pinker
- 33 . Musicologists
- 34 . Greek philosopher Pythagoras
- 35 . Schwartz, Howe, and Purves
- 36 . Marc Hauser and Josh McDermott
- 37 . Charles Rosen
- 38 . Sandra Trehub

List of Statements

- A Music exists outside of the world it is created in.
- B Music has a universal character despite cultural influences on it.
- C Music is a necessity for humans.
- D Music preference is related to the surrounding influences.
- E He discovered the mathematical basis of music.
- F Music doesn't enjoy the same status of research interest as language.
- G Humans and monkeys have similar traits in perceiving sound.

Questions 39-40

Choose the correct letter A, B, C or D.

Write your answers in boxes 39-40 on your answer sheet.

39 Why was the study of animal music inconclusive?

- A Animals don't have the same auditory system as humans.
- B Tests on animal music are limited.
- C Animals can't make up new tunes.
- D There aren't enough tests on a wide range of animals.

40 What is the main theme of this passage?

- A Language and learning
- B The evolution of music
- C The role of music in human society
- D Music for animals



The Sweet Scent of Success

A Innovation and entrepreneurship, in the right mix, can bring spectacular results and propel a business ahead of the pack. Across a diverse range of commercial successes, from the Hills Hoist clothes line to the Cochlear ear implant, it is hard to generalize beyond saying the creators tapped into something consumers could not wait to get their hands on. However, most ideas never make it to the market. Some ideas that innovators are spruiking to potential investors include new water-saving shower heads, a keyless locking system, ping-pong balls that keep pollution out of rainwater tanks, making teeth grow from stem cells inserted in the gum, and technology to stop LPG tanks from exploding. Grant Kearney, chief executive of the Innovation Xchange, which connects businesses to innovation networks, says he hears of great business ideas that he knows will never get on the market. "Ideas by themselves are absolutely useless," he says. "An idea only becomes innovation when it is connected to the right resources and capabilities".

B One of Australia's latest innovation successes stems from a lemon-scented bath-room cleaner called Shower Power, the formula for which was concocted in a factory in Yatala, Queensland. In 1995, Tom Quinn and John Heron bought a struggling cleaning products business, OzKleen, for 250,000. It was selling 100 different kinds of cleaning products, mainly in bulk. The business was in bad shape, the cleaning formulas were ineffective and environmentally harsh, and there were few regular clients. Now Shower Power is claimed to be the top-selling bathroom cleaning product in the country. In the past 12 months, almost four million bottles of OzKleen's Power products have been sold and the company forecasts 2004 sales of 10 million bottles. The company's sales in 2003 reached \$11 million, with 700k of business being exports. In particular, Shower Power is making big inroads on the British market.

C OzKleen's turnaround began when Quinn and Heron hired an industrial chemist to revitalize the product line. Market research showed that people were looking for a better cleaner for the bathroom, universally regarded as the hardest room in the home to clean. The company also wanted to make the product formulas more environmentally friendly. One of Tom Quinn's sons, Peter, aged 24 at the time, began working with the chemist on the formulas, looking at the potential for citrus-based cleaning products. He detested all the chlorine-based cleaning products that dominated the market. "We didn't want to use chlorine, simple as that," he says. "It offers bad working conditions and there's no money in it." Peter looked at citrus



ingredients, such as orange peel, to replace the petroleum by-products in cleaners. He is credited with finding the Shower Power formula. “The head,” he says. The company is the recipe is in a vault somewhere and in my sole owner of the intellectual property.

D To begin with, Shower Power was sold only in commercial quantities but Tom Quinn decided to sell it in 750ml bottles after the constant “raves” from customers at their retail store at Beenleigh, near Brisbane. Customers were traveling long distances to buy supplies. Others began writing to OzKleen to say how good Shower Power was. “We did a dummy label and went to see Woolworths,” Tom Quinn says. The Woolworths buyer took a bottle home and was able to remove a stain from her basin that had been impossible to shift. From that point on, she championed the product and OzKleen had its first supermarket order, for a palette of Shower Power worth \$3000. “We were over the moon,” says OzKleen’s financial controller, Belinda McDonnell.

E Shower Power was released in Australian supermarkets in 1997 and became the top-selling product in its category within six months. It was all hands on deck at the factory, labeling and bottling Shower Power to keep up with demand. OzKleen ditched all other products and rebuilt the business around Shower Power. This stage, recalls McDonnell, was very tough. “It was hand-to-mouth, cash flow was very difficult,” she says. OzKleen had to pay new-line fees to supermarket chains, which also squeezed margins.

F OzKleen’s next big break came when the daughter of a Coles Myer executive used the product while on holidays in Queensland and convinced her father that Shower Power should be in Coles supermarkets. Despite the product success, Peter Quinn says the company was wary of how long the sales would last and hesitated to spend money on upgrading the manufacturing process. As a result, he remembers long periods of working round the clock to keep up with orders. Small tanks were still being used, so batches were small and bottles were labelled and filled manually. The privately owned OzKleen relied on cash flow to expand. “The equipment could not keep up with demand,” Peter Quinn says. Eventually a new bottling machine was bought for \$50,000 in the hope of streamlining production, but he says: “We got ripped off.” Since then, he has been developing a new automated bottling machine that can control the amount of foam produced in the liquid, so that bottles can be filled more effectively - “I love coming up with new ideas.” The machine is being patented.

G Peter Quinn says OzKleen’s approach to research and development is open slather. “If I need it, I get it. It is about doing something simple that no one else is doing. Most



of these things are just sitting in front of people ... it's just seeing the opportunities.” With a tried and tested product, OzKleen is expanding overseas and developing more Power-brand house-hold products. Tom Quinn, who previously ran a real estate agency, says: “We are competing with the same market all over the world, the cleaning products are sold everywhere.” Shower Power, known as Bath Power in Britain, was launched four years ago with the help of an export development grant from the Federal Government. “We wanted to do it straight away because we realised we had the same opportunities worldwide.” OzKleen is already number three in the British market, and the next stop is France. The Power range includes cleaning products for carpets, kitchens and pre-wash stain removal. The Quinn and Heron families are still involved. OzKleen has been approached with offers to buy the company, but Tom Quinn says he is happy with things as they are. “We’re having too much fun.”

Questions 1-7

Reading Passage 1 has six paragraphs, A-G.

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 1-7 on your answer sheet.

NB You may use any letter more than once.

- 1 . Description of one family member persuading another of selling cleaning products
- 2 . An account of the cooperation of all factory staff to cope with sales increase
- 3 . An account of the creation of the formula of Shower Power
- 4 . An account of buying the original OzKleen company
- 5 . Description of Shower Power’s international expansion
- 6 . The reason of changing the packaging size of Shower Power
- 7 . An example of some innovative ideas

Questions 8-11

Look at the following people and list of statements below.

Match each person with the correct statement.

Write the correct letter A-E in boxes 8-11 on your answer sheet

List of Statement

- A Described his story of selling his product to a chain store
- B Explained there was a shortage of money when sales suddenly increased
- C Believe innovations need support to succeed
- D Believes new products like Shower Power may incur risks
- E Says business won’t succeed with innovations

8 . Grant Kearney

9 . Tom Quinn

10 . Peter Quinn

11 . Belinda McDonnell



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Questions 12-13

Choose the correct letter A, B, C or D.

Write your answers in boxes 12-13 on your answer sheet.

12 Tom Quinn changed the bottle size to 750ml to make Shower Power

A Easier to package.

B Appealing to individual customers.

C Popular in foreign markets.

D Attractive to supermarkets.

13 Why did Tom Quinn decide not to sell OzKleen?

A No one wanted to buy OzKleen.

B New products were being developed in OzKleen.

C He couldn't make an agreement on the price with the buyer.

D He wanted to keep things unchanged.

Optimism and Health

Mindset is all. How you start the year will set the template for the rest, and two scientifically backed character traits hold the key: optimism and resilience (if the prospect leaves you feeling pessimistically spineless, the good news is that you can significantly boost both of these qualities).

Faced with 12 months of plummeting economics and rising human distress, staunchly maintaining a rosy view might seem deludedly Pollyannaish. But here we encounter the optimism paradox. As Brice Pitt, an emeritus professor of the psychiatry of old age at Imperial College, London, told me: "Optimists are unrealistic. Depressive people see things as they really are, but that is a disadvantage from an evolutionary point of view. Optimism is a piece of evolutionary equipment that carried us through millennia of setbacks."

Optimists have plenty to be happy about. In other words, if you can convince yourself that things will get better, the odds of it happening will improve - because you keep on playing the game. In this light, optimism "is a habitual way of explaining your setbacks to yourself", reports Martin Seligman, the psychology professor and author of Learned Optimism. The research shows that when times get tough, optimists do better than pessimists - they succeed better at work, respond better to stress, suffer fewer depressive episodes, and achieve more personal goals.

Studies also show that belief can help with the financial pinch. Chad Wallens, a social



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forecaster at the Henley Centre who surveyed middle-class Britons' beliefs about income, has found that "the people who feel wealthiest, and those who feel poorest, actually have almost the same amount of money at their disposal. Their attitudes and behaviour patterns, however, are different from one another."

Optimists have something else to be cheerful about - in general, they are more robust. For example, a study of 660 volunteers by the Yale University psychologist Dr. Becca Levy found that thinking positively adds an average of seven years to your life. Other American research claims to have identified a physical mechanism behind this. A Harvard Medical School study of 670 men found that the optimists have significantly better lung function. The lead author, Dr. Rosalind Wright, believes that attitude somehow strengthens the immune system. "Preliminary studies on heart patients suggest that, by changing a person's outlook, you can improve their mortality risk," she says.

Few studies have tried to ascertain the proportion of optimists in the world. But a 1995 nationwide survey conducted by the American magazine *Adweek* found that about half the population counted themselves as optimists, with women slightly more apt than men (53 per cent versus 48 per cent) to see the sunny side.

Of course, there is no guarantee that optimism will insulate you from the crunch's worst effects, but the best strategy is still to keep smiling and thank your lucky stars. Because (as every good sports coach knows) adversity is character-forming - so long as you practise the skills of resilience. Research among tycoons and business leaders shows that the path to success is often littered with failure: a record of sackings, bankruptcies and blistering castigation. But instead of curling into a foetal ball beneath the coffee table, they resiliently pick themselves up, learn from their pratfalls and march boldly towards the next opportunity.

The American Psychological Association defines resilience as the ability to adapt in the face of adversity, trauma or tragedy. A resilient person may go through difficulty and uncertainty, but he or she will doggedly bounce back.

Optimism is one of the central traits required in building resilience, say Yale University investigators in the *Annual Review of Clinical Psychology*. They add that resilient people learn to hold on to their sense of humour and this can help them to keep a flexible attitude when big changes of plan are warranted. The ability to accept your lot with equanimity also plays an important role, the study adds.

One of the best ways to acquire resilience is through experiencing a difficult childhood, the sociologist Steven Stack reports in the *Journal of Social Psychology*. For example, short men are less likely to commit suicide than tall guys, he says, because shorties develop psychological defence skills to handle the bullies and mickey-taking that their lack of stature attracts. By contrast, those who enjoyed



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adversity-free youths can get derailed by setbacks later on because they've never been inoculated against aggro.

If you are handicapped by having had a happy childhood, then practising proactive optimism can help you to become more resilient. Studies of resilient people show that they take more risks; 'they court failure and learn not to fear it.

And despite being thick-skinned, resilient types are also more open than average to other people. Bouncing through knock-backs is all part of the process.

It's about optimistic risk-taking - being confident that people will like you. Simply smiling and being warm to people can help. It's an altruistic path to self-interest - and if it achieves nothing else, it will reinforce an age-old adage: hard times can bring out the best in you.

Questions 14-17

Complete the summary below using NO MORE THAN TWO WORDS from Reading Passage 2 for each answer.

A study group from Yale University had discovered that optimism can stretch one's life length by 14____years. And another group from Harvard thinks they have found the biological basis - optimists have better 15_____because an optimist outlook boosts one's 16_____. The study on 17_____was cited as evidence in support of this claim.

Questions 18-22

Complete each sentence with the correct ending A-H.

18. Brice Pitt believes
19. The research at Henley Centre discovers
20. The study conducted by Adweek finds
21. The Annual Review of Clinical Psychology reports
22. Steven Stack says in his report

- A material wealth doesn't necessarily create happiness.
- B optimists tend to be unrealistic about human evolution.
- C optimism is advantageous for human evolution.
- D adversity is the breeding ground of resilience.
- E feelings of optimism vary according to gender.
- F good humour means good flexibility.
- G evenness of mind under stress is important to building resilience.
- H having an optimistic outlook is a habit.



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Questions 23-26

Do the following statements agree with the claims of the writer in Reading Passage?

In boxes 23-26 on your answer sheet write

YES if the statement agrees with the views of the writer

NO if the statement contradicts the views of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

- 23. The benefits of optimism on health have been long known.
- 24. Optimists have better relationships with people than pessimists.
- 25. People with happy childhoods might not be able to practise optimism.
- 26. Resilient people are often open, and even thickskinned.

Internal Market: Selling the inside

When you think of marketing, you more than likely think of marketing to your customers: How can you persuade more people to buy what you sell? But another "market" is just as important: your employees, the very people who can make the brand come alive for your customers. Yet in our work helping executives develop and carry out branding campaigns, my colleagues and I have found that companies very often ignore this critical constituency.

Why is internal marketing so important? First, because it's the best way to help employees make a powerful emotional connection to the products and services you sell. Without that connection, employees are likely to undermine the expectations set by your advertising. In some cases, this is because they simply don't understand what you have promised the public, so they end up working at cross-purposes. In other cases, it may be they don't actually believe in the brand and feel disengaged or, worse, hostile toward the company. We've found that when people care about and believe in the brand, they're motivated to work harder and their loyalty to the company increases. Employees are united and inspired by a common sense of purpose and identity.

Unfortunately, in most companies, internal marketing is done poorly, if at all. While executives recognise the need to keep people informed about the company's strategy and direction, few understand the need to convince employees of the brand's power—they take it as a given.

Employees need to hear the same messages that you send out to the marketplace. At most companies, however, internal and external communications are often mismatched. This can be very confusing, and it threatens employees' perceptions of the company's integrity: They are told one thing by management but observe that a



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different message is being sent to the public. One health insurance company, for instance, advertised that the welfare of patients was the company's number one priority, while employees were told that their main goal was to increase the value of their stock options through cost reductions. And one major financial services institution told customers that it was making a major shift in focus from being a financial retailer to a financial adviser, but, a year later, research showed that the customer experience with the company had not changed. It turned out that company leaders had not made an effort to sell the change internally, so employees were still churning out transactions and hadn't changed their behavior to match their new adviser role.

Enabling employees to deliver on customer expectations is important, of course, but it's not the only reason a company needs to match internal and external messages. Another reason is to help push the company to achieve goals that might otherwise be out of reach. In 1997, when IBM launched its e-business campaign (which is widely credited for turning around the company's image), it chose to ignore research that suggested consumers were unprepared to embrace IBM as a leader in e-business. Although to the outside world this looked like an external marketing effort, IBM was also using the campaign to align employees

around the idea of the Internet as the future of technology. The internal campaign changed the way employees thought about everything they did, from how they named products to how they organised staff to how they approached selling. The campaign was successful largely because it gave employees a sense of direction and purpose, which in turn restored their confidence in IBM's ability to predict the future and lead the technology industry. Today, research shows that people are four times more likely to associate the term "e-business" with IBM than with its nearest competitor.

Perhaps even more important, by taking employees into account, a company can avoid creating a message that doesn't resonate with staff or, worse, one that builds resentment. In 1996, United Airlines shelved its "Come Fly the Friendly Skies" slogan when presented with a survey that revealed the depth of customer resentment toward the airline industry. In an effort to own up to the industry's shortcomings, United launched a new campaign, "Rising," in which it sought to differentiate itself by acknowledging poor service and promising incremental improvements such as better meals. While this was a logical premise for the campaign given the tenor of the times, a campaign focusing on customers' distaste for flying was deeply discouraging to the staff. Employee resentment, ultimately made it impossible for United to deliver the improvements it was promising, which in turn undermined the "Rising" pledge. Three years later, United decided employee opposition was under-mining its success and pulled the campaign. It has since moved to a more inclusive brand message with the line "United," which both audiences can embrace. Here, a fundamental principle of advertising—find and address a customer concern—failed United because it did not consider the internal market.



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When it comes to execution, the most common and effective way to link internal and external marketing campaigns is to create external advertising that targets both audiences. IBM used this tactic very effectively when it launched its e-business campaign. It took out an eight-page ad in the Wall Street Journal declaring its new vision, a message directed at both customers and internal stakeholders. This is an expensive way to capture attention, but if used sparingly, it is the most powerful form of communication; in fact, you need do it only once for everyone in the company to read it. There's a symbolic advantage as well. Such a tactic signals that the company is taking its pledge very seriously; it also signals transparency—the same message going out to both audiences.

Advertising isn't the only way to link internal and external marketing. At Nike, a number of senior executives now hold the additional title of "Corporate Storyteller." They deliberately avoid stories of financial successes and concentrate on parables of "just doing it," reflecting and reinforcing the company's ad campaigns. One tale, for example, recalls how legendary coach and Nike cofounder Bill Bowerman, in an effort to build a better shoe for his team, poured rubber into the family waffle iron, giving birth to the prototype of Nike's famous Waffle Sole. By talking about such inventive moves, the company hopes to keep the spirit of innovation that characterises its ad campaigns alive and well within the company.

But while their messages must be aligned, companies must also keep external promises a little ahead of internal realities. Such promises provide incentives for employees and give them something to live up to. In the 1980s, Ford turned "Quality Is Job 1" from an internal rallying cry into a consumer slogan in response to the threat from cheaper, more reliable Japanese cars. It did so before the claim was fully justified, but by placing it in the public arena, it gave employees an incentive to match the Japanese. If the promise is pushed too far ahead, however, it loses credibility. When a beleaguered British Rail launched a campaign announcing service improvements under the banner "We're Getting There," it did so prematurely. By drawing attention to the gap between the promise and the reality, it prompted destructive press coverage. This, in turn, demoralised staff, who had been legitimately proud of the service advances they had made.

Questions 27-32

Complete each sentence with the correct ending, A-E, below.

Write the correct letter, A-E, in boxes 27-32 on your answer sheet.

NB You can use any letter more than once.

- 27. A health company
- 28. A financial institution
- 29. A computer company
- 30. An airline
- 31. A sport shoe company



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32. A railway company

- A alienated its employees by its apologetic branding campaign.
- B attracted negative publicity through its advertising campaign.
- C produced conflicting image between its employees and the general public.
- D successfully used an advertising campaign to inspire employees
- E draws on the legends of the company spirit.

Questions 33-40

Do the following statements agree with the claims of the writer in Reading Passage 3?

In boxes 33-40 on your answer sheet, write

- | | |
|-----------|--|
| YES | if the statement agrees with the views of the writer |
| NO | if the statement contradicts the views of the writer |
| NOT GIVEN | if it is impossible to say what the writer thinks about this |

33. A strong conviction in the brand can contribute to higher job performance.

34. It is common for companies to overlook the necessity for internal communication.

35. Consumers were ready to view IBM as a leader in e-business before the advertising campaign.

36. United Airlines' failure in its branding campaign was due to the bad advice of an advertisement agency.

37. United Airlines eventually abolished its campaign to boost image as the result of a market research.

38. It is an expensive mistake for IBM to launch its new e-business campaign.

39. Nike employees claimed that they were inspired by their company tales.

40. A slight difference between internal and external promises can create a sense of purpose.

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Morse Code

Morse code is being replaced by a new satellite-based system for sending dis-tress calls at sea. Its dots and dashes have had a good run for their money.

A "Calling all. This is our last cry before our eternal silence." Surprisingly this message, which flashed over the airwaves in the dots and dashes of Morse code on January 31st 1997, was not a desperate transmission by a radio operator on a sinking



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ship. Rather, it was a message signal-ling the end of the use of Morse code for distress calls in French waters. Since 1992 countries around the world have been decommissioning their Morse equipment with similar (if less poetic) sign-offs, as the world's shipping switches over to a new satellite-based arrangement, the Global Maritime Distress and Safety System. The final deadline for the switch-over to GMDSS is February 1st, a date that is widely seen as the end of art era.

B The code has, however, had a good history. Appropriately for a technology commonly associated with radio operators on sinking ships, the idea of Morse code is said to have occurred to Samuel Morse while he was on board a ship crossing the Atlantic. At the time Morse Was a painter and occasional inventor, but when another of the ships passengers informed him of recent advances in electrical theory, Morse was suddenly taken with the idea of building an electric telegraph to send messages in codes. Other inventors had been trying to do just that for the best part of a century. Morse succeeded and is now remembered as "the father of the tele-graph" partly thanks to his single-mindedness—it was 12 years, for example, before he secured money from Congress to build his first telegraph line—but also for technical reasons.

C Compared with rival electric telegraph designs, such as the needle telegraph developed by William Cooke and Charles Wheatstone in Britain, Morses design was very simple: it required little more than a "key" (essentially, a spring-loaded switch) to send messages, a clicking "sounder" to receive them, and a wire to link the two. But although Morses hardware was simple, there was a catch: in order to use his equipment, operators had to learn the special code of dots and dashes that still bears his name. Originally, Morse had not intended to use combinations of dots and dashes to represent individual letters. His first code, sketched in his notebook during that transatlantic voyage, used dots and dashes to represent the digits 0 to 9. Morses idea was that messages would consist of strings of numbers corresponding to words and phrases in a special numbered dictionary. But Morse later abandoned this scheme and, with the help of an associate, Alfred Vail, devised the Morse alphabet, which could be used to spell out messages a letter at a time in dots and dashes.

D At first, the need to learn this complicated-looking code made Morses telegraph seem impossibly tricky compared with other, more user-friendly designs, Cookes and Wheatstones telegraph, for example, used five needles to pick out letters on a diamond-shaped grid. But although this meant that anyone could use it, it also required five wires between telegraph stations. Morses telegraph needed only one. And some people, it soon transpired, had a natural facility for Morse code.

E As electric telegraphy took off in the early 1850s, the Morse telegraph quickly became dominant. It was adopted as the European standard in 1851, allowing direct connections between the telegraph networks of different countries. (Britain chose not to participate, sticking with needle telegraphs for a few more years.) By this time Morse code had been revised to allow for accents and other foreign characters,



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resulting in a split between American and International Morse that continues to this day.

F On international submarine cables, left and right swings of a light-beam reflected from a tiny rotating mirror were used to represent dots and dashes. Meanwhile a distinct telegraphic sub-culture was emerging, with its own customs and vocabulary, and a hierarchy based on the speed at which operators could send and receive Morse code. First-class operators, who could send and receive at speeds of up to 45 words a minute, handled press traffic, securing the best-paid jobs in big cities. At the bottom of the pile were slow, inexperienced rural operators, many of whom worked the wires as part-timers. As their Morse code improved, however, rural operators found that their new-found skill was a passport to better pay in a city job. Telegraphers soon, swelled the ranks of the emerging middle classes. Telegraphy was also deemed suitable work for women. By 1870, a third of the operators in the Western Union office in New York, the largest telegraph office in America, were female.

G In a dramatic ceremony in 1871, Morse himself said goodbye to the global community of telegraphers he had brought into being. After a lavish banquet and many adulatory speeches, Morse sat down behind an operators table and, placing his finger on a key connected to every telegraph wire in America, tapped out his final farewell to a standing ovation. By the time of his death in 1872, the world was well and truly wired: more than 650,000 miles of telegraph line and 30,000 miles of submarine cable were throbbing with Morse code; and 20,000 towns and villages were connected to the global network. Just as the Internet is today often called an "information superhighway", the telegraph was described in its day as an "instantaneous highway of thought",

H But by the 1890s the Morse telegraph's heyday as a cutting-edge technology was coming to an end, with the invention of the telephone and the rise of automatic telegraphs, precursors of the teleprinter, neither of which required specialist skills to operate. Morse code, however, was about to be given a new lease of life thanks to another new technology: wireless. Following the invention of radiotelegraphy by Guglielmo Marconi in 1896, its potential for use at sea quickly became apparent. For the first time, ships could communicate with each other, and with the shore, whatever the weather and even when out of visual range. In 1897 Marconi successfully sent Morse code messages between a shore station and an Italian warship 19km (12 miles) away. By 1910, Morse radio equipment was commonplace on ships.

Questions 1-8

Reading passage 1 has eight paragraphs, A-H.

Choose the correct heading for paragraphs A-H from the list of headings below. Write the correct number, i-xi, in boxes 1-8 on your answer sheet.

List of Headings

- i The advantage of Morse's invention



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- ii A suitable job for women
- iii Morse's invention was developed
- iv Sea rescue after the invention of radiotelegraphy
- v The emergence of many job opportunities
- vi Standard and variations
- vii Application of Morse code in a new technology
- viii The discovery of electricity
- ix International expansion of Morse Code
- x The beginning of an end
- xi The move of using code to convey information
- 1. Paragraph A
- 2. Paragraph B
- 3. Paragraph C
- 4. Paragraph D
- 5. Paragraph E
- 6. Paragraph F
- 7. Paragraph G
- 8. Paragraph H

Questions 9-13

Do the following statements agree with the information given in Reading Passage 1?

In boxes 9-13 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

- 9. Morse had already been famous as an inventor before his invention of Morse code.
- 10. Morse waited a long time before receiving support from the Congress.
- 11. Morse code is difficult to learn compared with other designs.
- 12. Companies and firms prefer to employ telegraphy operators from rural areas.
- 13. Morse died from overwork.



Wealth in A Cold Climate

Latitude is crucial to a nation's economic strength.

A Dr William Masters was reading a book about mosquitoes when inspiration struck. "There was this anecdote about the great yellow fever epidemic that hit Philadelphia in 1793," Masters recalls. "This epidemic decimated the city until the first frost came." The inclement weather froze out the insects, allowing Philadelphia to recover.

B If weather could be the key to a city's fortunes, Masters thought, then why not to the historical fortunes of nations? And could frost lie at the heart of one of the most enduring economic mysteries of all—why are almost all the wealthy, industrialised nations to be found at latitudes above 40 degrees? After two years of research, he thinks that he has found a piece of the puzzle. Masters, an agricultural economist from Purdue University in Indiana, and Margaret McMillan at Tufts University, Boston, show that annual frosts are among the factors that distinguish rich nations from poor ones. Their study is published this month in the *Journal of Economic Growth*. The pair speculate that cold snaps have two main benefits - they freeze pests that would otherwise destroy crops, and also freeze organisms, such as mosquitoes, that carry disease. The result is agricultural abundance and a big workforce.

C The academics took two sets of information. The first was average income for countries, the second climate data from the University of East Anglia. They found a curious tally between the sets. Countries having five or more frosty days a month are uniformly rich, those with fewer than five are impoverished. The authors speculate that the five-day figure is important; it could be the minimum time needed to kill pests in the soil. Masters says: "For example, Finland is a small country that is growing quickly, but Bolivia is a small country that isn't growing at all. Perhaps climate has something to do with that." In fact, limited frosts bring huge benefits to farmers. The chills kill insects or render them inactive; cold weather slows the break-up of plant and animal material in the soil, allowing it to become richer; and frosts ensure a build-up of moisture in the ground for spring, reducing dependence on seasonal rains. There are exceptions to the "cold equals rich" argument. There are well-heeled tropical places such as Hong Kong and Singapore, a result of their superior trading positions. Like-wise, not all European countries are moneyed in the former communist colonies, economic potential was crushed by politics.

D Masters stresses that climate will never be the overriding factor - the wealth of nations is too complicated to be attributable to just one factor. Climate, he feels, somehow combines with other factors such as the presence of institutions, including governments, and access to trading routes to determine whether a country will do well. Traditionally, Masters says, economists thought that institutions had the biggest effect on the economy, because they brought order to a country in the form of, for example,



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laws and property rights. With order, so the thinking went, came affluence. “But there are some problems that even countries with institutions have not been able to get around,” he says. “My feeling is that, as countries get richer, they get better institutions. And the accumulation of wealth and improvement in governing institutions are both helped by a favourable environment, including climate.”

E This does not mean, he insists, that tropical countries are beyond economic help and destined to remain penniless. Instead, richer countries should change the way in which foreign aid is given. Instead of aid being geared towards improving governance, it should be spent on technology to improve agriculture and to combat disease. Masters cites one example: “There are regions in India that have been provided with irrigation, agricultural productivity has gone up and there has been an improvement in health.” Supplying vaccines against tropical diseases and developing crop varieties that can grow in the tropics would break the poverty cycle.

F Other minds have applied themselves to the split between poor and rich nations, citing anthropological, climatic and zoological reasons for why temperate nations are the most affluent. In 350 BC, Aristotle observed that “those who live in a cold climate...are full of spirit”. Jared Diamond, from the University of California at Los Angeles, pointed out in his book *Guns, Germs and Steel* that Eurasia is broadly aligned east-west, while Africa and the Americas are aligned north-south. So, in Europe, crops can spread quickly across latitudes because climates are similar. One of the first domesticated crops, einkorn wheat, spread quickly from the Middle East into Europe; it took twice as long for corn to spread from Mexico to what is now the eastern United States. This easy movement along similar latitudes in Eurasia would also have meant a faster dissemination of other technologies such as the wheel and writing, Diamond speculates. The region also boasted domesticated livestock, which could provide meat, wool and motive power in the fields. Blessed with such natural advantages, Eurasia was bound to take off economically.

G John Gallup and Jeffrey Sachs, two US economists, have also pointed out striking correlations between the geographical location of countries and their wealth. They note that tropical countries between 23.45 degrees north and south of the equator are nearly all poor. In an article for the *Harvard International Review*, they concluded that “development surely seems to favour the temperate-zone economies, especially those in the northern hemisphere, and those that have managed to avoid both socialism and the ravages of war”. But Masters cautions against geographical determinism, the idea that tropical countries are beyond hope: “Human health and agriculture can be made better through scientific and technological research,” he says, “so we shouldn’t be writing off these countries. Take Singapore: without air conditioning, it wouldn’t be rich.”

Questions 14-20

Reading Passage 2 has seven paragraphs, A-G.



Choose the most suitable heading for paragraphs A-G from the list of headings below.
Write the appropriate number, i-x, in boxes 14-20 on your answer sheet.

List of Headings

- i The positive correlation between climate and wealth
- ii Other factors besides climate that influence wealth
- iii Inspiration from reading a book
- iv Other researchers' results do not rule out exceptional cases
- v Different attributes between Eurasia and Africa
- vi Low temperature benefits people and crops
- vii The importance of institution in traditional views
- viii The spread of crops in Europe, Asia and other places
- ix The best way to use aid
- x Confusions and exceptions

- 14. Paragraph A
- 15. Paragraph B
- 16. Paragraph C
- 17. Paragraph D
- 18. Paragraph E
- 19. Paragraph F
- 20. Paragraph G

Questions 21-26

Complete the summary below,

Choose NO MORE THAN TWO WORDS from the passage for each answer.

Write your answers in boxes 21-26 on your answer sheet.

Dr William Masters read a book saying that a(an) 21_____ epidemic which struck an American city hundreds of years ago was terminated by a cold frost. And academics found that there is a connection between climate and country's wealth as in the rich but small country of 22_____. Yet besides excellent surroundings and climate, one country still needs to improve their 23_____ to achieve long prosperity. Thanks to resembling weather conditions across latitude in the continent of 24_____, crops such as 25_____ is bound to spread faster than from South America to the North. Other researchers also noted that even though geographical factors are important, tropical country such as 26_____ still became rich due to scientific advancement.



High speed photography

A Photography gained the interest of many scientists and artists from its inception. Scientists have used photography to record and study movements, such as Eadweard Muybridge's study of human and animal locomotion in 1887. Artists are equally interested by these aspects but also try to explore avenues other than the photo-mechanical representation of reality, such as the pictorialist movement. Military, police, and security forces use photography for surveillance, recognition and data storage. Photography is used by amateurs to preserve memories, to capture special moments, to tell stories, to send messages, and as a source of entertainment. Various technological improvements and techniques have even allowed for visualising events that are too fast or too slow for the human eye.

B One of such techniques is called fast motion or professionally known as time-lapse. Time-lapse photography is the perfect technique for capturing events and movements in the natural world that occur over a timescale too slow for human perception to follow. The life cycle of a mushroom, for example, is incredibly subtle to the human eye. To present its growth in front of audiences, the principle applied is a simple one: a series of photographs are taken and used in sequence to make a moving-image film, but since each frame is taken with a lapse at a time interval between each shot, when played back at normal speed, a continuous action is produced and it appears to speed up. Put simply: we are shrinking time. Objects and events that would normally take several minutes, days or even months can be viewed to completion in seconds having been sped up by factors of tens to millions.

C Another commonly used technique is high-speed photography, the science of taking pictures of very fast phenomena. High-speed photography can be considered to be the opposite of time-lapse photography. One of the many applications is found in biology studies to study birds, bats and even spider silk. Imagine a hummingbird hovering almost completely still in the air, feeding on nectar. With every flap, its wings bend, flex and change shape. These subtle movements precisely control the lift its wings generate, making it an excellent hoverer. But a hummingbird flaps its wings up to 80 times every second. The only way to truly capture this motion is with cameras that will, in effect, slow down time. To do this, a greater length of film is taken at a high sampling frequency or frame rate, which is much faster than it will be projected on screen. When replayed at normal speed, time appears to be slowed down proportionately. That is why high-speed cameras have become such a mainstay of biology.

D In common usage, high-speed photography can also refer to the use of high-speed



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cameras that the photograph itself may be taken in a way as to appear to freeze the motion, especially to reduce motion blur. It requires a sensor with good sensitivity and either a very good shuttering system or a very fast strobe light. The recent National Geographic footage—captured last summer during an intensive three-day shoot at the Cincinnati Zoo—is unprecedented in its clarity and detail. “I’ve watched cheetahs run for 30 years,” said Cathryn Milker, founder of the zoo’s Cat Ambassador Program. “But I saw things in that super slow-motion video that I’ve never seen before.” The slow-motion video is entrancing. Every part of the sprinting cat’s anatomy—supple limbs, rippling muscles, hyperflexible spine—works together in a symphony of speed, revealing the fluid grace of the world’s fastest land animal.

E But things can’t get any more complicated in the case of filming a frog catching its prey. Frogs can snatch up prey in a few thousandths of a second—striking out with elastic tongues. Biologists would love to see how a frog’s tongue roll out, adhere to prey, and roll back into the frog’s mouth. But this all happened too fast, 50 times faster than an eye blink. So naturally people thought of using high-speed camera to capture this fantastic movement in slow motion. Yet one problem still remains—viewers would be bored if they watch the frog swim in slow motion for too long. So how to skip this? The solution is a simple one—adjust the playback speed, which is also called by some the film speed adjustment. The film will originally be shot at a high frame (often 300 frames per second, because it can be converted to much lower frame rates without major issues), but at later editing stage this high frame rate will only be preserved for the prey catching part, while the swimming part will be converted to the normal speed at 24 frames per second. Voila, the scientists can now sit back and enjoy watching without having to go through the pain of waiting.

F Sometimes taking a good picture or shooting a good film is not all about technology, but patience, like in the case of bat. Bats are small, dark-colored; they fly fast and are active only at night. To capture bats on film, one must use some type of camera-tripping device. Photographers or film-makers often place camera near the bat cave, on the path of the flying bats. The camera must be hard-wired with a tripping device so that every time a bat breaks the tripping beam the camera fires and it will keep doing so through the night until the camera’s battery runs out. Though highly-advanced tripping device can now allow for unmanned shooting, it still may take several nights to get a truly high quality film.

G Is it science? Is it art? Since the technique was first pioneered around two hundred years ago, photography has developed to a state where it is almost unrecognisable. Some people would even say the future of photography will be nothing like how we imagine it. No matter what future it may hold, photography will continue to develop as it has been repeatedly demonstrated in many aspects of our life that “a picture is worth a thousand words.”



Questions 27-30

Look at the following organisms (Questions 27-30) and the list of features below. Match each organism with the correct feature, A-D.

Write the correct letter, A-D, in boxes 27-30 on your answer sheet.

- 27. Mushroom
- 28. Hummingbird
- 29. Frog
- 30. Bat

- A too fast to be perceived
- B film at the place where the animal will pass
- C too slow to be visible to human eyes
- D adjust the filming speed to make it interesting

Questions 31-35

Complete the summary below.

Choose NO MORE THAN THREE WORDS from the passage for each answer.

Write your answers in boxes 31-35 on your answer sheet.

Fast motion (professionally known as time-lapse photography) and slow motion (or high-speed photography) are two commonest techniques of photography. To present before audiences something that occurs naturally slow, photographers take each picture at a 31 _____ before another picture. When these pictures are finally shown on screen in sequence at a normal motion picture rate, audiences see a 32 _____ that is faster than what it naturally is. This technique can make audiences feel as if 33 _____ is shrunk. On the other hand, to demonstrate how fast things move, the movement is exposed on a 34 _____ of film, and then projected on screen at normal playback speed. This makes viewers feel time is 35 _____

Questions 36-40

Reading Passage 3 has seven paragraphs, A-G.

Which paragraph contains the following information?

Write the correct letter, A-G, in boxes 36-40 on your answer sheet.

- 36. a description of photography's application in various fields
- 37. a reference to why high-speed photography has a significant role in biology
- 38. a traditional wisdom that assures readers of the prospects of photography
- 39. a reference to how film is processed before final release
- 40. a description of filming shooting without human effort



The Impact of the Potato

Jeff Chapman relates the story of history the most important vegetable

A The potato was first cultivated in South America between three and seven thousand years ago, though scientists believe they may have grown wild in the region as long as 13,000 years ago. The genetic patterns of potato distribution indicate that the potato probably originated in the mountainous west-central region of the continent.

B Early Spanish chroniclers who misused the Indian word batata (sweet potato) as the name for the potato noted the importance of the tuber to the Incan Empire. The Incas had learned to preserve the potato for storage by dehydrating and mashing potatoes into a substance called Chuchu could be stored in a room for up to 10 years, providing excellent insurance against possible crop failures. As well as using the food as a staple crop, the Incas thought potatoes made childbirth easier and used it to treat injuries.

C The Spanish conquistadors first encountered the potato when they arrived in Peru in 1532 in search of gold, and noted Inca miners eating chuchu. At the time the Spaniards failed to realize that the potato represented a far more important treasure than either silver or gold, but they did gradually begin to use potatoes as basic rations aboard their ships. After the arrival of the potato in Spain in 1570, a few Spanish farmers began to cultivate them on a small scale, mostly as food for livestock.

D Throughout Europe, potatoes were regarded with suspicion, distaste and fear. Generally considered to be unfit for human consumption, they were used only as animal fodder and sustenance for the starving. In northern Europe, potatoes were primarily grown in botanical gardens as an exotic novelty. Even peasants refused to eat from a plant that produced ugly, misshapen tubers and that had come from a heathen civilization. Some felt that the potato plant's resemblance to plants in the nightshade family hinted that it was the creation of witches or devils.

E In meat-loving England, farmers and urban workers regarded potatoes with extreme distaste. In 1662, the Royal Society recommended the cultivation of the tuber to the English government and the nation, but this recommendation had little impact. Potatoes did not become a staple until, during the food shortages associated with the Revolutionary Wars, the English government began to officially encourage potato cultivation. In 1795, the Board of Agriculture issued a pamphlet entitled "Hints Respecting the Culture and Use of Potatoes"; this was followed shortly by pro-potato editorials and potato recipes in The Times. Gradually, the lower classes began to follow the lead of the upper classes.

F A similar pattern emerged across the English Channel in the Netherlands, Belgium and France. While the potato slowly gained ground in eastern France (where it was



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often the only crop remaining after marauding soldiers plundered wheat fields and vineyards), it did not achieve widespread acceptance until the late 1700s. The peasants remained suspicious, in spite of a 1771 paper from the Facult de Paris testifying that the potato was not harmful but beneficial. The people began to overcome their distaste when the plant received the royal seal of approval: Louis XVI began to sport a potato flower in his buttonhole, and Marie-Antoinette wore the purple potato blossom in her hair.

G Frederick the Great of Prussia saw the potato's potential to help feed his nation and lower the price of bread, but faced the challenge of overcoming the people's prejudice against the plant. When he issued a 1774 order for his subjects to grow potatoes as protection against famine, the town of Kolberg replied: "The things have neither smell nor taste, not even the dogs will eat them, so what use are they to us?" Trying a less direct approach to encourage his subjects to begin planting potatoes, Frederick used a bit of reverse psychology: he planted a royal field of potato plants and stationed a heavy guard to protect this field from thieves. Nearby peasants naturally assumed that anything worth guarding was worth stealing, and so snuck into the field and snatched the plants for their home gardens. Of course, this was entirely in line with Frederick's wishes.

H Historians debate whether the potato was primarily a cause or an effect of the huge population boom in industrial-era England and Wales. Prior to 1800, the English diet had consisted primarily of meat, supplemented by bread, butter and cheese. Few vegetables were consumed, most vegetables being regarded as nutritionally worthless and potentially harmful. This view began to change gradually in the late 1700s. The Industrial Revolution was drawing an ever increasing percentage of the populace into crowded cities, where only the richest could afford homes with ovens or coal storage rooms, and people were working 12-16 hour days which left them with little time or energy to prepare food. High yielding, easily prepared potato crops were the obvious solution to England's food problems.

I Whereas most of their neighbors regarded the potato with suspicion and had to be persuaded to use it by the upper classes, the Irish peasantry embraced the tuber more passionately than anyone since the Incas. The potato was well suited to the Irish the soil and climate, and its high yield suited the most important concern of most Irish farmers: to feed their families.

J The most dramatic example of the potato's potential to alter population patterns occurred in Ireland, where the potato had become a staple by 1800. The Irish population doubled to eight million between 1780 and 1841, this without any significant expansion of industry or reform of agricultural techniques beyond the widespread cultivation of the potato. Though Irish landholding practices were primitive in comparison with those of England, the potato's high yields allowed even the poorest farmers to produce more healthy food than they needed with scarcely any



investment or hard labor. Even children could easily plant, harvest and cook potatoes, which of course required no threshing, curing or grinding. The abundance provided by potatoes greatly decreased infant mortality and encouraged early marriage.

Questions 1-5

Do the following statements agree with the views of the writer in Reading Passage 1?

In boxes 1-5 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

1. The early Spanish called potato as the Incan name 'Chuchu'
2. The purposes of Spanish coming to Peru were to find out potatoes
3. The Spanish believed that the potato has the same nutrients as other vegetables
4. Peasants at that time did not like to eat potatoes because they were ugly
5. The popularity of potatoes in the UK was due to food shortages during the war

Questions 6-13

Complete the sentences below with NO MORE THAN ONE WORD from the passage 1 for each answer.

Write your answers in boxes 6-13 on your answer sheet.

In France, people started to overcome their disgusting about potatoes because the King put a potato 6 _____ in his button hole.

Frederick realized the potential of potato but he had to handle the 7 _____ against potatoes from ordinary people.

The King of Prussia adopted some 8 _____ psychology to make people accept potatoes.

Before 1800, the English people preferred eating 9 _____ with bread, butter and cheese.

The obvious way to deal with England food problems were high yielding potato 10 _____

The Irish 11 _____ and climate suited potatoes well.

Between 1780 and 1841, based on the 12 _____ of the potatoes, the Irish population doubled to eight million.

The potato's high yields help the poorest farmers to produce more healthy food almost without 13 _____



Coastal Archaeology of Britain

A The recognition of the wealth and diversity of England's coastal archaeology has been one of the most important developments of recent years. Some elements of this enormous resource have long been known. The so-called 'submerged forests' off the coasts of England, sometimes with clear evidence of the human activity, had attracted the interest of antiquarians since at least the eighteenth century, but serious and systematic attention has been given to the archaeological potential of the coast only since the early 1980s.

B It is possible to trace a variety of causes for this concentration of effort and interest. In the 1980s and 1990s scientific research into climate change and its environmental impact spilled over into a much broader public debate as awareness of these issues grew; the prospect of rising sea levels over the next century, and their impact on current coastal environments, has been a particular focus for concern. At the same time archaeologists were beginning to recognize that the destruction caused by natural processes of coastal erosion and by human activity was having an increasing impact on the archaeological resource of the coast.

C The dominant process affecting the physical form of England in the post-glacial period has been rising in the altitude of sea level relative to the land, as the glaciers melted and the landmass readjusted. The encroachment of the sea, the loss of huge areas of land now under the North Sea and the English Channel, and especially the loss of the land bridge between England and France, which finally made Britain an island, must have been immensely significant factors in the lives of our prehistoric ancestors. Yet the way in which prehistoric communities adjusted to these environmental changes has seldom been a major theme in discussions of the period. One factor contributing to this has been that, although the rise in relative sea level is comparatively well documented, we know little about the constant reconfiguration of the coastline. This was affected by many processes, mostly quite, which have not yet been adequately researched. The detailed reconstruction of coastline histories and the changing environments available for human use will be an important theme for future research.

D So great has been the rise in sea level and the consequent regression of the coast that much of the archaeological evidence now exposed in the coastal zone. Whether being eroded or exposed as a buried land surface, is derived from what was originally terrestrial occupation. Its current location in the coastal zone is the product of later unrelated processes, and it can tell us little about past adaptations to the sea. Estimates of its significance will need to be made in the context of other related evidence from



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dry land sites. Nevertheless, its physical environment means that preservation is often excellent, for example in the case of the Neolithic structure excavated at the Stumble in Essex.

E In some cases these buried land surfaces do contain evidence for human exploitation of what was a coastal environment, and elsewhere along the modern coast there is similar evidence. Where the evidence does relate to past human exploitation of the resources and the opportunities offered by the sea and the coast, it is both diverse and as yet little understood. We are not yet in a position to make even preliminary estimates of answers to such fundamental questions as the extent to which the sea and the coast affected human life in the past, what percentage of the population at any time lived within reach of the sea, or whether human settlements in coastal environments showed a distinct character from those inland.

F The most striking evidence for use of the sea is in the form of boats, yet we still have much to learn about their production and use. Most of the known wrecks around our coast are not unexpectedly of post-medieval date, and offer an unparalleled opportunity for research which has yet been little used. The prehistoric sewn-plank boats such as those from the Humber estuary and Dover all seem to belong to the second millennium BC; after this there is a gap in the record of a millennium, which cannot yet be explained before boats reappear, but it built using a very different technology. Boatbuilding must have been an extremely important activity around much of our coast, yet we know almost nothing about it. Boats were some of the most complex artefacts produced by pre-modern societies, and further research on their production and use make an important contribution to our understanding of past attitudes to technology and technological change.

G Boats need landing places, yet here again our knowledge is very patchy. In many cases the natural shores and beaches would have sufficed, leaving little or no archaeological trace, but especially in later periods, many ports and harbors, as well as smaller facilities such as quays, wharves, and jetties, were built. Despite a growth of interest in the waterfront archaeology of some of our more important Roman and medieval towns, very little attention has been paid to the multitude of smaller landing places. Redevelopment of harbor sites and other development and natural pressures along the coast are subject these important locations to unprecedented threats, yet few surveys of such sites have been undertaken.

H One of the most important revelations of recent research has been the extent of industrial activity along the coast. Fishing and salt production are among the better documented activities, but even here our knowledge is patchy. Many forms of fishing will leave little archaeological trace, and one of the surprises of recent survey has



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been the extent of past investment in facilities for procuring fish and shellfish. Elaborate wooden fish weirs, often of considerable extent and responsive to aerial photography in shallow water, have been identified in areas such as Essex and the Severn estuary. The production of salt, especially in the late Iron Age and early Roman periods, has been recognized for some time, especially in the Thames estuary and around the Solent and Poole Harbor, but the reasons for the decline of that industry and the nature of later coastal salt working are much less well understood. Other industries were also located along the coast, either because the raw materials outcropped there or for ease of working and transport: mineral resources such as sand, gravel, stone, coal, ironstone, and alum were all exploited. These industries are poorly documented, but their remains are sometimes extensive and striking.

I Some appreciation of the variety and importance of the archaeological remains preserved in the coastal zone, albeit only in preliminary form, can thus be gained from recent work, but the complexity of the problem of managing that resource is also being realized. The problem arises not only from the scale and variety of the archaeological remains, but also from two other sources: the very varied natural and human threats to the resource, and the complex web of organizations with authority over, or interests in, the coastal zone. Human threats include the redevelopment of historic towns and old dockland areas, and the increased importance of the coast for the leisure and tourism industries, resulting in pressure for the increased provision of facilities such as marinas. The larger size of ferries has also caused an increase in the damage caused by their wash to fragile deposits in the intertidal zone. The most significant natural threat is the predicted rise in sea level over the next century especially in the south and east of England. Its impact on archaeology is not easy to predict, and though it is likely to be highly localized, it will be at a scale much larger than that of most archaeological sites. Thus protecting one site may simply result in transposing the threat to a point further along the coast. The management of the archaeological remains will have to be considered in a much longer time scale and a much wider geographical scale than is common in the case of dry land sites, and this will pose a serious challenge for archaeologists.

Questions 14-16

Choose the correct letter A, B, C or D.

Write your answers in boxes 14-16 on your answer sheet.

14 What has caused public interest in coastal archaeology in recent years?

- A The rapid development of England's coastal archaeology
- B The rising awareness of climate change
- C The discovery of an underwater forest
- D The systematic research conducted on coastal archaeological findings

15 What does the passage say about the evidence of boats?

- A There's enough knowledge of the boat building technology of the prehistoric



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people.

B Many of the boats discovered were found in harbours.

C The use of boats had not been recorded for a thousand years.

D Boats were first used for fishing.

16 What can be discovered from the air?

A Salt mines

B Roman towns

C Harbours

D Fisheries

Questions 17-23

Do the following statements agree with the information given in Reading Passage 2?

In boxes 17-23 on your answer sheet write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

17. England lost much of its land after the ice age due to the rising sea level.

18. The coastline of England has changed periodically.

19. Coastal archaeological evidence may be well protected by sea water.

20. The design of boats used by pre-modern people was very simple.

21. Similar boats were also discovered in many other European countries.

22. There are few documents relating to mineral exploitation.

23. Large passenger boats are causing increasing damage to the seashore.

Questions 24-26

Choose THREE letters A-G.

Write your answers in boxes 24-26 on your answer sheet.

Which THREE of the following statements are mentioned in the passage?

A How coastal archaeology was originally discovered

B It is difficult to understand how many people lived close to the sea.

C How much the prehistoric communities understand the climate change

D Our knowledge of boat evidence is limited.

E Some fishing ground was converted to ports.

F Human development threatens the archaeological remains.

G Coastal archaeology will become more important in the future

Living Dunes

When you think of a sand dune, you probably picture a barren pile of lifeless sand.



But sand dunes are actually dynamic natural structures. They grow, shift and travel. They crawl with living things. Some sand dunes even sing.

A Although no more than a pile of wind-blown sand, dunes can roll over trees and buildings, march relentlessly across highways, devour vehicles on its path, and threaten crops and factories in Africa, the Middle East, and China. In some places, killer dunes even roll in and swallow up towns. Entire villages have disappeared under the sand. In a few instances the government built new villages for those displaced only to find that new villages themselves were buried several years later. Preventing sand dunes from overwhelming cities and agricultural areas has become a priority for the United Nations Environment Program.

B Some of the most significant experimental measurements on sand movement were performed by Ralph Bagnold, a British engineer who worked in Egypt prior to World War II. Bagnold investigated the physics of particles moving through the atmosphere and deposited by wind. He recognised two basic dune types, the crescentic dune, which he called “barchan,” and the linear dune, which he called longitudinal or “sief” (Arabic for “sword”). The crescentic barchan dune is the most common type of sand dune. As its name suggests, this dune is shaped like a crescent moon with points at each end, and it is usually wider than it is long. Some types of barchan dunes move faster over desert surfaces than any other type of dune. The linear dune is straighter than the crescentic dune with ridges as its prominent feature. Unlike crescentic dunes, linear dunes are longer than they are wide—in fact, some are more than 100 miles (about 160 kilometers) long. Dunes can also be comprised of smaller dunes of different types, called complex dunes.

C Despite the complicated dynamics of dune formation, Bagnold noted that a sand dune generally needs the following three things to form: a large amount of loose sand in an area with little vegetation—usually on the coast or in a dried-up river, lake or sea bed; a wind or breeze to move the grains of sand; and an obstacle, which could be as small as a rock or as big as a tree, that causes the sand to lose momentum and settle. Where these three variables merge, a sand dune forms.

D As the wind picks up the sand, the sand travels, but generally only about an inch or two above the ground, until an obstacle causes it to stop. The heaviest grains settle against the obstacle, and a small ridge or bump forms. The lighter grains deposit themselves on the other side of the obstacle. Wind continues to move sand up to the top of the pile until the pile is so steep that it collapses under its own weight. The collapsing sand comes to rest when it reaches just the right steepness to keep the dune stable. The repeating cycle of sand inching up the windward side to the dune crest, then slipping down the dune’s slip face allows the dune to inch forward, migrating in the direction the wind blows.

E Depending on the speed and direction of the wind and the weight of the local sand,



dunes will develop into different shapes and sizes. Stronger winds tend to make taller dunes; gentler winds tend to spread them out. If the direction of the wind generally is the same over the years, dunes gradually shift in that direction. But a dune is “a curiously dynamic creature”, wrote Farouk ElBaz in National Geographic. Once formed, a dune can grow, change shape, move with the wind and even breed new dunes. Some of these offspring may be carried on the back of the mother dune. Others are born and race downwind, outpacing their parents.

F Sand dunes even can be heard ‘singing’ in more than 30 locations worldwide, and in each place the sounds have their own characteristic frequency, or note. When the thirteenth century explorer Marco Polo encountered the weird and wonderful noises made by desert sand dunes, he attributed them to evil spirits. The sound is unearthly. The volume is also unnerving. Adding to the tone’s other worldliness is the inability of the human ear to localise the source of the noise. Stéphane Douady of the French national research agency CNRS and his colleagues have been delving deeper into dunes in Morocco, Chile, China and Oman, and believe they can now explain the exact mechanism behind this acoustic phenomenon.

G The group hauled sand back to the laboratory and set it up in channels with automated pushing plates. The sands still sang, proving that the dune itself was not needed to act as a resonating body for the sound, as some researchers had theorised. To make the booming sound, the grains have to be of a small range of sizes, all alike in shape: well-rounded. Douady’s key discovery was that this synchronised frequency—which determines the tone of sound—is the result of the grain size. The larger the grain, the lower the key. He has successfully predicted the notes emitted by dunes in Morocco, Chile and the US simply by measuring the size of the grains they contain. Douady also discovered that the singing grains had some kind of varnish or a smooth coating of various minerals: silicon, iron and manganese, which probably formed on the sand when the dunes once lay beneath an ancient ocean. But in the muted grains this coat had been worn away, which explains why only some dunes can sing. He admits he is unsure exactly what role the coating plays in producing the noise. The mysterious dunes, it seems, aren’t quite ready yet to give up all of their secrets.

Questions 27-33

Reading passage 3 has seven paragraphs, A-G.

Choose the correct heading for paragraphs A-G from the list of headings below. Write the correct number, i-x, in boxes 27-33 on your answer sheet.

List of Headings

- | | |
|-----|--|
| i | Shaping and reforming |
| ii | Causes of desertification |
| iii | Need combination of specific conditions |
| iv | Potential threat to industry and communication |



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- v An old superstition demystified
- vi Differences and similarities
- vii A continuous cycling process
- viii Habitat for rare species
- ix Replicating the process in laboratory
- x Commonest type of dune

- 27 Paragraph A
- 28 Paragraph B
- 29 Paragraph C
- 30 Paragraph D
- 31 Paragraph E
- 32 Paragraph F
- 33 Paragraph G

Questions 34-36

Complete the sentences below.

Choose ONE WORD ONLY from the passage for each answer.

Write your answers in boxes 34-36 on your answer sheet.

34_____dune is said to have long ridges that can extend hundreds of miles.

According to Bagnold, an 35_____is needed to stop the sand from moving before a dune can form.

Stéphane Douady believes the singing of dunes is not a spiritual phenomenon, but purely 36_____.

Questions 37-40

Complete the summary below.

Choose ONE WORD ONLY from the passage for each answer.

Write your answers in boxes 37-40 on your answer sheet.

There are many different types of dunes, two of which are most commonly found in deserts throughout the world, the linear dune and the 37_____dune, some times also known as the crescentic dune. It's been long known that in some places dunes can even sing and the answer lies in the sand itself. To produce singing sand in lab, all the sands must have similar 38_____. And scientists have discovered that the size of the sand can affect the 39_____of the sound. But the function of the varnish composed by a mixture of 40_____ still remains puzzling.



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What the Managers Really Do?

When students graduate and first enter the workforce, the most common choice is to find an entry-level position. This can be a job such as an unpaid internship, an assistant, a secretary, or a junior partner position. Traditionally, we start with simpler jobs and work our way up. Young professionals start out with a plan to become senior partners, associates, or even managers of a workplace. However, these promotions can be few and far between, leaving many young professionals unfamiliar with management experience. An important step is understanding the role and responsibilities of a person in a managing position. Managers are organisational members who are responsible for the work performance of other organisational members. Managers have formal authority to use organisational resources and to make decisions. Managers at different levels of the organisation engage in different amounts of time on the four managerial functions of planning, organising, leading, and controlling.

However, as many professionals already know, managing styles can be very different depending on where you work. Some managing styles are strictly hierarchical. Other managing styles can be more casual and relaxed, where the manager may act more like a team member rather than a strict boss. Many researchers have created a more scientific approach in studying these different approaches to managing. In the 1960s, researcher Henry Mintzberg created a seminal organisational model using three categories. These categories represent three major functional approaches, which are designated as interpersonal, informational and decisional.

Introduced Category 1: INTERPERSONAL ROLES. Interpersonal roles require managers to direct and supervise employees and the organisation. The figurehead is typically a top of middle manager. This manager may communicate future organisational goals or ethical guidelines to employees at company meetings. They also attend ribbon-cutting ceremonies, host receptions, presentations and other activities associated with the figurehead role. A leader acts as an example for other employees to follow, gives commands and directions to subordinates, makes decisions, and mobilises employee support. They are also responsible for the selection and training of employees. Managers must be leaders at all levels of the organisation; often lower-level managers look to top management for this leadership example. In the role of liaison, a manager must coordinate the work of others in different work units, establish alliances between others, and work to share resources. This role is particularly critical for middle managers, who must often compete with other managers for important resources, yet must maintain successful working relationships with them for long time periods.

Introduced Category 2: INFORMATIONAL ROLES. Informational roles are those in which managers obtain and transmit information. These roles have changed



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dramatically as technology has improved. The monitor evaluates the performance of others and takes corrective action to improve

that performance. Monitors also watch for changes in the environment and within the company that may affect individual and organisational performance. Monitoring occurs at all levels of management. The role of disseminator requires that managers inform employees of changes that affect them and the organisation. They also communicate the company's vision and purpose.

Introduced Category 3: DECISIONAL ROLES. Decisional roles require managers to plan strategy and utilise resources. There are four specific roles that are decisional. The entrepreneur role requires the manager to assign resources to develop innovative goods and services, or to expand a business. The disturbance handler corrects unanticipated problems facing the organisation from the internal or external environment. The third decisional role, that of resource allocator, involves determining which work units will get which resources. Top managers are likely to make large, overall budget decisions, while middle managers may make more specific allocations. Finally, the negotiator works with others, such as suppliers, distributors, or labor unions, to reach agreements regarding products and services.

Although Mintzberg's initial research in 1960s helped categorise manager approaches, Mintzberg was still concerned about research involving other roles in the workplace. Mintzberg considered expanding his research to other roles, such as the role of disseminator, figurehead, liaison and spokesperson. Each role would have different special characteristics, and a new categorisation system would have to be made for each role to understand it properly.

While Mintzberg's initial research was helpful in starting the conversation, there has since been criticism of his methods from other researchers. Some criticisms of the work were that even though there were multiple categories, the role of manager is still more complex. There are still many manager roles that are not as traditional and are not captured in Mintzberg's original three categories. In addition, sometimes, Mintzberg's research was not always effective. The research, when applied to real-life situations, did not always improve the management process in real-life practice.

These two criticisms against Mintzberg's research method raised some questions about whether or not the research was useful to how we understand "managers" in today's world. However, even if the criticisms against Mintzberg's work are true, it does not mean that the original research from the 1960s is completely useless. Those researchers did not say Mintzberg's research is invalid. His research has two positive functions to the further research.

The first positive function is Mintzberg provided a useful functional approach to analyse management. And he used this approach to provide a clear concept of the role



of manager to the researcher. When researching human behavior, it is important to be concise about the subject of the research. Mintzberg's research has helped other researchers clearly define what a "manager" is, because in real-life situations, the "manager" is not always the same position title. Mintzberg's definitions added clarity and precision to future research on the topic.

The second positive function is Mintzberg's research could be regarded as a good beginning to give a new insight to further research on this field in the future. Scientific research is always a gradual process. Just because Mintzberg's initial research had certain flaws, does not mean it is useless to other researchers. Researchers who are interested in studying the workplace in a systematic way have older research to look back on. A researcher doesn't have to start from the very beginning— older research like Mintzberg's has shown what methods work well and what methods are not as appropriate for workplace dynamics. As more young professionals enter the job market, this research will continue to study and change the way we think about the modern workplace.

Questions 1-6

Look at the following descriptions or deeds (Questions 1-6) and the list of categories below.

Match each description or deed with the correct category, A,B or C.

Write the correct letter, A, B, or C, in boxes 1-6 on your answer sheet.

NB You may use any letter more than once.

List of Categories

A INTERPERSONAL ROLES

B INFORMATIONAL ROLES

C DECISIONAL ROLES

1. the development of business scheme
2. presiding at formal events
3. using employees and funds
4. getting and passing message on to related persons
5. relating the information to employees and organisation
6. recruiting the staff

Questions 7-8

Choose TWO letters, A-E.

Write the correct letters in boxes 7-8 on your answer sheet.

Which TWO positive functions about Mintzberg's research are mentioned in the last two paragraphs?

A offers waterproof categories of managers

B provides a clear concept to define the role of a manager

C helps new graduates to design their career



- D suggests ways for managers to do their job better
E makes a fresh way for further research

Questions 9-13

Do the following statements agree with the information given in Reading Passage 1?
In boxes 9-13 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

9. Young professionals can easily know management experience in the workplace.
10. Mintzberg's theory broke well-established notions about managing styles.
11. Mintzberg got a large amount of research funds for his contribution.
12. All managers do the same work.
13. Mintzberg's theory is valuable for future studies.

Storytelling, From Prehistoric Craves To Modern Cinemas

A It was told, we suppose, to people crouched around a fire: a tale of adventure, most likely—relating some close encounter with death: a remarkable hunt, an escape from mortal danger; a vision, or something else out of the ordinary. Whatever its thread, the weaving of this story was done with a prime purpose. The listeners must be kept listening. They must not fall asleep. So, as the story went on, its audience should be sustained by one question above all: What happens next?

B The first fireside stories in human history can never be known. They were kept in the heads of those who told them. This method of storage is not necessarily inefficient. From documented oral traditions in Australia, the Balkans and other parts of the world we know that specialised storytellers and poets can recite from memory literally thousands of lines, in verse or prose, verbatim - word for word. But while memory is rightly considered an art in itself, it is clear that a primary purpose of making symbols is to have a system of reminders or mnemonic cues - signs that assist us to recall certain information in the mind's eye.

C In some Polynesian communities, a notched memory stick may help to guide a storyteller through successive stages of recitation. But in other parts of the world, the activity of storytelling historically resulted in the development or even the invention of writing systems. One theory about the arrival of literacy in ancient Greece, for example, argues that the epic tales about the Trojan War and the wanderings of



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Odysseus traditionally attributed to Homer were just so enchanting to hear that they had to be preserved. So the Greeks, c. 750-700BC. borrowed an alphabet from their neighbors in the eastern Mediterranean, the Phoenicians.

D The custom of recording stories on parchment and other materials can be traced in many manifestations around the world, from the priestly papyrus archive of ancient Egypt to the birch-bark scrolls on which the North American Ojibway Indians set down their creation myth. It is a well-trying and universal practice: so much so that to this day storytime is probably most often associated with words on paper. The formal practice of narrating a story aloud would seem-so we assume-to have given way to newspapers, novels and comic strips. This, however, is not the case. Statistically it is doubtful that the majority of humans currently rely upon the written word to get access to stories. So what is the alternative source?

E Each year, over 7 billion people will go to watch the latest offering from Hollywood. Bollywood and beyond. The supreme storyteller of today is cinema. The movies, as distinct from still photography, seem to be an essentially modern phenomenon. This is an illusion, for there are, as we shall see, certain ways in which the medium of film is indebted to very old precedents of arranging 'sequences' of images. But any account of visual storytelling must begin with the recognition that all storytelling beats with a deeply atavistic pulse: that is, a 'good story' relies upon formal patterns of plot and characterisation that have been embedded in the practice of storytelling over many generations.

F Thousands of scripts arrive every week at the offices of the major film studios. But aspiring screenwriters really need look no further for essential advice than the fourth-century BC Greek Philosopher Aristotle. He left some incomplete lecture notes on the art of telling stories in various literary and dramatic modes, a slim volume known as the Poetics. Though he can never have envisaged the popcorn-fuelled actuality of a multiplex cinema, Aristotle is almost prescient about the key elements required to get the crowds flocking to such a cultural hub. He analyzed the process with cool rationalism. When a story enchants us, we lose the sense of where we are; we are drawn into the story so thoroughly that we forget it is a story being told. This is, in Aristotle's phrase, 'the suspension of disbelief.'

G We know the feeling. If ever we have stayed in our seats, stunned with grief, as the credits roll by, or for days after seeing that vivid evocation of horror have been nervous about taking a shower at home, then we have suspended disbelief. We have been caught, or captivated, in the storyteller's web. Did it all really happen? We really thought so for a while. Aristotle must have witnessed often enough this suspension of disbelief. He taught at Athens, the city where theater developed as a primary form of civic ritual and recreation. Two theatrical types of storytelling, tragedy and comedy, caused Athenian audiences to lose themselves in sadness and laughter respectively. Tragedy, for Aristotle, was particularly potent in its capacity to enlist and then purge



the emotions of those watching the story unfold on the stage, so he tried to identify those factors in the storyteller's art that brought about such engagement. He had, as an obvious sample for analysis, not only the fifth-century BC masterpieces of Classical Greek tragedy written by Aeschylus, Sophocles and Euripides. Beyond them stood Homer, whose stories even then had canonical status: The Iliad and The Odyssey were already considered literary landmarks—stories by which all other stories should be measured. So what was the secret of Homer's narrative art?

It was not hard to find. Homer created credible heroes. His heroes belonged to the past, they were mighty and magnificent, yet they were not, in the end, fantasy figures. He made his heroes sulk, bicker, cheat and cry. They were, in short, characters—protagonists of a story that an audience would care about, would want to follow, would want to know what happens next. As Aristotle saw, the hero who shows a human side, some flaw or weakness to which mortals are prone, is intrinsically dramatic.

Questions 14-18

Reading Passage 2 has eight paragraphs, A-H.

Which paragraph contains the following information?

Write the correct letter, A-H, in boxes 14-18 on your answer sheet.

14. A misunderstanding of how people today get stories
15. The categorisation of stories
16. The fundamental aim of storytelling
17. A description of reciting stories without any assistance
18. How to make story characters attractive

Questions 19-22

Classify the following information as referring to

- A adopted the writing system from another country
- B used organic materials to record stories
- C used tools to help to tell stories

Write the correct letter, A, B or C in boxes 19-22 on your answer sheet.

19. Egyptians
20. Ojibway
21. Polynesians
22. Greek

Questions 23-26

Complete the sentences below with ONE WORD ONLY from the passage.

Write your answers in boxes 23-26 on your answer sheet.

Aristotle wrote a book on the art of storytelling called the 23_____.

Aristotle believed the most powerful type of story to move listeners is 24_____.

Aristotle viewed Homer's works as 25_____.



Aristotle believed attractive heroes should have some 26 _____.

Monkeys and Forests

AS AN EAST WIND blasts through a gap in the Cordillera de Tilaran, a rugged mountain range that splits northern Costa Rica in half, a female mantled howler monkey moves through the swaying trees of the forest canopy.

A. Ken Glander, a primatologist from Duke University, gazes into the canopy, tracking the female's movements. Holding a dart gun, he waits with infinite patience for the right moment to shoot. With great care, Glander aims and fires. Hit in the rump, the monkey wobbles. This howler belongs to a population that has lived for decades at Hacienda La Pacifica, a working cattle ranch in Guanacaste province. Other native primates -white-faced capuchin monkeys and spider monkeys - once were common in this area, too, but vanished after the Pan-American Highway was built nearby in the 1950s. Most of the surrounding land was clear-cut for pasture.

B. Howlers persist at La Pacifica, Glander explains, because they are leaf- eaters. They eat fruit, when it's available but, unlike capuchin and spider monkeys, do not depend on large areas of fruiting trees. "Howlers can survive anyplace you have half a dozen trees, because their eating habits are so flexible," he says. In forests, life is an arms race between trees and the myriad creatures that feed on leaves. Plants have evolved a variety of chemical defenses, ranging from bad-tasting tannins, which bind with plant-produced nutrients, rendering them indigestible, to deadly poisons, such as alkaloids and cyanide.

C. All primates, including humans, have some ability to handle plant toxins. "We can detoxify a dangerous poison known as caffeine, which is deadly to a lot of animals:" Glander says. For leaf-eaters, long-term exposure to a specific plant toxin can increase their ability to defuse the poison and absorb the leaf nutrients. The leaves that grow in regenerating forests, like those at La Pacifica, are actually more howler friendly than those produced by the undisturbed, centuries-old trees that survive farther south, in the Amazon Basin. In younger forests, trees put most of their limited energy into growing wood, leaves and fruit, so they produce much lower levels of toxin than do well-established, old-growth trees.

D. The value of maturing forests to primates is a subject of study at Santa Rosa



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National Park, about 35 miles northwest of Hacienda La Pacifica. The park hosts populations not only of mantled howlers but also of white-faced capuchins and spider monkeys. Yet the forests there are young, most of them less than 50 years old. Capuchins were the first to begin using the reborn forests, when the trees were as young as 14 years. Howlers, larger and heavier than capuchins, need somewhat older trees, with limbs that can support their greater body weight. A working ranch at Hacienda La Pacifica also explain their population boom in Santa Rosa. “Howlers are more resilient than capuchins and spider monkeys for several reasons,” Fedigan explains. “They can live within a small home range, as long as the trees have the right food for them. Spider monkeys, on the other hand, occupy a huge home range, so they can’t make it in fragmented habitat.”

E. Howlers also reproduce faster than do other monkey species in the area. Capuchins don’t bear their first young until about 7 years old, and spider monkeys do so even later, but howlers give birth for the first time at about 3.5 years of age. Also, while a female spider monkey will have a baby about once every four years, well-fed howlers can produce an infant every two years.

F. The leaves howlers eat hold plenty of water, so the monkeys can survive away from open streams and water holes. This ability gives them a real advantage over capuchin and spider monkeys, which have suffered during the long, ongoing drought in Guanacaste.

G. Growing human population pressures in Central and South America have led to persistent destruction of forests. During the 1990s, about 1.1 million acres of Central American forest were felled yearly. Alejandro Estrada, an ecologist at Estacion de Biologia Los Tuxtlas in Veracruz, Mexico, has been exploring how monkeys survive in a landscape increasingly shaped by humans. He and his colleagues recently studied the ecology of a group of mantled howler monkeys that thrive in a habitat completely altered by humans: a cacao plantation in Tabasco, Mexico. Like many varieties of coffee, cacao plants need shade to grow, so 40 years ago the landowners planted fig, monkey pod and other tall trees to form a protective canopy over their crop. The howlers moved in about 25 years ago after nearby forests were cut. This strange habitat, a hodgepodge of cultivated native and exotic plants, seems to support about as many monkeys as would a same-sized patch of wild forest. The howlers eat the leaves and fruit of the shade trees, leaving the valuable cacao pods alone, so the farmers tolerate them.

H. Estrada believes the monkeys bring underappreciated benefits to such farms, dispersing the seeds of fig and other shade trees and fertilizing the soil with feces. He



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points out that howler monkeys live in shade coffee and cacao plantations in Nicaragua and Costa Rica as well as in Mexico. Spider monkeys also forage in such plantations, though they need nearby areas of forest to survive in the long term. He hopes that farmers will begin to see the advantages of associating with wild monkeys, which includes potential ecotourism projects.

“Conservation is usually viewed as a conflict between agricultural practices and the need to preserve nature,” Estrada says. “We’re moving away from that vision and beginning to consider ways in which agricultural activities may become a tool for the conservation of primates in human-modified landscapes.”

Questions 27-32

The reading Passage has eight paragraphs A-H.

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 27-32 on your answer sheet.

- 27 . A reference of rate of reduction in forest habitats
- 28 . An area where only one species of monkey survived while other two species vanished
- 29 . A reason for howler monkey of choose new leaves as food over old ones
- 30 . Mention to howler monkey’s diet and eating habits
- 31 . A reference of asking farmers’ changing attitude toward wildlife
- 32 . The advantage for howler monkey’s flexibility living in a segmented habitat

Questions 33-35

Look at the following places and the list of descriptions below.

Match each description with the correct place, A-E.

Write the correct letter, A-E, in boxes 33-35 on your answer sheet.

A Hacienda La Pacifica

B Santa Rosa National Park

C A cacao plantation in Tabasco, Mexico

D Estacion de Biologia Los Tuxtlas in Veracruz, Mexico

E Amazon Basin

- 33 . A place where howler monkeys benefit to the local region’s agriculture
- 34 . A place where it is the original home for all three native monkeys
- 35 . A place where capuchins monkey comes for a better habitat

Questions 36-40

Complete the sentences below.

Choose NO MORE THAN TWO WORDS from the passage for each answer.



Write your answers in boxes 36-40 on your answer sheet.

The reasons why howler monkeys survive better in local region than other two species

Howlers live between in La Pacifica since they can feed themselves with leaves when 36_____is not easily found

Howlers have better ability to alleviate the 37_____, which old and young trees used to protect themselves

When compared to that of spider monkeys and capuchin monkeys, the 38_____rate of howlers is relatively faster (round for just every 2 years).

The monkeys can survive away from open streams and water holes as the leaves howlers eat hold high content of 39_____, which helps them to resist the continuous 40_____in Guanacaste.

Does An IQ Test Prove Creativity?

Everyone has creativity, some a lot more than others. The development of humans, and possibly the universe, depends on it. Yet creativity is an elusive creature. What do we mean by it? What is going on in our brains when ideas form? Does it feel the same for artists and scientists? We asked writers and neuroscientists, pop stars and AI gurus to try to deconstruct the creative process-and learn how we can all ignite the spark within.

A In the early 1970s, creativity was still seen as a type of intelligence. But when more subtle tests of IQ and creative skills were developed in the 1970s, particularly by the father of creativity testing, Paul Torrance, it became clear that the link was not so simple. Creative people are intelligent, in terms of IQ tests at least, but only averagely or just above. While it depends on the discipline, in general beyond a certain level IQ does not help boost creativity; it is necessary but not sufficient to make someone creative.

B Because of the difficulty of studying the actual process, most early attempts to study creativity concentrated on personality. According to creativity specialist Mark Runco of California State University, Fullerton, the “creative personality” tends to place a high value on aesthetic qualities and to have broad interests, providing lots of resources to draw on and knowledge to recombine into novel solutions. “Creatives” have an attraction to complexity and an ability to handle conflict. They are also usually highly self-motivated, perhaps even a little obsessive. Less creative people, on the other hand, tend to become irritated if they cannot immediately fit all the pieces together. They are less tolerant of confusion. Creativity comes to those who wait, but only to those who are happy to do so in a bit of a fog.



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C But there may be a price to pay for having a creative personality. For centuries, a link has been made between creativity and mental illness. Psychiatrist Jamison of Johns Hopkins University in Baltimore, Maryland, found that established artists are significantly more likely to have mood disorders. But she also suggests that a change of mood state might be the key to triggering a creative event, rather than the negative mood itself. Intelligence can help channel this thought style into great creativity, but when combined with emotional problems, lateral, divergent or open thinking can lead to mental illness instead.

D Jordan Peterson, a psychologist at the University of Toronto, Canada, believes he has identified a mechanism that could help explain this. He says that the brains of creative people seem more open to incoming stimuli than less creative types. Our senses are continuously feeding a mass of information into our brains, which have to block or ignore most of it to save us from being snowed under. Peterson calls this process latent inhibition, and argues that people who have less of it, and who have a reasonably high IQ with a good working memory can juggle more of the data, and so may be open to more possibilities and ideas. The downside of extremely low latent inhibition may be a confused thought style that predisposes people to mental illness. So for Peterson, mental illness is not a prerequisite for creativity, but it shares some cognitive traits.

E But what of the creative act itself? One of the first studies of the creative brain at work was by Colin Martindale, a psychologist from the University of Maine in Orono. Back in 1978, he used a network of scalp electrodes to record an electroencephalogram, a record of the pattern of brain waves, as people made up stories. Creativity has two stages: inspiration and elaboration, each characterised by very different states of mind. While people were dreaming up their stories, he found their brains were surprisingly quiet. The dominant activity was alpha waves, indicating a very low level of cortical arousal: a relaxed state, as though the conscious mind was quiet while the brain was making connections behind the scenes. It's the same sort of brain activity as in some stages of sleep, dreaming or rest, which could explain why sleep and relaxation can help people be creative. However, when these quiet minded people were asked to work on their stories, the alpha wave activity dropped off and the brain became busier, revealing increased cortical arousal, more corralling of activity and more organised thinking. Strikingly, it was the people who showed the biggest difference in brain activity between the inspiration and development stages who produced the most creative storylines. Nothing in their background brain activity marked them as creative or uncreative. "It's as if the less creative person can't shift gear," says Guy Claxton, a psychologist at the University of Bristol, UK. "Creativity requires different kinds of thinking. Very creative people move between these states intuitively." Creativity, it seems, is about mental flexibility: perhaps not a two-step process, but a toggling between two states. In a later study, Martindale found that communication between the sides of the brain is also important.



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F Paul Howard-Jones, who works with Claxton at Bristol, believes he has found another aspect of creativity. He asked people to make up a story based on three words and scanned their brains using functional magnetic resonance imaging. In one trial, people were asked not to try too hard and just report the most obvious story suggested by the words. In another, they were asked to be inventive. He also varied the words so it was easier or harder to link them. As people tried harder and came up with more creative tales, there was a lot more activity in a particular prefrontal brain region on the right-hand side. These regions are probably important in monitoring for conflict, helping us to filter out many of combining the words and allowing us to pull out just the desirable connections, Howard-Jones suggests. It shows that there is another side to creativity, he says. The story-making task, particularly when we are stretched, produces many options which we have to assess. So part of creativity is a conscious process of evaluating and analysing ideas. The test also shows that the more we try and are stretched, the more creative our minds can be.

G And creativity need not always be a solitary, tortured affair, according to Teresa Amabile of Harvard Business School. Though there is a slight association between solitary writing or painting and negative moods or emotional disturbances, scientific creativity and workplace creativity seem much more likely to occur when people are positive and buoyant. In a decade-long study of real businesses, to be published soon, Amabile found that positive moods relate positively to creativity in organisations, and that the relationship is a simple linear one. Creative thought also improves people's moods, her team found, so the process is circular. Time pressures, financial pressures and hard-earned bonus schemes on the other hand, do not boost workplace creativity: internal motivation, not coercion, produces the best work.

H Another often forgotten aspect of creativity is social. Vera John-Steiner of the University of New Mexico says that to be really creative you need strong social networks and trusting relationships, not just active neural networks. One vital characteristic of a highly creative person, she says, is that they have at least one other person in their life who doesn't think they are completely nuts

Questions 1-4

Do the following statements agree with the information given in Reading Passage 1?
In boxes 1-4 on your answer sheet, write

TRUE	if the statement agrees with the information
------	--

FALSE	if the statement contradicts the information
-------	--

NOT GIVEN	If there is no information on this
-----------	------------------------------------

1. High IQ guarantees better creative ability in one person than that who achieves an average score in an IQ test.

2. In a competitive society, individuals' language proficiency is more important than



other abilities.

3 . A wider range of resources and knowledge can be integrated by more creative people into bringing about creative approaches.

4 . A creative person not necessarily suffers more mental illness.

Questions 5-9

Use the information in the passage to match the people (listed A-F) with opinions or deeds below.

Write the appropriate letters A-F in boxes 5-9 on your answer sheet.

A Jamison

B Jordan Peterson

C Guy Claxton

D Howard-Jone

E Teresa Amabile

F Vera John-Steiner

5 . Instead of producing the negative mood, a shift of mood state might be the one important factor of inducing a creative thinking.

6 . Where the more positive moods individuals achieve, there is higher creativity in organizations.

7 . Good interpersonal relationship and trust contribute to a person with more creativity.

8 . Creativity demands an ability that can easily change among different kinds of thinking.

9 . Certain creative mind can be upgraded if we are put into more practice in assessing and processing ideas.

Questions 10-13

Complete the summary paragraph described below. In boxes 10-13 on your answer sheet, write the correct answer with NO MORE THAN THREE WORDS.

But what of the creative act itself? In 1978, Colin Martindale made records of pattern of brain waves as people made up stories by applying a system constituted of many 10_____. The two phrases of creativity, such as 11_____were found. While people were still planning their stories, their brains shows little active sign and the mental activity was showed a very relaxed state as the same sort of brain activity as in sleep, dreaming or rest. However, experiment proved the signal of 12_____went down and the brain became busier, revealing increased cortical arousal, when these people who were in a laidback state were required to produce their stories. Strikingly, it was found the person who was perceived to have the greatest 13_____in brain activity between two stages, produced storylines with highest level of creativity.



How Well Do We Concentrate?

A

Do you read while listening to music? Do you like to watch TV while finishing your homework? People who have these kinds of habits are called multi-taskers. Multitaskers are able to complete two tasks at the same time by dividing their focus. However, Thomas Lehman, a researcher in Psychology, believes people never really do multiple things simultaneously. Maybe a person is reading while listening to music, but in reality, the brain can only focus on one task. Reading the words in a book will cause you to ignore some of the words of the music. When people think they are accomplishing two different tasks efficiently, what they are really doing is dividing their focus. While listening to music, people become less able to focus on their surroundings. For example, we all have experience of times when we talk with friends and they are not responding properly. Maybe they are listening to someone else talk, or maybe they are reading a text on their smart phone and don't hear what you are saying. Lehman called this phenomenon "email voice"

B

The world has been changed by computers and its spin offs like smart-phones or cellphones. Now that most individuals have a personal device, like a smart-phone or a laptop, they are frequently reading, watching or listening to virtual information. This raises the occurrence of multitasking in our day to day life. Now when you work, you work with your typewriter, your cellphone, and some colleagues who may drop by at any time to speak with you. In professional meetings, when one normally focuses and listens to one another, people are more likely to have a cell phone in their lap, reading or communicating silently with more people than ever, even inventions such as the cordless phone has increased multitasking. In the old days, a traditional wall phone would ring, and then the housewife would have to stop her activities to answer it. When it rang, the housewife will sit down with her legs up, and chat, with no laundry or sweeping or answering the door. In the modern era, our technology is convenient enough to not interrupt our daily tasks.

C

Earl Miller, an expert at the Massachusetts Institute of Technology, studied the prefrontal cortex, which controls the brain while a person is multitasking. According to his studies, the size of this cortex varies between species. He found that for humans,



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the size of this part constitutes one third of the brain, while it is only 4 to 5 percent in dogs, and about 15% in monkeys. Given that this cortex is larger on a human, it allows a human to be more flexible and accurate in his or her multitasking.. However, Miller wanted to look further into whether the cortex was truly processing information about two different tasks simultaneously. He designed an experiment where he presents visual stimulants to his subjects in a way that mimics multi-tasking. Miller then attached sensors to the patients' heads to pick up the electric patterns of the brain. This sensor would show if the brain particles, called neurons, were truly processing two different tasks. What he found is that the brain neurons only lit up in singular areas one at a time, and never simultaneously.

D

Davis Meyer, a professor of University of Michigan, studied the young adults in a similar experiment. He instructed them to simultaneously do math problems and classify simple words into different categories. For this experiment, Meyer found that when you think you are doing several jobs at the same time, you are actually switching between jobs. Even though the people tried to do the tasks at the same time, and both tasks were eventually accomplished, overall, the task took more time than if the person focused on a single task one at a time.

E

People sacrifice efficiency when multitasking, Gloria Mark set office workers as his subjects. He found that they were constantly multitasking. He observed that nearly every 11 minutes people at work were disrupted. He found that doing different jobs at the same time may actually save time. However, despite the fact that they are faster, it does not mean they are more efficient. And we are equally likely to self-interrupt as be interrupted by outside sources. He found that in office nearly every 12 minutes an employee would stop and with no reason at all, check a website on their computer, call someone or write an email. If they concentrated for more than 20 minutes, they would feel distressed. He suggested that the average person may suffer from a short concentration span. This short attention span might be natural, but others suggest that new technology may be the problem. With cellphones and computers at our sides at all times, people will never run out of distractions. The format of media, such as advertisements, music, news articles and TV shows are also shortening, so people are used to paying attention to information for a very short time.

F

So even though focusing on one single task is the most efficient way for our brains to work, it is not practical to use this method in real life. According to human nature, people feel more comfortable and efficient in environments with a variety of tasks, Edward Hallowell said that people are losing a lot of efficiency in the workplace due



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to multitasking, outside distractions and self-distractions. As it matter of fact, the changes made to the workplace do not have to be dramatic. No one is suggesting we ban e-mail or make employees focus on only one task. However, certain common workplace tasks, such as group meetings, would be more efficient if we banned cell-phones, a common distraction. A person can also apply these tips to prevent self-distraction. Instead of arriving to your office and checking all of your e-mails for new tasks, a common workplace ritual, a person could dedicate an hour to a single task first thing in the morning. Self-timing is a great way to reduce distraction and efficiently finish tasks one by one, instead of slowing ourselves down with multi-tasking.

Questions 14-18

Reading Passage 2 has six paragraphs, A-F.

Which paragraph contains the following information?

Write the correct letter, A-F, in boxes 14-18 on your answer sheet.

- 14. a reference to a domestic situation that does not require multitasking
- 15. a possible explanation of why we always do multitask together
- 16. a practical solution to multitask in work environment
- 17. relating multitasking to the size of prefrontal cortex
- 18. longer time spent doing two tasks at the same time than one at a time

Questions 19-23

Look at the following statements (Questions 19-23) and the list of scientists below.

Match each statement with the correct scientist, A-E.

Write the correct letter, A-E, in boxes 19-23 on your answer sheet.

NB You may use any letter more than once.

List of Scientists

A Thomas Lehman

B Earl Miller

C David Meyer

D Gloria Mark

E Edward Hallowell

- 19. When faced multiple visual stimulants, one can only concentrate on one of them.
- 20. Doing two things together may be faster but not better.
- 21. People never really do two things together even if you think you do.
- 22. The causes of multitask lie in the environment.
- 23. Even minor changes in the workplace will improve work efficiency.

Questions 24-26

Complete the sentences below.



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Choose NO MORE THAN TWO WORDS from the passage for each answer.

Write your answers in boxes 24-26 on your answer sheet.

A term used to refer to a situation when you are reading a text and cannot focus on your surroundings is 24_____

The 25_____part of the brain controls multitasking.

The practical solution of multitask in work is not to allow use of cellphone in 26_____.

Robert Louis Stevenson

A Scottish novelist, poet, essayist, and travel writer, Robert Louis Stevenson was born at 8 Howard Place, Edinburgh, Scotland, on 13 November 1850. It has been more than 100 years since his death. Stevenson was a writer who caused conflicting opinions about his works. On one hand, he was often highly praised for his expert prose and style by many English-language critics. On the other hand, others criticised the religious themes in his works, often misunderstanding Stevenson's own religious beliefs. Since his death a century before, critics and biographers have disagreed on the legacy of Stevenson's writing. Two biographers, KF and CP, wrote a biography about Stevenson with a clear focus. They chose not to criticise aspects of Stevenson's personal life. Instead, they focused on his writing, and gave high praise to his writing style and skill.

The literary pendulum has swung these days. Different critics have different opinions towards Robert Louis Stevenson's works. Though today, Stevenson is one of the most translated authors in the world, his works have sustained a wide variety of negative criticism throughout his life. It was like a complete reversal of polarity—from highly positive to slightly less positive to clearly negative; after being highly praised as a great writer, he became an example of an author with corrupt ethics and lack of moral. Many literary critics passed his works off as children's stories or horror stories, and thought to have little social value in an educational setting. Stevenson's works were often excluded from literature curriculum because of its controversial nature. These debates remain, and many critics still assert that despite his skill, his literary works still lack moral value.

One of the main reasons why Stevenson's literary works attracted so much criticism was due to the genre of his writing. Stevenson mainly wrote adventure stories, which was part of a popular and entertaining writing fad at the time. Many of us believe adventure stories are exciting, offers engaging characters, action, and mystery but ultimately can't teach moral principles. The plot points are one-dimensional and



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rarely offer a deeper moral meaning, instead focusing on exciting and shocking plot twists and thrilling events. His works were even criticised by fellow authors. Though Stevenson's works have deeply influenced Oscar Wilde, Wilde often joked that Stevenson would have written better works if he wasn't born in Scotland. Other authors came to Stevenson's defence, including Galsworthy who claimed that Stevenson is a greater writer than Thomas Hardy.

Despite Wilde's criticism, Stevenson's Scottish identity was an integral part of his written works. Although Stevenson's works were not popular in Scotland when he was alive, many modern Scottish literary critics claim that Sir Walter Scott and Robert Louis Stevenson are the most influential writers in the history of Scotland. While many critics exalt Sir Walter Scott as a literary genius because of his technical ability, others argue that Stevenson deserves the same recognition for his natural ability to capture stories and characters in words. Many of Scott's works were taken more seriously as literature for their depth due to their tragic themes, but fans of Stevenson praise his unique style of story-telling and capture of human nature. Stevenson's works, unlike other British authors, captured the unique day to day life of average Scottish people. Many literary critics point to this as a flaw of his works. According to the critics, truly important literature should transcend local culture and stories. However, many critics praise the local taste of his literature. To this day, Stevenson's works provide valuable insight to life in Scotland during the 19th century.

Despite much debate of Stevenson's writing topics, his writing was not the only source of attention for critics. Stevenson's personal life often attracted a lot of attention from his fans and critics alike. Some even argue that his personal life eventually outshone his writing. Stevenson had been plagued with health problems his whole life, and often had to live in much warmer climates than the cold, dreary weather of Scotland in order to recover. So he took his family to a south pacific island Samoa, which was a controversial decision at that time. However, Stevenson did not regret the decision. The sea air and thrill of adventure complimented the themes of his writing, and for a time restored his health. From there, Stevenson gained a love of travelling, and for nearly three years he wandered the eastern and central Pacific. Much of his works reflected this love of travel and adventure that Stevenson experienced in the Pacific islands. It was as a result of this biographical attention that the feeling grew that interest in Stevenson's life had taken the place of interest in his works. Whether critics focus on his writing subjects, his religious beliefs, or his eccentric lifestyle of travel and adventure, people from the past and present have different opinions about Stevenson as an author. Today, he remains a controversial yet widely popular figure in Western literature.

Questions 27-31

Choose the correct letter, A, B, C or D.

Write the correct letter in boxes 27-31 on your answer sheet.

27 Stevenson's biographers KF and CP



- A underestimated the role of family played in Stevenson's life.
B overestimated the writer's works in the literature history.
C exaggerated Stevenson's religious belief in his works.
D elevated Stevenson's role as a writer.
- 28 The main point of the second paragraph is
A the public give a more fair criticism to Stevenson's works.
B recent criticism has been justified.
C the style of Stevenson's works overweigh his faults in his life.
D Stevenson's works' drawback is lack of ethical nature.
- 29 According to the author, adventure stories
A do not provide plot twists well.
B cannot be used by writers to show moral values.
C are more fashionable art form.
D can be found in other's works but not in Stevenson's.
- 30 What does the author say about Stevenson's works?
A They describe the life of people in Scotland.
B They are commonly regarded as real literature.
C They were popular during Stevenson's life.
D They transcend the local culture and stories.
- 31 The lifestyle of Stevenson
A made his family envy him so much.
B should be responsible for his death.
C gained more attention from the public than his works.
D didn't well prepare his life in Samoa.

Questions 32-35

Do the following statements agree with the information given in Reading Passage 3?

In boxes 32-35 on you answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN If there is no information on this

- 32 . Although Oscar Wilde admired Robert Louis Stevenson very much, he believed Stevenson could have written greater works.
- 33 . Robert Louis Stevenson encouraged Oscar Wilde to start writing at first.
- 34 . Galsworthy thought Hardy is greater writer than Stevenson is.
- 35 . Critics only paid attention to Robert Louis Stevenson's writing topics.

Questions 36-40

Complete the notes using the list of words, A-I, below.

Write the correct letter, A-I, in boxes 36-40 on your answer sheet.



Sir Walter Scott and Robert Louis Stevenson

A lot of people believe that Sir Walter Scott and Robert Louis Stevenson are the most influential writer in the history of Scotland, but Sir Walter Scott is more proficient in 36_____, while Stevenson has better 37_____. Scott's books illustrate 38_____ especially in terms of tragedy, but a lot of readers prefer Stevenson's 39_____. What's more, Stevenson's understanding of 40_____ made his works have the most unique expression of Scottish people.

A natural ability

B romance

C colorful language

D critical acclaim

E humor

F technical control

G storytelling

H depth

I human nature

The Connection Between Culture and Thought

A

The world's population has surpassed 7 billion and continues to grow. Across the globe, humans have many differences. These differences can be influenced by factors such as geography, climate, politics, nationality, and many more. Culture is one such aspect that can change the way people behave.

B

Your culture may influence your clothing, your language, and many aspects of your life. But is culture influential enough to change the way an individual thinks? It has long been believed that people from different cultures would think differently. For



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example, a young boy from a farm would talk about cows while a boy from New York will talk about cars. If two young children from different countries are asked about their thoughts about a painting, they would answer differently because of their cultural backgrounds.

C

In recent years, there has been new research that changed this long-held belief; However, this new research is not the first to explore the idea that culture can change the way we think. Earlier research has provided valuable insight to the question. One of the earliest research projects was carried out in the Soviet Union. This project was designed to find out whether culture would affect people's way of thought processing. The researchers focused on how living environment and nationality might influence how people think. The experiment led by Bessett aimed to question such awareness of cognitive psychology. Bessett conducted several versions of the experiment to test different cognitive processes.

D

One experiment led by Bessett and Masuku showed an animated video picturing a big fish swimming among smaller fish and other sea creatures. Subjects were asked to describe the scene. The Japanese participants tended to focus on the aquatic background, such as the plants and colour of the water, as well as the relationship between the big and small fish. American participants tended to focus on individual fishes, mainly the larger, more unique looking fish. The experiment suggested that members of Eastern cultures focus more on the overall picture, while members of Western culture focus more on the individuals.

E

In another experiment performed by Bessett and Choi, the subjects were presented with some very convincing evidence for a position. Both the Korean and the American showed strong support. And after they were given some evidence opposing the position, the Korean started to modify or decreased their support. However, the American began to give more support to the former argument. This project suggested that in Korean culture, support for arguments is based on context. Ideas and conclusions are changeable and flexible, so an individual may be more willing to change his or her mind. For Americans, they were less willing to change their original conclusion.

F

Bessett and Ara devised an experiment to test the thought processing of both oriental and occidental worlds. Test subject was given an argument "All animals with furs



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hibernate. Rabbit has fur. Therefore, rabbit hibernate”. People from the eastern world questioned the argument as not being logical, because in their knowledge some furry animals just don’t hibernate. But the American think the statement is right. They assume the logic deduction is based on a correct argument, thus the conclusion is right since the logic is right.

G

From these early experiments in the Soviet Union, one might conclude that our original premise— that culture can impact the way we think—was still correct. However, recent research criticises this view, as well as Bessett’s early experiments. Though these experiments changed the original belief on thought processing, how much does it result from all factors needs further discussion. Fischer thinks Bessett’s experiments provide valuable information because his research only provides qualitative descriptions, not results from controlled environment. Chang partly agrees with him, because there are some social factors that might influence the results.

H

Another criticism of Bessett’s experiments is that culture was studied as a sub-factor of nationality. The experiments assumed that culture would be the same among all members of a nationality. For example, every American that participated in the experiments could be assumed to have the same culture. In reality, culture is much more complicated than nationality. These early experiments did not control for other factors, such as socioeconomic status, education, ethnicity, and regional differences in culture. All of these factors could have a big effect on the individual’s response.

I

A third criticism of Bessett’s experiment is that the content itself should have been more abstract, such as a puzzle or an IQ test. With objective content, such as nature and animals, people from different countries of the world might have different preconceived ideas about these animals. Prior knowledge based on geographic location would further complicate the results. A test that is more abstract, or more quantitative, would provide a more controlled study of how cognitive processing works for different groups of people.

J

The research on culture’s effect on cognitive processing still goes on today, and while some criticisms exist of Bessett’s early studies, the projects still provide valuable insight. It is important for future research projects to control carefully for the variables, such as culture. Something like culture is complex and difficult to define. It can also be influenced by many other variables, such as geography or education styles. When



studying a variable like culture, it is critical that the researcher create a clear definition for what is—and what is not—considered culture.

K

Another important aspect of modern research is the ethical impact of the research. A researcher must consider carefully whether the results of the research will negatively impact any of the groups involved. In an increasingly globalised job economy, generalisations made about nationalities can be harmful to prospective employees. This information could also impact the way tests and university admissions standards are designed, which would potentially favor one group or create a disadvantage for another. When conducting any research about culture and nationality, researchers should consider all possible effects, positive or negative, that their conclusions may have when published for the world to see.

Questions 1-5

Reading Passage 1 has eleven paragraphs, A-K.

Which paragraph contains the following information?

Write the correct letter, A-K, in boxes 1-5 on your answer sheet.

NB You may use any letter more than once.

- 1 All people have the same reaction to a certain point of view.
- 2 Qualitative descriptions are valuable in exploring thought processing.
- 3 Different cultures will affect the description of the same scene.
- 4 We thought of young people as widely different at different geographical locations.
- 5 Eastern people are less likely to stick to their argument.

Questions 6-9

Look at the following statements (Questions 6-9) and the list of researchers below.

Match each statement with the correct researcher, A-C.

Write the correct letter, A-C, in boxes 6-9 on your answer sheet.

NB You may use any letter more than once.

A Bessett & Masuku

B Bessett & Choi

C Bessett & Ara

- 6 Geographical location affects people's position on certain arguments.
- 7 Animated images reveal different process strategies.
- 8 Eastern people challenge a deduction because they knew it is not true.
- 9 Eastern people find more difficulty when asked to identify the same object.

Questions 10-13

Complete the sentences below.

Choose NO MORE THAN TWO WORDS from the passage for each answer

Write your answers in boxes 10-13 on your answer sheet.



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Researchers in the Soviet Union wanted to find out how 10 _____ and nationality will control the way people think.

Bessett and Ara's experiment shows, for Americans, so long as the logic deduction is based on a correct argument, the 11 _____ should be right.

Fischer thinks Bessett's research is quite valuable because it is conducted in a 12 _____ way rather than in controlled environment.

Future researchers on culture's effect on cognitive processing should start with a 13 _____ of culture as a variable.

Implication of False Belief Experiments

A

A considerable amount of research since the mid 1980s has been concerned with what has been termed children's theory of mind. This involves children's ability to understand that people can have different beliefs and representations of the world— a capacity that is shown by four years of age. Furthermore, this ability appears to be absent in children with autism. The ability to work out that another person is thinking is clearly an important aspect of both cognitive and social development. Furthermore, one important explanation for autism is that children suffering from this condition do not have a theory of mind(TOM). Consequently, the development of children's TOM has attracted considerable attention.

B

Wimmer and Perner devised a 'false belief task' to address this question. They used some toys to act out the following story. Maxi left some chocolate in a blue cupboard before he went out. When he was away his mother moved the chocolate to a green cupboard. Children were asked to predict where Maxi will look for his chocolate when he returns. Most children under four years gave the incorrect answer, that Maxi will look in the green cupboard. Those over four years tended to give the correct answer, that Maxi will look in the blue cupboard. The incorrect answers indicated that the younger children did not understand that Maxi's beliefs and representations no longer matched the actual state of the world, and they failed to appreciate that Maxi will act on the basis of his beliefs rather than the way that the world is actually organised.

C

A simpler version of the Maxi task was devised by Baron-Cohen to take account of



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criticisms that younger children may have been affected by the complexity and too much information of the story in the task described above. For example, the child is shown two dolls, Sally and Anne, who have a basket and a box, respectively. Sally also has a marble, which she places in her basket, and then leaves to take a walk. While she is out of the room, Anne takes the marble from the basket, eventually putting it in the box. Sally returns, and child is then asked where Sally will look for the marble. The child passes the task if she answers that Sally will look in the basket, where she put the marble; the child fails the task if she answers that Sally will look in the box, where the child knows the marble is hidden, even though Sally cannot know, since she did not see it hidden there. In order to pass the task, the child must be able to understand that another's mental representation of the situation is different from their own, and the child must be able to predict behavior based on that understanding. The results of research using false-belief tasks have been fairly consistent: most normally-developing children are unable to pass the tasks until around age four.

D

Leslie argues that, before 18 months, children treat the world in a literal way and rarely demonstrate pretence. He also argues that it is necessary for the cognitive system to distinguish between what is pretend and what is real. If children were not able to do this, they would not be able to distinguish between imagination and reality. Leslie suggested that this pretend play becomes possible because of the presence of a decoupler that copies primary representations to secondary representations. For example, children, when pretending a banana is a telephone, would make a secondary representation of a banana. They would manipulate this representation and they would use their stored knowledge of 'telephone' to build on this pretence.

E

There is also evidence that social processes play a part in the development of TOM. Meins and her colleagues have found that what they term mind mindedness in maternal speech to six-month old infants is related to both security of attachment and to TOM abilities. Mind Mindedness involves speech that discusses infants' feelings and explains their behaviour in terms of mental stages(e.g. 'you1 re feeling hungry')

F

Lewis investigated older children living in extended families in Crete and Cyprus. They found that children who socially interact with more adults, who have more friends. And who have more older siblings tend to pass TOM tasks at a slightly earlier age than other children. Furthermore, because young children are more likely to talk about their thoughts and feelings with peers than with their mothers, peer interaction may provide a special impetus to the development of a TOM. A similar point has been made by Dunn, who argues that peer interaction is more likely to contain pretend play



and that it is likely to be more challenging because other children, unlike adults, do not make large adaptations to the communicative needs of other children.

G

In addition, there has been concern that some aspects of the TOM approach underestimate children's understanding of other people. After all, infants will point to objects apparently in an effort to change a person's direction of gaze and interest; they can interact quite effectively with other people; they will express their ideas in opposition to the wishes of others; and they will show empathy for the feeling of others. Schatz studied the spontaneous speech of three-year-old and found that these children used mental terms, and used them in circumstances where there was a contrast between, for example, not being sure where an object was located and finding it, or between pretending and reality. Thus the social abilities of children indicate that they are aware of the difference between mental states and external reality at ages younger than four.

H

A different explanation has been put forward by Harris. He proposed that children use 'simulation'. This involves putting yourself in the other person's position, and then trying to predict what the other person would do. Thus success on false belief tasks can be explained by children trying to imagine what they would do if they were a character in the stories, rather than children being able to appreciate the beliefs of other people. Such thinking about situations that do not exist involves what is termed counterfactual reasoning.

I

A different explanation has been put forward by Harris. He proposed that children use "simulation". This involves putting yourself in the other person's position, and then trying to predict what the other person would do. Thus, success on false belief tasks can be explained by children trying to imagine what they would do if they were a character in the stories, rather than children being able to appreciate the beliefs of other people. Such thinking about situations that do not exist involves what is termed counterfactual reasoning.

Questions 14-20

Look at the following statements (Questions 14-20) and the list of researchers below.

Match each statement with the correct researcher, A-G.

Write the correct letter. A-G. in boxes 14-20 on your answer sheet.

List of Researchers

A Baron-Cohen



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B Meins

C Wimmer and Perner

D Lewis E Dunn F Schatz G Harris

E Dunn

F Schatz

G Harris

14 gave an alternative explanation that children may not be understanding other's belief

15 found that children under certain age can tell difference between reality and mentality

16 conducted a well-known experiment and drew conclusion that young children were unable to comprehend the real state of the world

17 found that children who get along with adults often comparatively got through the test more easily

18 revised an easier experiment to rule out the possibility that children might be influenced by sophisticated reasoning

19 related social factor such as mother-child communication to capability act in TOM

20 explained children are less likely to tell something interactive to their mother than to their friends

Questions 21-26

Complete the summary below.

Choose ONE WORD ONLY from the passage for each answer.

Write your answers in boxes 21-26 on your answer sheet.

In 1980s, research studies were designed to test the subject called Theory of Mind that if children have the ability to represent the reality. First experiments were carried out on this subject on a boy. And questions had been made on where the boy can find the location of the 21_____. But it was accused that it had excessive 22_____. So second modified experiment was conducted involving two dolls, and most children passed the test at the age of 23_____. Then Lewis and Dunn researched 24_____ children in a certain place, and found children who have more interaction such as more conversation with 25_____ have better performance in the test, and peer interaction is 26_____ because of consisting pretending elements.



The Biggest Impact

A. In 1980 a team of researchers were analysing soil samples at what was then known as the KT boundary. The K is misleading, as it actually refers to the cretaceous era, while the T refers to the tertiary era. What made geologists originally place a division in that distant time, some 65 million years ago, was the mass extinction which then occurred, seeing over two thirds of all land and sea life disappear, including the dinosaurs — or more strictly, all non-birdlike dinosaurs (since birds are now considered dinosaurs' descendants). Whilst this was not the biggest extinction of all, it is definitely the most famous. But what caused it?

B. The researchers discovered that sedimentary layers at the KT boundary contained a concentration of iridium many times higher than what normally occurs — up to 120 times. Most iridium disappeared when the Earth was molten, sinking into its metallic core. However, this element is abundant in asteroids and comets, which led to an intriguing hypothesis — that an asteroid or comet had struck the Earth, causing the mass extinction. The object would have vaporised almost immediately upon impact, throwing its iridium-rich contents into the atmosphere, from where it eventually settled across the entire planet. The problem was, an asteroid large enough to do this would have left traces of its impact in the Earth's crust, and at that time there were no known signs. Or were there?

C. In actual fact, in the 1960s, a contractor named Baltosser working for a Mexican state-owned oil company had looked at a gravity map of the Yucatán Peninsula, near the Gulf of Mexico. He noticed a large arc-shape, showing a symmetry that was impossible to naturally occur. Company policy forbade him from releasing his findings, and so the secret lay until 1978, when two geophysicists, Camargo and Penfield, working for the same company, discovered it again. In the search for possible oil-drilling sites, they had been examining magnetic surveys in the Gulf of Mexico, which revealed an underwater arc. The two arcs, sea-based and land-based, matched perfectly, showing a circle 180 kilometers wide, centred on the coastal village of Chicxulub, and so it became known as the Chicxulub Crater.

D. In 1981, Camargo and Penfield released their findings, but the world was not listening. It took over ten years, and much more evidence (rock samples, drilling cores, and dating of the seabed rocks to the magic figure of 65 million years), before scientists began to accept the findings, although widespread skepticism existed, and still remains, to some extent, today. It is occasionally argued that the impact was not the sole reason for the mass extinction, or that there were other contemporaneous impacts, or that extensive volcanism or climate and sea-level change were the real causes. It was perhaps this that led, in 2010, to an international panel of over 40 scientists being convened in order to specifically address the evidence. They concluded that an asteroid impact, as evidenced by the Chicxulub Crater, was indeed



the cause of the mass extinction.

E. Trying to picture that event, the most powerful ever in the Earth’s history, strains the imagination. It begins with a 10-15 kilometer wide rock appearing from nowhere, almost instantaneously vaporising, and releasing over two million times the energy of an atomic bomb. The most immediate effect is a cloud of super-heated dust, ash, and steam expanding outwards, igniting fires, and broiling everything in its path. A split second later follows a series of shock waves, traveling across the surface of the globe, triggering earthquakes and volcanic eruptions. Next there is a ‘mega-tsunami’, thousands of meters high, ripping coastlines apart and stirring up the oceans. Then, in the next few weeks, the huge amounts of carbon dioxide from the vaporisation of carbonate rock heats the Earth, but with the atmosphere choked with dust for years, sunlight is blocked, killing off plants, ultimately plunging the Earth into winter and the entire biosphere into absolute chaos.

F. The surprising fact is not that so many creatures became extinct, but that so many survived! With global disruption to plant communities, the herbivorous dinosaurs died quickly, and their predators soon followed. Sea-based life suffered disastrously, and all giant marine reptiles disappeared, yet the ancestors of the crocodile survived. It is theorised that, like modern crocodiles, they were semi-aquatic and thus were able to shelter in the water from fires and blast damage, and yet could scavenge on land amongst the abundance of dead animals for years afterwards. Similarly, insects, worms, and molluscs could all feed on dead plant and animal matter, allowing those that fed on these creatures to survive. Consequently, insectivores, scavengers, or those with omnivorous eating habits, including mammals and smaller bird-like reptiles, were preserved.

G. Thus, the dinosaurs as we know them, after 135 million years as the dominant land animal, were all but gone. This allowed mammals, then only small burrowing cat-like creatures (attributes which had also helped ensure their survival throughout the disaster), to emerge from the undergrowth, diversity, and eventually rule the land. In an ironic consequence, that class of animal ultimately led to species Homo sapiens, or human beings. So, were it not for that disastrous extinction 65 million years ago, we would not be here today.

Questions 27-32

Reading Passage Three has seven paragraphs, A-G.

List of Heading	
i	The situation in the sea
ii	The first piece of evidence
iii	A fortunate consequence



iv	Preservation strategies
v	Company procedures
vi	The mystery of the border
vii	A first-hand view
viii	An unexpected element
ix	A final decision
x	Heated debate

Choose the correct heading for Paragraphs B-G from the list of headings.

Write the correct number, i-x, for each answer.

Example	Answer
Paragraph A	vi.....

- 27 . Paragraph B
- 28 . Paragraph C
- 29 . Paragraph D
- 30 . Paragraph E
- 31 . Paragraph F
- 32 . Paragraph G

Questions 33-37

Complete the summary of the first half of the passage.

Choose NO MORE THAN TWO WORDS from the passage for each answer.

Baltosser, a contractor, was the first to identify the 33_____, but could not reveal this information because of 34_____. Years later, the discovery of 35_____ at the KT boundary added further evidence, after which Camargo and Penfield finally showed the world Baltosser's discovery. Nevertheless, they had to overcome 36_____ until an international commission confirmed this event as the real reason for the 37_____ which followed.

Questions 38-40

Choose the correct letter, A, B, C, or D.

38 After the asteroid struck the Earth:

- A the shock wave was followed by the object's vaporisation.
- B the Earth warmed before going cold.
- C the tsunami caused earthquakes and volcanic eruptions.
- D the eruptions plunged the atmosphere into chaos.

39 In the aftermath of the asteroid strike

- A all the dinosaurs died.



- B all reptiles died.
C the dead animals were important.
D the water allowed shelter for mammals.
- 40 Mammals of that time survived because they
- A consumed dead animals and plant
B were large and strong.
C lived in the shadows of trees.
D were a special class of animals.

BIOMETRICS

A

The term “biometrics” is derived from the Greek words bio (life) and metric (to measure). It refers to technologies for measuring and analysing a person’s physiological or behavioural characteristics, such as fingerprints, irises, voice patterns, facial patterns and hand measurements, for identification and verification purposes. One of the earliest known examples of biometrics in practice was a form of fingerprinting used in China in the 14th century. Chinese merchants stamped children’s palm prints and footprints on paper with ink to distinguish the young children from one another. This method of biometrics is still being practised today.

B

Until the late 1800s, identification largely relied upon ‘photographic memory.’ In the 1890s, an anthropologist and police desk clerk in Paris named Alphonse Bertillon sought to fix the problem of identifying convicted criminals and turned biometrics into a distinct field of study. He developed a method of multiple body measurements which was named after him – Bertillonage. Bertillon based his system on the claim that the measurement of adult bones does not change after the age of 20. He also introduced a cataloguing system, which enabled the filing and checking of records quite quickly. His system was used by police authorities throughout the world, until 1903, when two identical measurements were obtained for two different persons at Fort Leavenworth prison. The prison switched to fingerprinting the following day and the rest of the world soon followed, abandoning Bertillonage forever. After the failure of Bertillonage, the police started using fingerprinting, which was developed by Richard Edward Henry of Scotland Yard, essentially reverting to the same methods used by the Chinese for years.

C



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In the past three decades, biometrics has moved from a single method (fingerprinting) to more than ten different methods. Hundreds of companies are involved with this development and continue to improve their methods as the technology available to them advances. As the industry grows, however, so does the public concern over privacy issues. Laws and regulations continue to be drafted and standards are beginning to be developed. While no other biometric has yet reached a wide range of use of fingerprinting, some are beginning to be used in both legal and business areas.

D

Identification and verification have long been in practice by presenting a personal document, such as a licence, ID card or a passport. It may also require personal information such as passwords or PINs. For security reasons, often two, or all three, of these systems are combined but as times progress, we are in constant need for more secure and accurate measures. Authentication by biometric verification is becoming increasingly common in corporate and public security systems, consumer electronics and point-of-sale applications. In addition to security, the driving force behind biometric verification has been convenience. Already, many European countries are introducing a biometric passport which will carry a paper-thin computer chip to store the facial image and at least one additional biometric identifier. This will help to counter fraudulent efforts to obtain duplicate passports and will verify the identity of the holder against the document.

E

Identification and verification are mainly used today in the fight against crime with the methods of fingerprint and DNA analysis. It is also used in security for granting access rights by voice pattern recognition. Additionally, it is used for personal comfort by identifying a person and changing personal settings accordingly, as in setting car seats by facial recognition. Starting in early 2000, the use of biometrics in schools has become widespread, particularly in the UK and USA. A number of justifications are given for such practices, including combatting truancy and replacing library cards or meal cards with fingerprinting systems. Opponents of school biometrics have raised privacy concerns against the creation of databases that would progressively include the entire population.

F

Biometric devices consist of a reader or scanning device, software that converts the gathered information into digital form, and a database that stores the biometric data for comparison with previous records. When converting the biometric input, the software identifies specific points of data as match points. The match points are processed using an algorithm into a value that can be compared with biometric data in



the database. There are two types of biometrics: behavioural and physical. Behavioural biometrics are generally used for verification while physical biometrics can be used for either identification or verification.

G

Iris-pattern and retina-pattern authentication methods are already employed in some bank automatic teller machines. Voice waveform recognition, a method of verification that has been used for many years with tape recordings in telephone wiretaps, is now being used for access to proprietary databanks in research facilities. Facial-recognition technology has been used by law enforcement to pick out individuals in large crowds with considerable reliability. Hand geometry is being used in the industry to provide physical access to buildings. Earlobe geometry has been used to disprove the identity of individuals who claim to be someone they are not (identity theft). Signature comparison is not as reliable, all be itself, as other biometric verification methods but offers an extra layer of verification when used in conjunction with one or more other methods. No matter what biometric methodology is used, the identification verification process remains the same. A record of a person's unique characteristic is captured and kept in a database. Later on, when identification verification is required, a new record is captured and compared with the previous record in the database. If the data in the new record matches that in the database record, the person's identity is confirmed.

H

As technology advances, and time goes on, more and more private companies and public utilities will use biometrics for safe, accurate identification. However, these advances will raise many concerns throughout society, where many may not be educated on the methods. Some believe this technology can cause physical harm to an individual using it, or that instruments used are unsanitary. For example, there are concerns that retina scanners might not always be clean. There are also concerns as to whether our personal information taken through biometric methods can be misused, tampered with, or sold, eg. by criminals stealing, rearranging or copying the biometric data. Also, the data obtained using biometrics can be used in unauthorised ways without the individual's consent. Much still remains to be seen in the effectiveness of biometric verification before we can identify it as the safest system for identification.

Questions 27-31

Reading Passage 3 has eight paragraphs, A-H.

Which paragraph contains the following information?

Write the correct letter A-H in boxes 27-31 on your answer sheet.

- 27 possible health hazards associated with the use of biometrics
- 28 convicted criminals were not the first to be identified by the use of biometrics
- 29 the application of mathematics in assessing biometric data



- 30 despite its limitations, biometrics has become a commercial field of activity
31 some biometric methods are useful only in conjunction with others

Questions 32-34

Complete the sentences below.

Choose NO MORE THAN TWO WORDS from the passage for each answer.

- 32 Members of the public are becoming increasingly worried about the _____ that may accompany the use of biometrics.
33 Biometrics can be used to improve the _____ of drivers and passengers.
34 Regardless of the technology used, it has one common purpose: to find somebody's _____ and store it on the computer.

Questions 35-40

Complete the summary with the list of words A-L below.

Write the correct letter A-L in boxes 35-40 on your answer sheet.

BIOMETRICS

As long ago as the 14th century, the Chinese made use of biometrics in order to tell young children apart, but it was only in the 1890s when it was first used by the authorities as a means of 35 _____ in criminal cases. The system developed by the Frenchman Bertillon – that of measuring adult bones – was flawed, however, and so police adopted 36 _____ as a more reliable way of identifying suspects. Governments, companies and even schools employ biometric technology to ensure, for example, that people do not enter a country illegally, gain access to certain buildings, or assume someone else's 37 _____. Apart from security, another important 38 _____ behind biometric verification has been 39 _____. The use of biometrics, however, has its critics, who say that the data collected could be used for different purposes without our 40 _____.

- | | | |
|------------------|------------------|-----------------|
| A identification | B security | C convenience |
| D scanning | E fingerprinting | F identity |
| G violation | H measuring | I justification |
| J approval | K factor | L apprehension |

The Beginning of Football!

A

Football as we now know it developed in Britain in the 19th century, but the game is far older than this. In fact, the term has historically been applied to games played on



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foot, as opposed to those played on horseback, so 'football' hasn't always involved kicking a ball. It has generally been played by men, though at the end of the 17th century, games were played between married and single women in a town in Scotland. The married women regularly won.

B

The very earliest form of football for which we have evidence is the 'tsu'chu', which was played in China and may date back 3,000 years. It was performed in front of the Emperor during festivities to mark his birthday. It involved kicking a leather ball through a 30-40 cm opening into a small net fixed onto long bamboo canes – a feat that demanded great skill and excellent technique.

C

Another form of the game, also originating from the Far East, was the Japanese 'kemari' which dates from about the fifth century and is still played today. This is a type of circular football game, a more dignified and ceremonious experience requiring certain skills, but not competitive in the way the Chinese game was, nor is there the slightest sign of struggle for possession of the ball. The players had to pass the ball to each other, in a relatively small space, trying not to let it touch the ground.

D

The Romans had a much livelier game, 'harpastum'. Each team member had his own specific tactical assignment took a noisy interest in the proceedings and the score. The role of the feet was so small as scarcely to be of consequence. The game remained popular for 700 or 800 years, but, although it was taken to England, it is doubtful whether it can be considered as a forerunner of contemporary football.

E

The game that flourished in Britain from the 8th to the 19th centuries was substantially different from all the previously known forms – more disorganised, more violent, more spontaneous and usually played by an indefinite number of players. Frequently, the games took the forms of a heated contest between whole villages. Kicking opponents were allowed, as in fact was almost everything else.

F

There was tremendous enthusiasm for football, even though the authorities repeatedly intervened to restrict it, as a public nuisance. In the 14th and 15th centuries, England, Scotland and France all made football punishable by law, because of the disorder that commonly accompanied it, or because the well-loved recreation prevented subjects



from practicing more useful military disciplines. None of these efforts had much effect.

G

The English passion for football was particularly strong in the 16th century, influenced by the popularity of the rather better organised Italian game of ‘calcio’. English football was as rough as ever, but it found a prominent supporter in the school headmaster Richard Mulcaster. He pointed out that it had positive educational value and promoted health and strength. Mulcaster claimed that all that was needed was to refine it a little, limit the number of participants in each team and, more importantly, have a referee to oversee the game.

H

The game persisted in a disorganised form until the early 19th century, when a number of influential English schools developed their own adaptations. In some, including Rugby School, the ball could be touched with the hands or carried; opponents could be tripped up and even kicked. It was recognised in educational circles that, as a team game, football helped to develop such fine qualities as loyalty, selflessness, cooperation, subordination and deference to the team spirit. A ‘games cult’ developed in schools and some form of football became an obligatory part of the curriculum.

I

In 1863, developments reached a climax. At Cambridge University, an initiative began to establish some uniform standards and rules that would be accepted by everyone, but there were essentially two camps: the minority – Rugby School and some others – wished to continue with their own form of the game, in particular allowing players to carry the ball. In October of the same year, eleven London clubs and schools sent representatives to establish a set of fundamental rules to govern the matches played amongst them. This meeting marked the birth of the Football Association.

J

The dispute concerning kicking and tripping opponents and carrying the ball was discussed thoroughly at this and subsequent meetings, until eventually, on 8 December, the die-hard exponents of the Rugby style withdrew, marking a final split between rugby and football. Within eight years, the Football Association already had 50 member clubs, and the first football competition in the world was started – the FA Cup.

Questions 1-7



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Reading Passage 1 has ten paragraphs A-J.

Choose the correct headings for paragraphs D-J from the list of headings below.

Write the correct number i-x in boxes 1-7 on your answer sheet.

List of Headings

- i Limited success in suppressing the game
- ii Opposition to the role of football in schools
- iii A way of developing moral values
- iv Football matches between countries
- v A game that has survived
- vi Separation into two sports
- vii Proposals for minor improvements
- viii Attempts to standardize the game
- ix Probably not an early version of football
- x A chaotic activity with virtually no rules

Example Paragraph C Answer v

- 1 Paragraph D
- 2 Paragraph E
- 3 Paragraph F
- 4 Paragraph G
- 5 Paragraph H
- 6 Paragraph I
- 7 Paragraph J

Questions 8-13

Complete each sentence with the correct ending A-L from the box below.

Write the correct letter A-F in boxes 8-13 on your answer sheet.

- 8 Tsu'chu
- 9 Kemari
- 10 Harpastum
- 11 From the 8th centuries, football in the British Isles
- 12 In the past, the authorities legitimately despised the football and acted on the belief that football.
- 13 When it was accepted in academic settings, football.

- A was seen as something to be encouraged in the young.
- B involved individual players having different responsibilities.
- C was influenced by a game from another country.
- D was a cooperative effort by all the players.
- E distracted people from more important activities.
- F was played by teams of a fixed size.
- G was less popular than it later became.
- H was often played by one community against another.
- I formed part of a celebration.



THE DEVELOPING WORLD

A

THE DEVELOPING WORLD – the economically underdeveloped countries of Asia, Africa, Oceania and Latin America – is considered as an entity with common characteristics, such as poverty, high birth rates, and economic dependence on the advanced countries. Until recently, the developing world was known as ‘the third world’. The French demographer Alfred Sauvy coined the expression (in French) in 1952 by analogy with the ‘third estate’ – the commoners of France before and during the French Revolution – as opposed to priests and nobles, comprising the first and second estates respectively. ‘Like the third estate’, wrote Sauvy, ‘the third world is nothing, and it wants to be something’. The term, therefore, implies that the third world is exploited, much as the third estate was exploited and that, like the third estate, its destiny is a revolutionary one. It conveys as well a second idea, also discussed by Sauvy – that of non-alignment, for the developing world belongs neither to the industrialised capitalist world nor to the industrialised former communist bloc. The expression ‘third world’ was used at the 1955 conference of Afro-Asian countries held in Bandung, Indonesia. In 1956 a group of social scientists associated with Sauvy’s National Institute of Demographic Studies, in Paris, published a book called ‘Le Tiers-Monde’. Three years later, the French economist Francois Perroux launched a new journal, on problems of underdevelopment, with the same title. By the end of the 1950s, the term was frequently employed in the French media to refer to the underdeveloped countries of Asia, Africa, Oceania and Latin America. Present-day politicians and social commentators, however, now use the term ‘developing world’ in a politically correct effort to dispel the negative connotations of ‘third world’.

B

Countries in the developing world have a number of common traits: distorted and highly dependent economies devoted to producing primary products for the developed world; traditional, rural social structures; high population growth and widespread poverty. Nevertheless, the developing world is sharply differentiated, for it includes countries on various levels of economic development. And despite the poverty of the countryside and the urban shantytowns, the ruling elites of most third world countries are wealthy.

C



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This combination of conditions in Asia, Africa, Oceania and Latin America is linked to the absorption of the developing world into the international capitalist economy, by way of conquest or indirect domination. The main economic consequence of Western domination was the creation, for the first time in history, of a world market. By setting up sub-economies linked to the West throughout the developing world, and by introducing other modern institutions, industrial capitalism disrupted traditional economies and, indeed, societies. This disruption led to underdevelopment.

D

Because the economies of underdeveloped countries have been geared to the needs of industrialised countries, they often comprise only a few modern economic activities, such as mining or the cultivation of plantation crops. Control over these activities has often remained in the hands of large foreign firms. The prices of developing world products are usually determined by large buyers in the economically dominant countries of the West, and trade with the West provides almost all the developing world's income. Throughout the colonial period, outright exploitation severely limited the accumulation of capital within the foreign-dominated countries. Even after decolonization (in the 1950s, 1960s, and 1970s), the economies of the developing world grew slowly, or not at all, owing largely to the deterioration of the 'terms of trade' – the relationship between the cost of the goods a nation must import from abroad and its income from the exports it sends to foreign countries. Terms of trade are said to deteriorate when the cost of imports rises faster than income from exports. Since buyers in the industrialised countries determined the prices of most products involved in international trade, the worsening position of the developing world was scarcely surprising. Only the oil-producing countries – after 1973 – succeeded in escaping the effects of Western domination of the world economy.

E

No study of the developing world could hope to assess its future prospects without taking into account population growth. While the mortality rate from poverty-related diseases continues to cause international concern, the birth rate continues to rise at unprecedented levels. This population explosion in the developing world will surely prevent any substantial improvements in living standards, as well as threaten people in stagnant economies with worsening poverty and starvation levels.

Questions 14-18

Reading Passage 2 has five paragraphs, A-E.

Choose the most suitable heading for each paragraph from the list of headings below.

Write the appropriate number i-viii in boxes 14-18 on your answer sheet.

List of Headings

i The great divide between rich and poor.



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- ii The status and destiny of the developing world follow a European precedent.
- iii Economic progress in the developing world slowed down by political unrest.
- iv More people, less food.
- v Western countries refuse to acknowledge their history of colonisation.
- vi Open trade is the main reason these countries become impoverished.
- vii Rivalry in the developing world between capitalist and former communist bloc countries.
- viii Prices and conditions set by outsiders.
- 14 Paragraph A
- 15 Paragraph B
- 16 Paragraph C
- 17 Paragraph D
- 18 Paragraph E

Questions 19-22

Do the following statements agree with the information given in Reading Passage 2?

In boxes 19-22 on your answer sheet, write

YES if the statement agrees with the information

NO if the statement contradicts the information

NOT GIVEN if there is no information on this

- 19 Agriculture still plays a role in the economy of developing countries.
- 20 The population of the developing world increases at such a fast rate because they constantly need to renew the labour force.
- 21 Countries that spend more on imports than they earn from exports can experience problems.
- 22 Like the developing world, oil-rich countries are also a victim of dominance by Western powers.

Questions 23-26

Complete each sentence with the correct ending A-F below.

Write the correct letter A-F in space 23-26 below.

- 23 Countries in the developing world _____
- 24 The term 'the third world' implies _____
- 25 One factor that is prevalent in the developing world is _____
- 26 One consequence of the terms of trade was _____
- A economic dependence on developed countries.
- B that decolonization took a long time to achieve.
- C dictate the needs of industrialised countries.
- D share common characteristics.
- E that many economies stagnated.
- F a society that wants something it does not have.



Father of modern management

A

Peter Drucker was one of the most important management thinkers of the past hundred years. He wrote about 40 book and thousands of articles and he never rested in his mission to persuade the world that management matters. “Management is an organ of institutions ... the organ that converts a mob into an organisation, and human efforts into performance.” Did he succeed? The range of his influence was extraordinary. Wherever people grapple with tricky management problems, from big organizations to small ones, from the public sector to the private, and increasingly in the voluntary sector, you can find Drucker’s fingerprints.

B

His first two books – *The End of Economic Man* (1939) and *The Future of Industrial Man* (1942) – had their admirers, including Winston Churchill, but they annoyed academic critics by ranging so widely over so many different subjects. Still, the second of these books attracted attention with its passionate insistence that companies had a social dimension as well as an economic purpose. His third book, *The Concept of the Corporation*, became an instant bestseller and has remained in print ever since.

C

The two most interesting arguments in *The Concept of the Corporation* actually had little to do with the decentralization fad. They were to dominate his work. The first had to do with “empowering” workers. Drucker believed in treating workers as resources rather than just as costs. He was a harsh critic of the assembly-line system of production that then dominated the manufacturing sector – partly because assembly lines moved at the speed of the slowest and partly because they failed to engage the creativity of individual workers. The second argument had to do with the rise of knowledge workers. Drucker argued that the world is moving from an “economy of goods” to an economy of “knowledge” – and from a society dominated by an industrial proletariat to one dominated by brain workers. He insisted that this had profound implications for both managers and politicians. Managers had to stop treating workers like cogs in a huge inhuman machine and start treating them as brain workers. In turn, politicians had to realise that knowledge, and hence education, was the single most important resource for any advanced society. Yet Drucker also thought that this economy had implications for knowledge workers themselves. They had to come to terms with the fact that they were neither “bosses” nor “workers”, but something in between: entrepreneurs who had responsibility for developing their most important resource, brainpower, and who also needed to take more control of their own careers, including their pension plans.

D

However, there was also a hard side to his work. Drucker was responsible for inventing one of the rational school of management’s most successful products –



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“management by objectives”. In one of his most substantial works, *The Practice of Management* (1954), he emphasised the importance of managers and corporations setting clear long-term objectives and then translating those long-term objectives into more immediate goals. He argued that firms should have an elite corps of general managers, who set these long-term objectives, and then a group of more specialised managers. For his critics, this was a retreat from his earlier emphasis on the soft side of management. For Drucker it was all perfectly consistent: if you rely too much on empowerment you risk anarchy, whereas if you rely too much on command-and-control you sacrifice creativity. The trick is for managers to set long-term goals, but then allow their employees to work out ways of achieving those goals. If Drucker helped make management a global industry, he also helped push it beyond its business base. He was emphatically a management thinker, not just a business one. He believed that management is “the defining organ of all modern institutions”, not just corporations.

E

There are three persistent criticisms of Drucker’s work. The first is that he focused on big organisations rather than small ones. *The Concept of the Corporation* was in many ways a fanfare to big organisations. As Drucker said, “We know today that in modern industrial production, particularly in modern mass production, the small unit is not only inefficient, it cannot produce at all.” The book helped to launch the “big organisation boom” that dominated business thinking for the next 20 years. The second criticism is that Drucker’s enthusiasm for management by objectives helped to lead the business down a dead end. They prefer to allow ideas, including ideas for long-term strategies, to bubble up from the bottom and middle of the organisations rather than being imposed from on high. Thirdly, Drucker is criticised for being a maverick who has increasingly been left behind by the increasing rigour of his chosen field. There is no single area of academic management theory that he made his own.

F

There is some truth in the first two arguments. Drucker never wrote anything as good as *The Concept of the Corporation* on entrepreneurial start-ups. Drucker’s work on management by objectives sits uneasily with his earlier and later writings on the importance of knowledge workers and self-directed teams. But the third argument is short-sighted and unfair because it ignores Drucker’s pioneering role in creating the modern profession of management. He produced one of the first systematic studies of a big company. He pioneered the idea that ideas can help galvanise companies. The biggest problem with evaluating Drucker’s influence is that so many of his ideas have passed into conventional wisdom. In other words, he is the victim of his own success. His writings on the importance of knowledge workers and empowerment may sound a little banal today. But they certainly weren’t banal when he first dreamed them up in the 1940s, or when they were first put in to practice in the Anglo-Saxon world in the 1980s. Moreover, Drucker continued to produce new ideas up until his 90s. His work on the management of voluntary organisations remained at the cutting edge.

Questions 17-32



Reading Passage 2 has six paragraphs, A-F

Choose the correct heading for each paragraph from the list below.

Write the correct number, i-ix, in box 17-32 on your answer sheet.

List of Headings

- i The popularity and impact of Drucker's work
- ii Finding fault with Drucker
- iii The impact of economic globalisation
- iv Government regulation of business
- v Early publications of Drucker's
- vi Drucker's view of balanced management
- vii Drucker's rejection of big business
- viii An appreciations of the pros and cons of Drucker's work
- ix The changing role of the employee

27 Paragraph A

28 Paragraph B

29 Paragraph C

30 Paragraph D

31 Paragraph E

32 Paragraph F

Questions 33-36

Do the following statements agree with the information given in Reading Passage 2?

In boxes 20-23 on your answer sheet, write

YES if the statement agrees with what is stated in the passage

NO if the statement counters to what is stated in the passage

NOT GIVEN if there is no relevant information given in the passage

33 Drucker believed the employees should enjoy the same status as the employers in a company

34 Drucker argued the managers and politicians will dominate the economy during a social transition

35 Drucker support that workers are not simply put themselves just in the employment relationship and should develop their resources of intelligence voluntarily

36 Drucker's work on the management is out of date in moderns days

Questions 37-38

Choose TWO letters from A-E.

Write your answers in boxes 24 and 25 on your answer sheet.

Which TWO of the following are true of Drucker's views?

A High-rank executives and workers should be put in balanced positions in management practice

B Young executives should be given chances to start from low-level jobs

C More emphasis should be laid on fostering the development of the union.

D Management should facilitate workers with tools of self-appraisal instead of



controlling them from the outside force

E Leaders should go beyond the scope of management details and strategically establish goals

Questions 39-40

Choose TWO letters from A-E.

Write your answers in boxes 26 and 27 on your answer sheet.

Which TWO of the following are mentioned in the passage as criticisms to Drucker and his views?

- A His lectures focus too much on big organisations and ignore the small ones.
- B His lectures are too broad and lack of being precise and accurate about the facts.
- C He put a source of objectives more on corporate executives but not on average workers.
- D He acted much like a maverick and did not set up his own management groups
- E He was overstating the case for knowledge workers when warning business to get prepared.

Growing of the Aging Society

A

American scientists say that the elderly are now healthier, happier and more independent. The results of a study that has taken place over a 14-year period will be released at the end of the month. The research will show that common health disorders suffered by the elderly are affecting fewer people and happening after in life.

B

Over the last 14 years, The National Long-term Health Care Survey has gathered data from more than 20,000 males and females over the age of 65 about their health and lifestyles. The group has analysed the results of data gathered in 1994 on conditions such as arthritis, high blood pressure and poor circulation; these were the most common medical complaints for this age group. The results show that these conditions are troubling a smaller proportion of people each year and decreasing very quickly. Other diseases suffered by the elderly including dementia, emphysema and arteriosclerosis are also affecting fewer people.



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C

According to Kenneth Manton, a demographer from Duke University in North Carolina, “the question of what should be considered normal ageing has really changed.” He also mentioned that diseases suffered by many people around the age of 65 in 1982 are now not occurring until people reach the age of 70-75.

D

It is clear that due to medical advances some diseases are not as prominent as they used to be. However, there was also be other factors influencing this change. For instance, improvements in childhood nutrition in the first quarter of the twentieth century gave many people a better start in life than was possible before.

E

The data also shows some negative changes in public health. The research suggests that the rise of respiratory conditions such as lung cancer and bronchitis may reflect changing smoking habits and an increase in air pollution. Manton says that as we have been exposed to worse and worse pollution, it is not surprising that some people over the age of 60 are suffering as a result.

F

Manton also found that better-educated people are likely to live longer. For instance, women of 65 with less than eight years of education are expected to live to around 82. Those who studied more could be able to live seven years longer. Whilst some of this can be attributed to better-educated people usually having a higher income, Manton believes it is mainly because they pay closer attention to their health.

G

Also, the survey estimated how independent people of 65 were and found a striking trend. In the 1994 survey, almost 80% of them were able to complete activities such as eating and dressing alone as well as handling difficult tasks, like cooking and managing their financial affairs. This situation indicates an important drop among disabled elderly people in the population. If 14 years ago, the apparent trends in the US had continued, researchers believe that there would be one million disabled elderly people in today's population. Manton shows the trend saved more than \$200 billion for the US's governments Medicare system, and it has suggested the elderly American population is less of a financial burden than expected.

H



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The growing number of independent elderly people is probably linked to the huge increase in home medical aids. For instance, the research shows the use of raising toilet seat covers and bath seats has increased by more than fifty per cent. Also, these developments about health benefits are reported by the MacArthur Foundation's research group for successful ageing. It found the elderly who are able to take care of themselves were more likely to stay healthy in their old age.

I

Retaining a certain level of daily physical activity may also help brain function, according to Carl Cotman, a neuroscientist at the University of California at Irvine. He found that rats exercising on a treadmill have higher levels of a brain-derived neurotrophic factor in their brains. He believes the hormone which holds neuron functions may prevent the active human's brain function from declining.

J

Teresa Seeman, a social epidemiologist at the University of Southern California in Los Angeles, was conducting the same research. She found a line between self-esteem and stress in people over 70. The elderly who did challenging activities such as driving have more control of their mind and have a lower level of the stress hormone cortisol in their brains. Chronically high levels of this hormone can cause heart disease.

K

However, an independent life may have negative points. Seeman knew that the elderly people that were living alone were able to retain higher levels of stress hormones even when sleeping. The research indicates that elderly people are happier if they can live an independent life but also acknowledge when they need help.

L

Seeman says, "With many cases of research about ageing, these results help common sense." Also, the situations show that we may be ignoring some of the simple factors. She mentions, "The sort of thing your grandmother always used to talk to you about seems to be exactly right."

Questions 1-6

Reading Passage 1 has twelve paragraphs, A-L.

Choose the correct heading for paragraphs B-G from the list of headings below.

Write the correct number, i-vii, in boxes 1-6 on your answer sheet.

List of Headings

i Disorders strike much later in life.



- ii Drawbacks in public health.
- iii Longevity based on high education.
- iv The elderly people of today got better nutrition when they were children.
- v The elderly are becoming more well off.
- vi Most of independent people over 65 complete activities themselves.
- vii Diseases have decreased recently.

Example Answer

Paragraph A v

- 1 Paragraph B
- 2 Paragraph C
- 3 Paragraph D
- 4 Paragraph E
- 5 Paragraph F
- 6 Paragraph G

Questions 7-13

Do the following statements reflect the claims of the writer in Reading Passage 1?

In boxes 7-13 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

- 7 Smoking habits are a crucial cause in some cancers.
- 8 The better-educated elderly people tend to live longer.
- 9 People over 65 can independently manage a variety of tasks.
- 10 Elderly people have overcome dementia as a result of home medical aids.
- 11 Continuing physical exercises is likely to assist digestive function.
- 12 People over 70 who still do challenging things such as driving are able to lower their level of the hormone cortisol which is linked to heart disease.
- 13 Isolation may cause a higher level of stress hormones.

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The Mysteries of Water

Section A

From the nature of dark matter and the origin of the universe to the research for a theory of everything, we come across many mysteries. Whilst these are all puzzles on a grand scale, there is another not quite so grand but equally confusing mystery of the physical world that you can observe from the comfort of your own kitchen. Simply fill a tall glass with chilled water, throw in an ice cube and leave it to stand. The fact that the ice cube floats is the first oddity. And the mystery deepens if you take a



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thermometer and measure the temperature of the water at various depths. At the top, near the ice cube, you'll find it to be around 0°C, but at the bottom, it should be about 4°C. That's why water is denser at 4°C than it is at any other temperature which is another strange feature that sets it apart from other liquids.

Section B

Water's odd but essential qualities don't stop there, for ice is less dense than water, and water is less dense at its freezing point than it is when it is slightly warmer. It freezes from the top down rather than the bottom up. So even during the ice ages, life kept going on to flourish on lake floors and in the deep ocean. Also, water has an extraordinary capacity to absorb up the heat, and this helps smooth out climatic changes that could otherwise lay waste to ecosystems. However, in spite of water's enormous importance to life, no single theory had been able to satisfactorily explain its mysterious qualities – until now. If we can believe physicists Anders Nilsson at Stanford University, California, and Lars Pettersson of Stockholm University, Sweden, we could, at last, be getting to the bottom of many of these anomalies.

Section C

Their disputed ideas expand on a theory proposed more than a century ago. According to Wilhelm Roentgen, the man who discovered the X-ray, the molecule in liquid water packs together not in just one way, as today's textbooks would have us believe, but in two different ways. The way its molecules are composed of two hydrogen atoms and one oxygen atom and how they interact with one another is essential to the understanding of water's mysteries. The oxygen atom has a slight negative charge whilst the hydrogen atoms share a compensating positive charge. Through this process, the hydrogen and oxygen atoms of neighbouring molecules are drawn to one another, forming a link called a hydrogen bond.

Section D

Hydrogen bonds are even weaker than the bonds that link the atoms within molecules together, so keep going to break and reform, but they are at their strongest when molecules are organized so that each hydrogen bond lines up with a molecular bond. The shaping of a water molecule is such that each H₂O molecule is surrounded by four neighbours organized in the shaping of a triangular pyramid better known as a tetrahedron. At least, that's the way the molecules organize themselves in ice. From the conventional view, liquid water has a similar, although less hard, structure, in which extra molecules are able to pack into some of the open gaps in the tetrahedral arrangement. It explains why liquid water is denser than ice – and it seems to comply with the results of various experiments that beams of X-rays, infrared light and neutrons are bounced off samples of water.

Section E

Some physicists had suggested that water placed under certain extreme conditions may separate into two different structures, but most had assumed it resumes a single



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structure under normal conditions. And then, 10 years ago, a change found by Pettersson and Nilsson called this idea into question. They were using X-ray absorption spectroscopy to research the amino acid glycine. The peaks in the X-ray absorption spectrum can shed light on the accurate nature of the target substance's chemical bonds on its structure. Critically, the researchers had got hold of a new, high-power X-ray source with which they could make more sensitive and precise measurements than had ever been possible. They soon knew that the water containing their glycine sample was producing a far more interesting spectrum than the amino acids did. Nilsson recalls, "What we saw there was sensational, so we had to get to the bottom of it."

Section F

The characteristic that sparked their interest was a peak point in the absorption spectrum that is not anticipated by the traditional model of liquid water. Actually, a paper published in 2004 concludes that at any given moment 85% of the hydrogen bonds in water must be weakened or broken. This is far more than the 10% anticipated by the textbook model. The hints of this finding are dramatic: it claims that a total rethink of the structure of water is needed. So, both Nilsson and Pettersson turned to other X-ray experiments to confirm these claims. Their first move was to enlist the aid of Shik Shin of the University of Tokyo who specialises in a technique called X-ray emission spectroscopy. The main thing about these spectra is that the shorter the wavelength of the X-ray in a substance's emission spectrum is, the looser the hydrogen bonding must be.

The team struck gold: the two peak spectrum of discharged X-ray might correspond to two separate structures. The researchers insisted that the spike of the longer-wavelength X-ray indicates the proportion of tetrahedrally organized molecules, whilst the shorter-wavelength peak reflects the proportion of disordered molecules. Critically, the shorter-wavelength peak in the X-ray emissions was the more intense of the two, suggesting that the loosely bound molecules must be more outstanding within the sample, an assertion that fitted the team's previous models. What's more, they also recognised that this peak shifts to an even shorter wavelength, as if the water was heated, the other peak remains more or less fixed.

Questions 14-18

Reading Passage 1 has six sections, A-F.

Choose the correct heading for sections B-F from the list of headings below.

Write the correct number, i-viii, in boxes 14-18 on your answer sheet.

List of Headings

- i a significant role to creatures
- ii spectrum's previous models
- iii a distinction of hydrogen bonds
- iv nature's mysteries in the small place
- v the effect of the spectrum on liquid water
- vi molecular composition of water



- vii water based on infrared light
viii on-body structure of water
Example Answer
Section A iv
14 Section B
15 Section C
16 Section D
17 Section E
18 Section F

Questions 19-26

Do the following statements reflect the claims of the writer in Reading Passage 2?

In boxes 19-26 on your answer sheet, write

YES if the statement reflects the opinion of the writer

NO if the statement contradicts the opinion of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

- 19 Water's temperature of the top and bottom is generally the same.
20 During the ice ages, there was life in the deep ocean because of warmth.
21 Wilhelm Roentgen discovered X-rays for water molecules.
22 Both hydrogen and oxygen's atoms are similar to a positive charge.
23 A single H₂O molecule is composed entirely of five-angled shape.
24 Pettersson and Nilsson were scrutinising the amino acid glycine by using X-ray absorption spectroscopy.
25 The water including glycine was making a superior spectrum to the amino acid.
26 The shorter wavelength is subjected to the longer wavelength.

The Beginnings of Art Therapy

Art therapy is a relative newcomer to the therapeutic field. Art therapy as a profession began in the mid-20th century, arising independently in English-speaking and European countries. Many of the early practitioners of art therapy acknowledged the influence of a variety of disciplines on their practices, ranging from psychoanalysis through to aesthetics and early childhood education. However, the roots of art as therapy go back as far as the late 18th century, when arts were used in the 'moral treatment' of psychiatric patients.

It wasn't until 1942, however, that the British artist Adrian Hill coined the term 'art therapy', as he was recovering from tuberculosis in a sanatorium. He discovered that therapeutic benefits could be derived from drawing and painting whilst recovering.



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Art, he claimed, could become therapeutic since it was capable of 'completely engrossing the mind... releasing the creative energy of the frequently inhibited patient'. This effect, argued Hill, could in turn help the patient as it would 'build up a strong defence against his misfortunes'.

In 1964, the British Association of Art Therapists was founded. Proponents of art therapy fell into one of two categories: those who believed that the therapeutic effect of art lay in its effectiveness as a psychoanalytic tool to assess a patient through their drawings and those who held the belief that art-making was an end in itself, the creative process acting therapeutically on the patient. The two practices, however, were not incompatible, a degree of overlap occurring between the two. A patient, for example, could produce work that could be analysed for content and forms of self-expression but which could also be a creative outlet at the same time.

Who Benefits from Art Therapy

Art therapy in all its forms has proved effective in the treatment of individuals suffering with a wide range of difficulties or disabilities. These include emotional, behavioural or mental health problems, learning or physical disabilities. These include emotional, behaviour or mental health problems, learning or physical disabilities, neurological conditions and physical illness. Therapy can be provided on a group or individual basis according to the clients' needs. Whether the approach adopted by the therapist is oriented towards a psychoanalytic or creative approach, the effect of therapy is multifold. Partaking in art therapy can raise a patient's self-awareness and enable them to deal with stress and traumatic experience. In addition, art therapy sessions can enhance a patient's cognitive abilities and help the patient enjoy the life-affirming pleasures of making art.

What an Art Therapy Session Involves

Typically, an art therapy session is fundamentally different from an art class in that the individual is encouraged to focus more on their internal feelings and to express them, rather than portray external objects. Although some traditional art classes may ask participants to draw from their imagination, in art therapy the patient's inner world of images, feelings, thoughts and ideas are always of primary importance to the experience. Any type of visual art and medium can be employed in the therapeutic process including painting, drawing, sculpture, photography and digital art.

Art therapy sessions are usually held by skilled and qualified professionals. The presence primarily of the therapist is to be in attendance, guiding and encouraging artistic expression in the patient, in accordance with the original meaning of the word for therapy derived from the Greek word 'therapeia', meaning 'being attentive to'.

The Regulation of Art Therapy



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Requirements for those wishing to become an art therapist vary from country to country. In the USA, where entry to the profession is highly regulated, a master's degree in art therapy is essential. In addition, those applying for such a post must have taken courses in a variety of studio art disciplines in order to demonstrate artistic proficiency. On completion of the master's degree, candidates also have to complete a minimum of 1000 hours of direct client contact post-graduation that is approved by the American Art Therapy Association (AATA).

However, whilst entry to the profession is strictly regulated in the USA, the same does not hold true for other countries. The problem is that art therapy is still considered a developing field. As such, until it becomes truly established as a therapy, its practice and application will remain unregulated in many countries for some time yet.

Questions 27-33

Do the following statements agree with the information given in the text?

For questions 27-33, write:

TRUE	if the statement agrees with the information
------	--

FALSE	if the statement contradicts the information
-------	--

NOT GIVEN	If there is no information on this
-----------	------------------------------------

27 The artist Adrian Hill was strongly influenced by psychoanalytic theories when formulating his ideas on art therapy.

28 Twentieth-century art therapy focuses on treating a client's mental or physical health problems rather than dealing with moral issues.

29 Approaches to art therapy can be broadly considered to be creative or psychoanalytic; however, practitioners tend to avoid combining the two schools of practice.

30 Clients who respond best to art therapy have a previous background in art.

31 Art therapy sessions are more concerned with expression through art than on the created art itself.

32 Many art therapists are insufficiently qualified as they are not aware of the regulations regarding the practice of art therapy.

33 Art therapy sessions involve limited interaction between therapist and client.

Questions 34-37

Complete the sentences below.

Choose NO MORE THAN THREE WORDS from the passage for each answer.

The early pioneers of art therapy admitted that their beliefs had been shaped by a 34 _____ influences.

Artist Adrian Hill realised the 35 _____ of art as therapy, and coined the term 'art therapy' in 1942.

Those supporting art therapy advised a psychoanalytic approach or alternatively one that placed more emphasis on the 36 _____ itself.



Whilst theories behind art therapy may differ, they are 37 _____ in practice.

Questions 38-40

Complete the summary with the list of words, A-F.

Write the correct letter, A-F, in spaces 38-40 below.

A	capable
B	strong
C	keen
D	inhibited
E	creative
F	therapeutic

How Art Therapy Evolved into Its Modern Form

Modern-day art therapy has its beginnings in the 1940s. Adrian Hill, one of its early pioneers, realised that art therapy was effective in helping patients create a 38 _____ resistance to psychological and social stresses. Hill considered that 39 _____ patients would particularly benefit from having an artistic outlet. Art therapy then developed into two types of practice, one emphasising a psychoanalytic approach and the other a more 40 _____ one. Today there is often an overlap between the two practices.

Education Philosophy

A Although we lack accurate statistics about child mortality in the preindustrial period, we do have evidence that in the 1660s, the mortality rate for children who died within 14 days of birth was as much as 30 per cent. Nearly all families suffered some premature death. Since all parents expected to bury some of their children, they found it difficult to invest in their newborn children. Moreover, to protect themselves from the emotional consequences of children's death, parents avoided making any emotional commitment to an infant. It is no wonder that we find mothers leaving their babies in gutters or referring to the death in the same paragraph as a reference to pickles.

B The 18th century witnessed the transformation from an agrarian economy to an industrial one - one of the vital social changes taking place in the Western world. An increasing number of people moved from their villages and small towns to big cities where life was quite different. Social supports which had previously existed in smaller



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communities were replaced by ruthless problems such as poverty, crime, substandard housing and disease. Due to the need for additional income to support the family, young children from the poorest families were forced into early employment and thus their childhood became painfully short. Children as young as 7 might be required to work full-time, subjected to unpleasant and unhealthy circumstances, from factories to prostitution. Although such a role has disappeared in most wealthy countries, the practice of childhood employment still remains a staple in underdeveloped countries and has rarely disappeared entirely.

C The lives of children underwent a drastic change during the 1800s in the United States. Previously, children from both rural and urban families were expected to participate in everyday labour due to the bulk of manual hard work. Nevertheless, thanks to the technological advances of the mid-1800s, coupled with the rise of the middle class and redefinition of roles of family members, work and home became less synonymous over time. People began to purchase toys and books for their children. When the country depended more upon machines, children in rural and urban areas were less likely to be required to work at home. Beginning from the Industrial Revolution and rising slowly over the course of the 19th century, this trend increased exponentially after the Civil War. John Locke, one of the most influential writers of his period, created the first clear and comprehensive statement of the 'environmental position' that family education determines a child's life, and via this, he became the father of modern learning theory. During the colonial period, his teachings about child care gained a lot of recognition in America.

D According to Jean Jacques Rousseau, who lived in an era of the American and French Revolution, people were 'noble savages' in the original state of nature, meaning they were innocent, free and uncorrupted. In 1762, Rousseau wrote a famous novel - Emile - to convey his educational philosophy through a story of a boy's education from infancy to adulthood. This work was based on his extensive observation of children and adolescents, their individuality, his developmental theory and on the memories of his own childhood. He contrasted children with adults and described their age-specific characteristics in terms of historical perspective and developmental psychology. Johan Heinrich Pestalozzi, living during the early stages of the Industrial Revolution, sought to develop schools to nurture children's all-round development. He agreed with Rousseau that humans were naturally good but were spoiled by a corrupt society. His approach to teaching consisted of both general and specific methods, and his theory was based upon establishing an emotionally healthy and homelike learning environment, which had to be in place before more specific instruction occurred.

E One of the best-documented cases of Pestalozzi's theory concerned a so-called feral child named Victor, who was captured in a small town in the south of France in 1800. Prepubescent, mute, naked, and perhaps 11 or 12 years old, Victor had been seen foraging for food in the gardens of the locals in the area, and sometimes accepted



people's direct offers of food before his final capture. Eventually, he was brought to Paris and expected to answer some profound questions about the nature of humanity, but that goal was quashed very soon. A young physician, Jean Marc Gaspard Itard, was optimistic about the future of Victor and initiated a five-year education plan to civilise him and teach him to speak. With a subsidy from the government, Itard recruited a local woman called Madame Guerin to assist him to provide a semblance of a home for Victor, and he spent an enormous amount of time and effort working with Victor. Itard's goal to teach Victor the basics of speech could never be fully achieved, but Victor had learnt some elementary forms of communication.

Although other educators were beginning to recognise the simple truth embedded in Rousseau's philosophy, it is not enough to identify the stages of children's development alone. There must be specific education geared towards those stages. One of the early examples was the invention of kindergarten, which was a word and a movement created by a German-born educator, Friedrich Froebel, in 1840. Froebel placed a high value on the importance of play in children's learning. His invention would spread around the world eventually in a variety of forms. Froebel's ideas were inspired through his cooperation with Johann Heinrich Pestalozzi. Froebel didn't introduce the notion of kindergarten until he was 58 years old, and he had been a teacher for four decades. The notion was a haven and a preparation for children who were about to enter the regimented educational system. The use of guided or structured play was a cornerstone of his kindergarten education because he believed that play was the most significant aspect of development at this time of life. Play served as a mechanism for a child to grow emotionally and to achieve a sense of self-worth. Meanwhile, teachers served to organise materials and a structured environment in which each child, as an individual, could achieve these goals. When Froebel died in 1852, dozens of kindergartens had been created in Germany. Kindergartens began to increase in Europe, and the movement eventually reached and flourished in the United States in the 20th century.

Questions 1-4

Reading Passage 1 has six paragraphs, A-F.

Choose the correct heading for paragraphs A and C-E from the list of headings below. Write the correct number, i-vii, in boxes 1-4 on your answer sheet.

	List of Headings
i	The evolution and development of educational concepts of different thinkers
ii	Why children had to work to alleviate the burden on family
iii	Why children are not highly valued
iv	An explanation for children dying in hospital at an early age
v	The first appearance of modern educational philosophy
vi	The application of a creative learning method on a wild child



vii The emergence and spread of the notion of kindergarten

1. Paragraph A

Example Answer

Paragraphs B ii

2. Paragraph C

3. Paragraph D

4. Paragraph E

Questions 5-8

Look at the following events (Questions 5-8) and the list of dates below.

Match each event with the correct date, A, B or C.

Write the correct letter, A, B or C, in boxes 5-8 on your answer sheet.

NB You may use any letter more than once.

	List of Dates
A	the 18th century (1700-1799)
B	the 19th century (1800-1899)
C	the 20th century (1900-1999)

5. the need for children to work

6. the rise of the middle class

7. the emergence of a kindergarten

8. the spread of kindergartens around the U.S.

Questions 9-13

Look at the following opinions or deeds (Questions 9-13) and the list of people below.

Match each opinion or deed with the correct person, A, B, C or D.

Write the correct letter, A, B, C or D, in boxes 9-13 on your answer sheet.

NB You may use any letter more than once.

	List of People
A	Jean Jacques Rousseau
B	Johan Heinrich Pestalozzi
C	Jean Marc Gaspard Itard
D	Friedrich Froebel

9. was not successful in proving a theory

10. observed children's records

11. suggested a setting for study which prioritized emotional comfort

12. proposed that corruption was not a characteristic of people's nature

13. was responsible for an increase in the number of a type of school



DRINKING FILTERED WATER

A The body is made up mainly of water. This means that the quality of water that we drink every day has an important effect on our health. Filtered water is healthier than tap water and some bottled water. This is because it is free of contaminants, that is, of substances that make it dirty or harmful. Substances that settle on the bottom of a glass of tap water and microorganisms that carry diseases (known as bacteria or germs) are examples of contaminants. Filtered water is also free of poisonous metals and chemicals that are common in tap water and even in some bottled water brands.

B The authorities know that normal tap water is full of contaminants and they use chemicals, such as chlorine and bromine in order to disinfect it. But such chemicals are hardly safe. Indeed, their use in water is associated with many different conditions and they are particularly dangerous for children and pregnant women. For example, consuming bromine for a long time may result in low blood pressure, which may then bring about poisoning of the brain, heart, kidneys and liver. Filtered water is typically free of such water disinfectant chemicals.

C Filtered water is also free of metals, such as mercury and lead. Mercury has ended up in our drinking water mainly because the dental mixtures used by dentists have not been disposed of safely for a long time. Scientists believe there is a connection between mercury in the water and many allergies and cancers as well as disorders, such as ADD, OCD, autism and depression.

D Lead, on the other hand, typically finds its way to our drinking water due to pipe leaks. Of course, modern pipes are not made of lead but pipes in old houses usually are. Lead is a well-known carcinogen and is associated with pregnancy problems and birth defects. This is another reason why children and pregnant women must drink filtered water.

E The benefits of water are well known. We all know, for example, that it helps to detoxify the body. So, the purer the water we drink, the easier it is for the body to rid itself of toxins. The result of drinking filtered water is that the body does not have to use as much of its energy on detoxification as it would when drinking unfiltered water. This means that drinking filtered water is good for our health in general. That is because the body can perform all of its functions much more easily and this results in improved metabolism, better weight management, improved joint lubrication as well as efficient skin hydration.

F There are many different ways to filter water and each type of filter targets different contaminants. For example, activated carbon water filters are very good at taking chlorine out. Ozone water filters, on the other hand, are particularly effective at removing germs.



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G For this reason, it is very important to know exactly what is in the water that we drink so that we can decide what type of water filter to use. A Consumer Confidence Report (CCR) should be useful for this purpose. This is a certificate that is issued by public water suppliers every year, listing the contaminants present in the water. If you know what these contaminants are, then it is easier to decide which type of water filter to get.

Questions 14-20

The text has seven paragraphs, A-G.

Which paragraph contains the following information?

Write the correct letter, A-G, next to questions 14-20.

- 14 . a short summary of the main points of the text
- 15 . a variety of methods used for water filtration
- 16 . making it easier for the body to get rid of dangerous chemicals
- 17 . finding out which contaminants your water filter should target
- 18 . allergies caused by dangerous metals
- 19 . a dangerous metal found in the plumbing of old buildings
- 20 . chemicals of cleaning products that destroy bacteria

Questions 21-26

Do the following statements agree with the information given in the text?

For questions 21-26, write:

TRUE	if the statement agrees with the information
------	--

FALSE	if the statement contradicts the information
-------	--

NOT GIVEN	If there is no information on this
-----------	------------------------------------

- 21 . The type of water you consume on a regular basis has a great impact on your overall health and wellness.
- 22 . Filtered water typically contains water disinfectant chemicals.
- 23 . Exposure to disinfectant chemicals is linked with poisoning of the vital organs.
- 24 . Drinking tap water helps minimise your exposure to harmful elements.
- 25 . People wearing artificial teeth are more likely to be contaminated.
- 26 . People who are depressed often suffer from dehydration.

ALL YOU NEED TO KNOW ABOUT SNOW

Types of Snowfall

Snow is typically the product of weather conditions in which an extratropical cyclone has formed. Extratropical cyclones bring extremely hazardous weather, such as high



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winds and heavy rain or snow, and are often referred to as windstorms in Europe. The band of precipitation associated with their warm front is often very extensive. When the warm front and cold front collide, snow can result on the poleward side of the precipitation band; that is, on the northern side in the Northern Hemisphere and on the southern side in the Southern Hemisphere.

Lake-effect snow is another kind of common snowfall. Although the name suggests a particular correspondence of this type of precipitation to lake features, in fact, all narrow bands of water may generate it. Lake-effect snow occurs when the water temperature is considerably higher than the air temperature of a cold front progressing over a large water mass. Warm moist air is then attracted upward at a relatively fast rate, condensing to form vertically oriented clouds. If the temperature difference between the body of water and the air above is significant, say, 13 degrees or more, this can result in heavy and prolonged snowfall.

Mountainous areas are also prone to experiencing heavy snowfall. Accumulations typically occur on the windward side of the mountain as precipitation is 'squeezed out' of the warm moist air as it is forced to ascend the slopes; the moisture condenses upon contact with the colder air found at higher altitudes and heavy snowfall can then occur if ground conditions are sufficiently cold.

How Snow Is Formed

Snow crystals, tiny supercooled cloud droplets, form at extremely low temperatures in the atmosphere. Temperatures lower than minus 35 degrees Celsius are required for this supercool moisture to freeze by itself. In warmer clouds, an aerosol particle such as clay or desert dust, or an ice nucleus is needed for the freezing to start.

Once a droplet of water has frozen, it starts to grow in the supersaturated environment of the cloud. Eventually, due to its size, the cloud will not be able to contain the ice crystal anymore. At this point the ice crystal will fall to the ground and, if it is not melted by warmer air at lower altitudes, it will do so as snow. Although the ice crystals that land on the ground are actually transparent, hollow imperfections in them mean that light is scattered and they often appear white in colour owing to diffuse reflection of the whole spectrum of light.

The Snowflake

The shape of a snowflake is determined by the atmospheric conditions present at the time of its formation, specifically temperature and humidity. Between 0 and -3 degrees Celsius, thin flat crystals called planar crystals grow. From -3 to 8, the crystals form needles or prisms with pencil-like shapes. The shape then reverts back to plate-like until after 22 degrees Celsius when column-like structures (needles and prisms, etc.) begin to form again. At temperatures of 22 degrees and below, as well as the column-like structures, more complex growth patterns also form.



Snowfall in the British Isles

Snowfall occurs frequently in the U.K., but the quantities are typically small and it seldom persists for very long. In recent years, a trend towards milder, wetter winters has been developing, though the 2010/11, 2011/12 and 2012/13 winters seem to have bucked the trend and, indeed, represent the longest period of consecutive cold winters for more than 50 years. For its latitude, the British Isles should see far more prolonged cold weather in winter and regular snowfall. However, the Gulf Stream, a mild Atlantic Ocean current, keeps the climate several degrees warmer than regions of similar latitude in other parts of the world. As a consequence, despite the occasional incident of prolonged cold, Britain's winters are typically not very severe.

Some parts of the isles see little, if any, snow from year to year. The most snow-prone are the Pennines, the Scottish Highlands, the Welsh Hills and the mountains of Northern Ireland. The Scottish Highlands boasts the isles' highest peaks and also their only winter ski resorts. For years, unreliable snowfall has threatened to close these resorts, though, having had three consecutive bumper seasons, there is now less pressure on the Scottish ski industry, which, not so long ago, was threatened with going out of existence.

Long-term weather forecasts for the British Isles are notoriously hard to get right; however, so far, three months before the official start of the meteorological winter in December, forecasters are predicting another winter of record-breaking low temperatures. They point to sunspot and geothermal activity, and changes in the strength of the Gulf Stream as key indicators of the fact that a cold winter is in prospect. Were their predictions to be realised, then this would point to the isles undergoing a subtle climatic change and a return to more severe winters in general.

Questions 27-33

Do the following statements agree with the information given in Reading Passage 3?

Write:

YES if the statement agrees with the views of the writer

NO if the statement contradicts the views of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

27 When a warm front from an extratropical cyclone meets a cold front, snow is more likely to occur on the poleward side of the weather system.

28 Lake-effect snow is aptly named, given that it is a weather phenomenon which is only associated with lakes.

29 Heavy snowfall is more likely to be seen on the side of the mountain that is exposed to high winds.

30 In the absence of dust or a similar particle to start the freezing process, supercool moisture will not freeze in a cloud whose mean temperature is -34 degrees or more.

31 The real colour of snow is the same as the colour snow appears to be to the human eye.



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32 Snowflakes shaped like a prism are more likely to form in milder weather than are flakes with more intricate growth patterns.

33 The thin flat crystals created at temperatures of between zero and minus three degrees Celsius are more voluminous than column-like crystals.

Questions 34-37

Complete the sentences below.

Choose NO MORE THAN THREE WORDS from the passage for each answer.

While snowfall is not an uncommon feature of a United Kingdom winter, it is rare for significant

34 _____ to accumulate, or for the snow to remain on the ground for any great length of time.

Three consecutive winters have failed to follow the 35 _____ towards milder, wetter weather.

Despite its latitude, the British Isles does not suffer from 36 _____ on account of the influence of the Gulf Stream.

Should next winter be another unusually cold one, this may be indicative of the fact that a 37 _____ is under way.

Questions 38-40

Complete the summary with the list of words A-F below.

Write the correct letter, A-F, in spaces 38-40 below.

The British Winter

Snowfall is not an uncommon sight in Britain during winter, but such weather rarely persists for very long. In fact, up until a few years ago, it looked like British winters were getting milder. However, this perception has changed fairly dramatically over the last three winters, which have been the longest 38 _____ period of cold winters for more than half a century. It is now feared that the climate of the British Isles is changing and that, should the next few winters be equally severe, this could signal a permanent move to more 39 _____ winter weather in general. Scientists blame sunspot and geothermal activity as well as the 40 _____ influence of the Gulf Stream, which, until now, has kept Britain's climate milder than that of areas of similar latitude throughout the rest of the world.

A uninterrupted

B mild

C strengthening

D waning

E extreme

F direct



DRIVING ON AIR

No matter how costly, hazardous, or polluting they are, nor how tedious it is to be stuck in traffic jams, cars are here to stay. In fact, the global car industry is worth a massive two trillion dollars a year.

Recently, Guy Negre, a French engineer on Renault's Formula One engines, designed and produced the Airpod – a vehicle which runs on air, is lightweight and compact and capable of reaching moderate speeds.

Since the transport sector constitutes one-seventh of all air pollution, Negre spent 15 years developing the Airpod, hoping to significantly reduce greenhouse gas emissions. Petrol-electric hybrids, already on the market, are touted as being environmentally friendly, yet he says they are barely less polluting than combustion-engine vehicles. The Airpod, on the other hand, produces just 10% of the carbon monoxide of other cars.

Major manufacturers are now considering hydrogen as a power source for vehicles, but this technology may be decades away. Meantime, according to Negre, electric vehicles remain impractical: batteries are expensive, and need replacement within five years; recharging takes several hours.

Negre's secondary aim in creating the Airpod was to bring cars within reach of consumers in the developing world. To date, his most impressive deal has been with an Indian car manufacturer which predicts the Airpod will retail for the price of an average motorcycle.

Currently, only three-wheeled Airpods are available, but Negre has a four-wheeled, five-door family saloon, plus vans, buses, taxis, boats, and aircraft on the drawing board.

So what is an Airpod? This small vehicle resembles an ordinary car except that it is made mostly from fiberglass – ten times as strong as steel but very light – meaning an Airpod weighs just 220 kilograms (484 lb). It has glass windows and an aluminium engine. However, it uses a joystick instead of a steering wheel, and it has backward-facing passenger seats and a front-opening door.

The 180cc engine of an Airpod allows it to reach a speed of around 70 kilometres per hour (kph) (43 mph), and it can drive for about 220 kilometres (137 miles) before refilling is necessary. It takes as little as 90 seconds to pump air into an Airpod from a high-speed compressor at a gas station, with air costing a mere 50 cents for a 220-kilometre journey. An on-board pump can refill the tank at home overnight.



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How does an Airpod work? Quite simply: air is released through pistons in the engine, which drive the wheels. Compressed air tanks store up to 175 litres (46 gallons) of air at about 180 times the pressure of an average car tyre. Passengers and passers-by might have concerns about explosions with such pressure, but, in the rare event of one, the thermoplastic tanks split to release air, rather than shattering and exploding. In fact, the same tanks are already installed on natural-gas buses.

For longer journeys, there is a battery-assisted hybrid Airpod, which Negre maintains is capable of reaching 80 kph (50 mph) and travelling around 1500 kilometres (930 miles) on four litres of petrol, although this version has yet to be manufactured or tested.

Still, in its infancy, the Airpod has both supporters and critics. Marcus Waardenberg, the organiser of an Airpod trial at a major Dutch airport, was impressed. 'The Airpods went over 40 kph (25 mph), were quiet and manoeuvrable. Refilling was fast and straightforward.' As a result, his company is replacing its fleet of electric service vehicles with Airpods.

Perhaps more significantly, AK Jagadeesh, from the Indian conglomerate, Tata, signed a \$60-million deal. 'We're going to use Airpod technology in Tata's Nano car,' he said.

Ulf Bossel, a sustainable energy consultant, commented that the Airpod easily reaches speeds of over 50 kph (31 mph). 'Initially, it could capture the second-car market. Then, there are those older people who can no longer afford conventional cars.' Both Europe and North America have ageing populations.

Bill Robertson, a motoring journalist, noted that the Airpod would suit large numbers of people who make two or three trips a day of fewer than ten kilometers, or who live in distant suburbs of big cities where public transport is poor. If the Airpod looked a little sexier, there would be the potential for it to make inroads into the golf buggy sector, which currently uses electric vehicles.

Among the detractors of the Airpod is the former champion racer, Martella Valentina, who would prefer a vehicle with a more robust engine. 'There are so many aggressive drivers out there,' she said. 'As a woman, I don't feel safe in an Airpod.' She added, 'Refilling overnight is a drag.'

The automotive engineer, Hamid Khan, concurs, expressing skepticism about sufficient energy storage under reasonable pressure to drive the car any distance, let alone the alleged 220 kilometres (135 miles) before refill. He insists this is unconfirmed by independent tests. Stopping and starting in typical city conditions would also lower the range even further, and more distressingly, safety data are lacking for crash testing. 'Negre claims fibreglass is stronger than steel, but the



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Airpod looks as though it would crumble under the wheels of a normal saloon,' commented Khan.

Nevertheless, Negre has signed deals to manufacture his car in the US, Latin America, India, and several European countries. Compressed air may no longer take a back seat to other power sources, and it is even conceivable that one day we may be flying in aircraft that fly on air.

DRIVING ON AIR

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Nevertheless, Negre has signed deals to manufacture his car in the US, Latin America, India, and several European countries. Compressed air may no longer take a back seat to other power sources, and it is even conceivable that one day we may be flying in aircraft that fly on air.

Questions 1-8

Complete the summary using the list of words, A-O, below:



Write the correct letter, A-O, in boxes 1-8 on your answer sheet.

The 1_____ of combustion-engine cars continues 2_____ there being problems with them. According to Negre, an automotive engineer and inventor, a(n) 3_____, the petrol-electric car, is really not much less 4_____. Negre believes his Airpod is far cleaner and cheaper, and will 5_____ drivers in the developing world in particular.

An Airpod is lighter than other cars at only 6_____ kilograms. The highest confirmed speed it can reach is around 7_____ kph. It can be refilled fast at a service station or more slowly at home. Some people may be worried about the high-pressure gas stored onboard an Airpod, but its tanks are safe and already in 8_____ on public buses.

- | | | |
|---------------|------------|-------------|
| A exist | B popular | C polluting |
| D 80 | E benefit | F although |
| G 70 | H polluted | I alternate |
| J alternative | K 220 | L use |
| M popularity | N 180 | O despite |

Questions 9-13

Look at the following statements and the list of people below.

Match each statement with the correct person: A, B, C, or D.

Write the correct letter, A, B, C, or D, in boxes 9-13 on your answer sheet.

- 9 He claims the hybrid Airpod can travel 1500 kilometres on four litres of petrol.
- 10 He imagines the Airpod will appeal to the elderly.
- 11 He doesn't think the Airpod will compete with golf buggies unless it changes its appearance.
- 12 He doesn't believe the Airpod can drive as far as its creator maintains.
- 13 He has agreed to the manufacture of the Airpod in a number of countries.

List of people

- A Bill Robertson
- B Guy Negre
- C Hamid Khan
- D Ulf Bossel

羊驼雅思

The Evolutionary Mystery: Crocodile Survives

A Even though crocodiles have existed for 200 million years, they're anything but primitive. As crocodiles' ancestors, crocodilia came to adapt to an aquatic lifestyle. When most of the other contemporary reptiles went extinct, crocodiles were able to make it because their bodies changed and they adapted better to the climate. They witnessed the rise and fall of the dinosaurs, which once ruled the planet, and even the 65 million years of alleged mammalian dominance didn't wipe them off. Nowadays,



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the crocodiles and alligators are not that different from their prehistoric ancestors, which proves that they were (and still are) incredibly adaptive.

B The first crocodile-like ancestors came into existence approximately 230 million years ago, and they had many of the features which make crocodiles natural and perfect stealth hunters: streamlined body, long tail, protective armour and long jaws. They are born with four short, webbed legs, but this does not mean that their capacity to move on the ground should ever be underestimated. When they move, they are so fast that you won't even have any chance to try making the same mistake again by getting too close, especially when they're hunting.

C Like other reptiles, crocodiles are poikilothermal animals (commonly known as coldblooded, whose body temperature changes with that of the surroundings) and consequently, require exposure to sunlight regularly to raise body temperature. When it is too hot, they would rather stay in water or shade. Compared with mammals and birds, crocodiles have a slower metabolism, which makes them less vulnerable to food shortage. In the most extreme case, a crocodile can slow its metabolism down even further, to the point that it would survive without food for a whole year, enabling them to outlive mammals in relatively volatile environments.

D Crocodiles have a highly efficient way to catch prey. The prey rarely realises there might be a crocodile under the water because the crocodile makes a move without any noise or great vibration when spotting its prey. It only keeps its eyes above the water level. As soon as it feels close enough to the victim, it jerks out of the water with its wide open jaws. Crocodiles are successful because they are capable of switching feeding methods. It chases after fish and snatches birds at the water surface, hides in the waterside bushes in anticipation of a gazelle, and when the chance to ambush presents itself, the crocodile dashes forward, knocks the animal out with its powerful tail and then drags the prey into the water to drown.

E In many crocodilian habitats, the hot season brings drought that dries up their hunting grounds, leaving it harder for them to regulate body temperatures. This actually allowed reptiles to rule. For instance, many crocodiles can protect themselves by digging holes and covering themselves in mud, waiting for months without consuming any food or water until the rains finally return. They transform into a quiescent state called aestivation.

F The majority of crocodilia are thought to go into aestivation during the dry season. In a six-year study by Kennett and Christian, the King Crocodiles, a species of Australian freshwater crocodiles, spent nearly four months a year underground without access to water resources. Doubly labelled water was applied to detect field metabolic rates and water flux, and during some years, plasma fluid samples were taken once a month to keep track of the effects of aestivation regarding the accumulation of nitrogenous wastes and electrolyte concentrations.



G The study discovered that the crocodiles' metabolic engines function slowly, creating waste and exhausting water and fat reserves. Waste is stored in the urine, becoming more and more concentrated. Nevertheless, the concentration of waste products in blood doesn't fluctuate much, allowing the crocodiles to carry on their normal functions. Besides, even though the crocodiles lost water reserves and body weight when underground, the losses were proportional; upon emerging, the aestivating animals had no dehydration and displayed no other harmful effects such as a slowed-down growth rate. The two researchers reckon that this capacity of crocodiles to get themselves through the harsh times and the long starvation periods is sure to be the answer to the crocodilian line's survival throughout history.

Questions 14-20

Reading Passage 2 has seven paragraphs, A-G.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, i-xi, in boxes 14-20 on your answer sheet.

	List of Headings
i	The positive impact of drought
ii	Research findings into crocodile survival
iii	The slow metabolism which makes the crocodile a unique animal
iv	The perfectly designed body for a great land roamer
v	Shifting eating habits and food intake
vi	A project on a special mechanism
vii	Regulating body temperature by the surrounding environment
viii	Body structure offers underwater aid to a successful predator
ix	A history of the supreme survivors
x	What makes the crocodile the fastest running animal on land
xi	Competition between crocodiles and other animals
14.	Paragraph A
15.	Paragraph B
16.	Paragraph C
17.	Paragraph D
18.	Paragraph E
19.	Paragraph F
20.	Paragraph G

Questions 21-26

Complete the summary below,

Choose NO MORE THAN TWO WORDS from the passage for each answer.

Write your answers in boxes 21-26 on your answer sheet.



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Aestivation

In many places inhabited by crocodilians, most types of crocodiles have evolved a successful scheme to survive in the drought brought by a 21_____. According to Kennett and Christian's six-year study of Australian freshwater crocodiles' aestivation, they found aestivating crocodiles spent around 22_____ of the year underground and had no access to 23_____. The amount of water in the body declined proportionately with 24_____; thus there is no sign of 25_____ and other health-damaging impact on the crocodiles even after an aestivation period. This super capacity helps crocodiles endure the tough drought without slowing their speed of 26_____.

Timekeeper: Invention of Marine Chronometer

A Up to the middle of the 18th century, the navigators were still unable to exactly identify the position at sea, so they might face a great number of risks such as the shipwreck or running out of supplies before arriving at the destination. Knowing one's position on the earth requires two simple but essential coordinates, one of which is the longitude.

B The longitude is a term that can be used to measure the distance that one has covered from one's home to another place around the world without the limitations of naturally occurring baseline like the equator. To determine longitude, navigators had no choice but to measure the angle with the naval sextant between Moon centre and a specific star— lunar distance—along with the height of both heavenly bodies. Together with the nautical almanac, Greenwich Mean Time (GMT) was determined, which could be adopted to calculate longitude because one hour in GMT means 15-degree longitude. Unfortunately, this approach laid great reliance on the weather conditions, which brought great inconvenience to the crew members. Therefore, another method was proposed, that is, the time difference between the home time and the local time served for the measurement. Theoretically, knowing the longitude position was quite simple, even for the people in the middle of the sea with no land in sight. The key element for calculating the distance travelled was to know, at the very moment, the accurate home time. But the greatest problem is: how can a sailor know the home time at sea?



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C The simple and again obvious answer is that one takes an accurate clock with him, which he sets to the home time before leaving. A comparison with the local time (easily identified by checking the position of the Sun) would indicate the time difference between the home time and the local time, and thus the distance from home was obtained. The truth was that nobody in the 18th century had ever managed to create a clock that could endure the violent shaking of a ship and the fluctuating temperature while still maintaining the accuracy of time for navigation.

D After 1714, as an attempt to find a solution to the problem, the British government offered a tremendous amount of £20,000, which were to be managed by the magnificently named 'Board of Longitude'. If timekeeper was the answer (and there could be other proposed solutions, since the money wasn't only offered for timekeeper), then the error of the required timekeeping for achieving this goal needed to be within 2.8 seconds a day, which was considered impossible for any clock or watch at sea, even when they were in their finest conditions.

E This award, worth about £2 million today, inspired the self-taught Yorkshire carpenter John Harrison to attempt a design for a practical marine clock. In the later stage of his early career, he worked alongside his younger brother James. The first big project of theirs was to build a turret clock for the stables at Brockelsby Park, which was revolutionary because it required no lubrication. Harrison designed a marine clock in 1730, and he travelled to London in seek of financial aid. He explained his ideas to Edmond Halley, the Astronomer Royal, who then introduced him to George Graham, Britain's first-class clockmaker. Graham provided him with financial aid for his early-stage work on sea clocks. It took Harrison five years to build Harrison Number One or H1. Later, he sought the improvement from alternate design and produced H4 with the giant clock appearance. Remarkable as it was, the Board of Longitude wouldn't grant him the prize for some time until it was adequately satisfied.

F Harrison had a principal contestant for the tempting prize at that time, an English mathematician called John Hadley, who developed the sextant. The sextant is the tool that people adopt to measure angles, such as the one between the Sun and the horizon, for a calculation of the location of ships or planes. In addition, his invention is significant since it can help determine longitude.

G Most chronometer forerunners of that particular generation were English, but that doesn't mean every achievement was made by them. One wonderful figure in the history is the Lancastrian Thomas Earnshaw, who created the ultimate form of chronometer escapement—the spring detent escapement—and made the final decision on format and productions system for the marine chronometer, which turns it into a genuine modern commercial product, as well as a safe and pragmatic way of navigation at sea over the next century and half.



Questions 27-31

Reading Passage 3 has seven paragraphs, A-G.

Which paragraph contains the following information?

Write the correct letter, A-G, in boxes 27-31 on your answer sheet.

NB You may use any letter more than once.

- 27. a description of Harrison's background
- 28. problems caused by poor ocean navigation
- 29. the person who gave financial support to Harrison
- 30. an analysis of the long-term importance of sea clock invention
- 31. the practical usage of longitude

Questions 32-34

Do the following statements agree with the information given in Reading Passage 1 ?

In boxes 32-34 on your answer sheet, write

TRUE	if the statement agrees with the information
FALSE	if the statement contradicts the information
NOT GIVEN	If there is no information on this

- 32. In theory, sailors can easily calculate their longitude position at sea.
- 33. To determine longitude, the measurement of the distance from the Moon to a given star is essential.
- 34. Greenwich Mean Time was set up by the English navigators.

Questions 35-40

Complete the sentences below.

Choose NO MORE THAN TWO WORDS AND/OR A NUMBER from the passage for each answer.

Write your answers in boxes 35-40 on your answer sheet.

Sailors were able to use the position of the Sun to calculate 35_____.

An invention that could win the competition would lose no more than 36_____ every day.

John and James Harrison's clock worked accurately without 37_____.

Harrison's main competitor's invention was known as 38_____.

Hadley's instrument can use 39_____ to make a calculation of location of ships or planes.

The modern version of Harrison's invention is called the 40_____.

参考答案

William Gilbert and Magnetism



- 2i
- 3vi
- 4x
- 5ix
- 6iv
- 7ii
- 8TRUE
- 9TRUE
- 10NOT GIVEN
- 11-13C,D,E

Being Left-handed in a Right-handed World

- 14C
- 15A
- 16B
- 17F
- 18D
- 19B
- 20F
- 21D
- 22A
- 23YES
- 24NO
- 25NOT GIVEN
- 26NOT GIVEN

Improving Patient Safety

- 27C
- 28D
- 29B
- 30A
- 31D
- 32C
- 33B
- 34D
- 35A
- 36E
- 37F
- 38B
- 39B
- 40A

Learning by Examples

- 1D



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- 2A
- 3C
- 4E
- 5FALSE
- 6TRUE
- 7TRUE
- 8FALSE
- 9less
- 10social
- 11watched
- 12observer
- 13Nutcracker

The Ant and the Mandarin

- 14C
- 15F
- 16A
- 17D
- 18E
- 19TRUE
- 20FALSE
- 21FALSE
- 22TRUE
- 23TRUE
- 24FALSE
- 25TRUE
- 26FALSE

What is a dinosaur?

- 27vi
- 28xi
- 29xiii
- 30vii
- 31iv
- 32v
- 33viii
- 34skeletal anatomy
- 35eosuchians
- 36two long bones
- 37B
- 38G
- 39H
- 40D



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The Mozart Effect

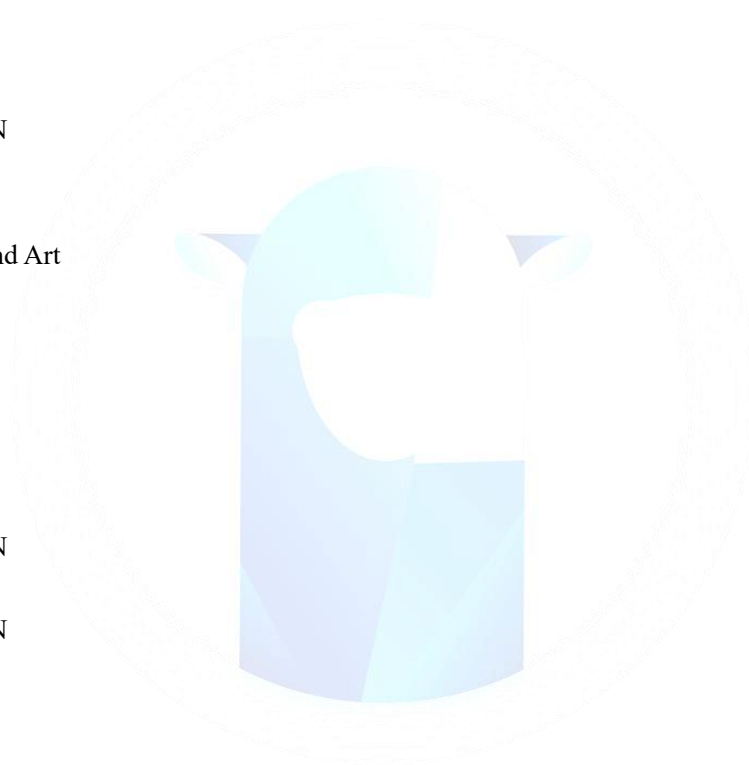
- 1D
- 2G
- 3B
- 4A
- 5F
- 6short/10-minute
- 7complex
- 8rats
- 9TRUE
- 10FALSE
- 11FALSE
- 12NOT GIVEN
- 13TRUE

Life-Casting and Art

- 14C
- 15E
- 16B
- 17F
- 18D
- 19NO
- 20NOT GIVEN
- 21NO
- 22NOT GIVEN
- 23NO
- 24YES
- 25B
- 26D

Music: Language We All Speak

- 27iii
- 28vii
- 29iv
- 30i
- 31viii
- 32F
- 33B
- 34E
- 35D
- 36G
- 37A
- 38C
- 39C



羊驼雅思



40C

The Sweet Scent of Success

1F

2E

3C

4B

5G

6D

7A

8C

9A

10D

11B

12B

13D

Optimism and Health

147/seven

15lung function

16immune system

17heart patients

18C

19A

20E

21G

22D

23NOT GIVEN

24NOT GIVEN

25YES

26YES

Internal Market: Selling the inside

27C

28C

29D

30A

31E

32B

33YES

34YES

35NO

36NO

37NOT GIVEN



羊驼雅思



更多雅思资料领取



更多雅思资料领取

38NO

39NOT GIVEN

40YES

Morse Code

1x

2xi

3iii

4i

5vi

6ii

7ix

8vii

9FALSE

10TRUE

11TRUE

12NOT GIVEN

13NOT GIVEN

Wealth in A Cold Climate

14iii

15vi

16i

17ii

18ix

19v

20iv

21yellow fever

22Finland

23institutions/governments

24Europe

25inkorn wheat

26Singapore

High speed photography

27C

28A

29D

30B

31lapse/time interval

32continuous action

33time

34greater length

35slowed down proportionately



36A
37C
38G
39E
40F

The Impact of the Potato

1FALSE
2FALSE
3NOT GIVEN
4TRUE
5TRUE
6flower
7prejudice
8reverse
9meat
10crops
11soil
12cultivation
13investment

Coastal Archaeology of Britain

14B
15C
16D
17TRUE
18FALSE
19TRUE
20FALSE
21NOT GIVEN
22TRUE
23TRUE
24-26B,D,F

Living Dunes

27iv
28x
29iii
30vii
31i
32v
33ix
34linear
35obstacle



羊驼雅思



- 36acoustic
- 37barchan
- 38shape
- 39tone
- 40minerals

What the Managers Really Do?

- 1C
- 2A
- 3C
- 4B
- 5B
- 6A
- 7-8B,E
- 9FALSE
- 10NOT GIVEN
- 11NOT GIVEN
- 12FALSE
- 13TRUE

Storytelling, From Prehistoric Craves To Modern Cinemas

- 14D
- 15G
- 16A
- 17B
- 18H
- 19B
- 20B
- 21C
- 22A
- 23Poetics
- 24tragedy
- 25landmarks(-stories)
- 26flaw/weakness

Monkeys and Forests

- 27G
- 28A
- 29C
- 30B
- 31H
- 32D
- 33C
- 34A



更多雅思资料领取



更多雅思资料领取

35B

36fruit

37plant toxins

38reproductive/birth

39water

40drought

Does An IQ Test Prove Creativity?

1FALSE

2NOT GIVEN

3TRUE

4TRUE

5A

6E

7F

8C

9D

10scalp electrodes

11inspiration and elaboration

12alpha wave activity

13difference

How Well Do We Concentrate?

14B

15E

16F

17C

18D

19B

20D

21A

22E

23E

24email voice

25prefrontal cortex

26group meetings

Robert Louis Stevenson

27D

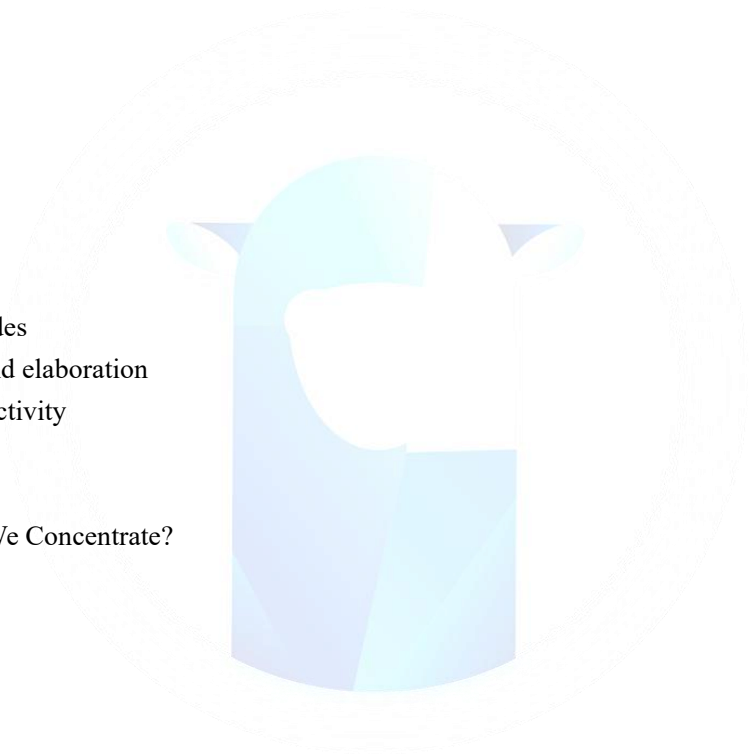
28D

29B

30A

31C

32TRUE





更多雅思资料领取



更多雅思资料领取

33NOT GIVEN

34FALSE

35FALSE

36F

37A

38H

39G

40I

The Connection Between Culture and Thought

1E

2G

3D

4B

5E

6B

7A

8C

9A

10living environment

11conclusion

12qualitative (descriptions)

13clear definition

Implication of False Belief Experiments

14G

15F

16C

17D

18A

19B

20E

21chocolate

22information

23four/4

24older

25adults

26challenging

The Biggest Impact

27viii

28ii

29ix

30vii

羊驼雅思



更多雅思资料领取



更多雅思资料领取

- 31iv
- 32iii
- 33Chicxulub Crater
- 34company policy
- 35iridium
- 36widespread skepticism
- 37mass extinction
- 38B
- 39C
- 40A

BIOMETRICS

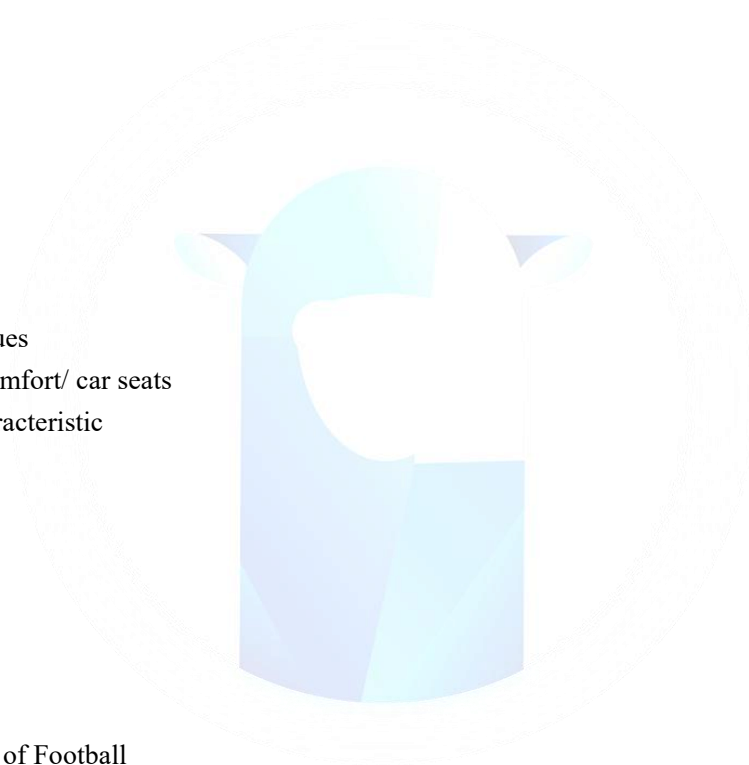
- 27. H
- 28. A
- 29. F
- 30. C
- 31. G
- 32. privacy issues
- 33. personal comfort/ car seats
- 34. unique characteristic
- 35. A
- 36. E
- 37. F
- 38. K
- 39. C
- 40. J

The Beginning of Football

- 1. ix
- 2. x
- 3. i
- 4. vii
- 5. iii
- 6. viii
- 7. vi
- 8. I
- 9. D
- 10. B
- 11. H
- 12. E
- 13. A

THE DEVELOPING WORLD

- 14. ii



羊驼雅思



更多雅思资料领取



更多雅思资料领取

- 15. i
- 16. vi
- 17. viii
- 18. iv
- 19. YES
- 20. NOT GIVEN
- 21. YES
- 22. NO
- 23. D
- 24. F
- 25. A
- 26. E

Father of modern management

- 27i
- 28v
- 29ix
- 30vi
- 31ii
- 32viii
- 33NOT GIVEN
- 34NOT GIVEN
- 35YES
- 36NO
- 37-38A,E
- 39-40 A,C

Growing of the Aging Society

- 1 vii
- 2 i
- 3 iv
- 4 ii
- 5 iii
- 6 vi
- 7 FALSE
- 8 TRUE
- 9 TRUE
- 10 NOT GIVEN
- 11 FALSE
- 12 TRUE
- 13 NOT GIVEN

The Mysteries of Water

- 14 i





更多雅思资料领取



更多雅思资料领取

- 15 vi
- 16 iii
- 17 viii
- 18 v
- 19 NO
- 20 YES
- 21 NOT GIVEN
- 22 NO
- 23 NO
- 24 YES
- 25 YES
- 26 NOT GIVEN

The Beginnings of Art Therapy

- 27 NOT GIVEN
- 28 TRUE
- 29 NOT GIVEN
- 30 NOT GIVEN
- 31 TRUE
- 32 FALSE
- 33 FALSE
- 34 variety of
- 35 therapeutic benefits
- 36 creative process
- 37 not incompatible
- 38 B
- 39 D
- 40 E

Education Philosophy

- 1 iii
- 2 v
- 3 i
- 4 vi
- 5 A
- 6 B
- 7 B
- 8 C
- 9 C
- 10 A
- 11 B
- 12 A
- 13 D



羊驼雅思



更多雅思资料领取



更多雅思资料领取

DRINKING FILTERED WATER

- 14A
- 15F
- 16E
- 17G
- 18C
- 19D
- 20B
- 21TRUE
- 22FALSE
- 23TRUE
- 24FALSE
- 25NOT GIVEN
- 26NOT GIVEN

ALL YOU NEED TO KNOW ABOUT SNOW

- 27YES
- 28NO
- 29YES
- 30YES
- 31NO
- 32YES
- 33NOT GIVEN
- 34quantities
- 35trend
- 36severe winters|very cold weather
- 37(subtle) climatic change
- 38A
- 39E
- 40D

DRIVING ON AIR

- 1 M
- 2 O
- 3 J
- 4 C
- 5 E
- 6 K
- 7 G
- 8 L
- 9 B
- 10 D
- 11 A
- 12 C

羊驼雅思



更多雅思资料领取



更多雅思资料领取

13 B

The Evolutionary Mystery: Crocodile Survives

14ix

15iv

16iii

17v

18i

19vi

20ii

21hot season/dry season

22four months

23water resources

24body weight

25dehydration

26growth

Timekeeper: Invention of Marine Chronometer

27E

28A

29E

30G

31B

32TRUE

33TRUE

34NOT GIVEN

35local time

36 2.8 seconds

37lubrication

38(a/the) sextant

39angles

40marine chronometer

羊驼雅思